ADVANCED NAVIGATION



Motus
Reference Manual

v2.13

2026-09-23

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1 Motus

Motus is a miniature high accuracy MEMS Inertial Measurement Unit (IMU) and Altitude Heading Reference System (AHRS). It features high accuracy MEMS accelerometers and gyroscopes, and magnetometers. Motus is calibrated for sensor errors over a wide temperature range.

This manual covers the Rugged and OEM variants of Motus.



Figure 1: Image: Motus Rugged and OEM variants

1.1 Foundational Knowledge

Before working with an Inertial Navigation System, you should have background knowledge of geospatial concepts, and principles of Inertial Navigation Systems.

Helpful resources include:

- **Inertial Navigation Systems Primer**
- **Glossary**

1.2 Disclaimer

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2 Revision History

Version	Date	Changes
2.13	2026-09-23	Updated to Motus Firmware 2.51, which includes ANPP improvements for aiding source configuration packets Updated to INS Manager 8.2, including Aiding Source Documented Motus in the Aiding Source Configuration Packet
2.12	2026-08-24	Updated to Motus Firmware 2.5, with updated ANPP packets Updated to INS Manager 8.1, including new Aiding Sources Configuration window Merged Rugged and OEM versions of the manual Added PCB Design Tips for OEM
2.11	2026-02-27	Minor doc corrections
2.10	2025-12-03	Update to INS Manager 8.0
2.9.3	2025-08-19	Clarification added to spec for voltage
2.9.2	2025-05-01	Typo re INS license upgrade
2.9.1	2025-04-14	Minor formatting changes and corrections
2.9	2024-03-11	Part Number Corrections
2.8	2024-03-25	Updated firmware changelog, see **2.2 Firmware Release Notes**
2.7	2024-02-20	Updated firmware changelog, see **2.2 Firmware Release Notes** Updated automatic magnetometer calibration explanation to reflect Firmware changes, see **8.6 Magnetic Calibration** ANPP packets 70 (Raw DVL Data Packet) and 185 (Installation Alignment Packet) updated Content migrated to new documentation platform Minor editorial changes
2.6	2023-11-14	Updated software changelog, see **Using Motus Manager** Updated automatic magnetometer calibration explanation to reflect Firmware changes, see **8.6 Magnetic Calibration** Updated Disclaimer
2.5	2023-02-20	Updated software changelog, see **Using Motus Manager** Updated software installation information to include Java 11 requirement, see **Using Motus Manager** and **4 Getting Started** Added instructions to run Manager on MacOS, see **Using Motus Manager**
2.4	2022-10-12	Added Device ID table, see **13.3 System Packets** Updated software changelog, see **Using Motus Manager** Updated firmware changelog, see **2.2 Firmware Release Notes**

Version	Date	Changes
		Updated product part numbers, see **5 Part Numbers and Ordering Options** Updated Log Converter Tool image and description, section 12.9.3 Added breakout cables to list of accessories, see **5 Part Numbers and Ordering Options** Updated connector pinout to include RS422, section 9.10 Added RS422 breakout cable information, section 9.12.2 Added ODU connector pin location diagram, section 9.10 Clarified geoid, ellipsoid and orthometric heights, section 5.7 Updated automatic magnetometer calibration explanation to reflect Firmware changes, section 11.5.3 Updated External Time Packet to show it is Read only, see **13.4.33 External Time Packet** Added note that Motus Manager only works on systems with x64 architecture, sections 7.2 and 10.1
2.3	2021-10-25	Updated firmware changelog, see **2.2 Firmware Release Notes** Updated hardware changelog, see **2.1 Hardware Changelog**
2.2	2021-07-12	Updated firmware changelog, see **2.2 Firmware Release Notes** Improved power supply specifics section 13.4 & 14.5
2.1	2021-01-22	Updated firmware changelog, see **2.2 Firmware Release Notes** Updated hardware changelog, see **2.1 Hardware Changelog** Updated OEM images section 13, 14 & 14.4
2.0	2020-10-28	Update for v2 Hardware, sections 9, v1 hardware now obsolete Update Manager section 12 Updated OEM drawings and detail, sections 13 and 14
1.1	2019-11-29	Updated firmware changelog, see **2.2 Firmware Release Notes** Updated hardware changelog, see **2.1 Hardware Changelog** Updated dynamic pin functions table, **12.3.1 Dynamic Pin Functions** Updated GPIO configuration packet, see **13.5.6 GPIO Configuration Packet** Added pressure depth transducer deprecated input function, see **12.3.1.21 Pressure Depth Transducer (deprecated)** Added Nortek DVL input function, see **12.3.1.42 Nortek DVL Input** Added Teledyne DVL input function, see **12.3.1.41 Linkquest USBL Input** Added Tritech USBL input function, see **12.3.1.41 Linkquest USBL Input** Added Linkquest DVL input function, see **12.3.1.40 Linkquest DVL Input** Added Linkquest USBL input functionality, see **12.3.1.41 Linkquest USBL Input** Added reverse alignment input functions, see **12.3.1.16 Reverse Alignment, Forward Low** Added zero angular velocity input function, see **12.3.1.40 Linkquest DVL Input** Clarified port baud rates in sections Updated read status for external air data packet, see **13.4.39 External Air Data Packet** Updated read status for external odometer packet, see **13.4.38 External Odometer Packet** Modified quaternion element labels Clarified geoid height packet description Removed external pitot pressure packet

Version	Date	Changes
		Updated external body velocity packet size and definition, see **13.4.28 External Body Velocity Packet** Added wheel speed sensor specifications Corrected Raw GNSS packet velocity units Updated GPIO output configuration packet Clarified negative acknowledgement packets Clarified 1PPS output voltages Updated specifications Clarified navigation specifications Corrected power supply drop out voltage Added odometer information to Installation chapter Added magnetics information to Installation chapter Updated operating temperature information Corrected IP rating of rugged enclosure Clarified raw GNSS data output packets Corrected system status and filter status flags Corrected raw sensors packet fields
1.0	2016-03-17	Initial Release

Table 1: Revision History

2.1 Hardware Changelog

Version	Date	Changes
2.3	2021-09-06	Pressure sensor update
2.2	2020-09-29	Internal manufacturing improvements
2.0	2020-09-04	Minor changes to the baseplate thickness. No increase to overall height, length or width Added alignment markings to baseplate Improved sensor performance Removal of 0.2 mm material from from the inner area of the base of the housing to make mounting onto uneven surfaces easier
1.4	2018-07-16	Internal manufacturing improvements No noticeable changes for customers
1.3	2017-09-07	Improvements to ADC synchronisation results in slightly improved performance under high dynamics
1.2	2016-10-05	Internal manufacturing improvements
1.0	2015-09-14	Initial release

Table 2: Hardware Changelog

2.2 Firmware Release Notes

Motus OEM uses the same firmware as Motus Rugged.

Version	Date	Changes
2.51	2026-08-28	**ANPP Improvements:** Added support for aiding source configuration packets.
2.5	2026-05-05	**Improved External Aiding Configuration:** Improved external aiding source lever-arm and alignment configurations and status reporting across aiding sources. **Expanded DVL Support:** Added support for Nortek Nucleus DVL and Water Linked DVL. **Train Vehicle Profile:** Added support for Train vehicle profile.
2.41	2024-03-18	Fixed Pressure Sensor reporting NaN values on v2.3 HW
2.4	2024-02-02	Enabled the pressure sensor for Motus hardware v2.3 and newer Improved stability and performance of automatic magnetometer calibration algorithm for some vehicle profiles, see 8.6 Magnetic Calibration Updated DVL input to use Odometer Offset Fix magnetometer sensor recovery and failure flag triggering when filter options packet is received Increased magnetometer sample rate to 100 Hz on newer hardware version devices Fix intermittent fault on start-up Extended external GNSS lever arm offset range to 50 m horizontal and 100 m vertical Fix Time Filter initialised when receiving 1PPS and Raw GNSS packet in the GPIO or Aux port Fix timestamp occasionally jumping backwards in NMEA0183 packets Fix transient Magnetometer Overrange flag on cold boot Fix intermittent filter resets Added check to prevent zero scaling in all magnetometer calibration algorithms Improved reliability of time calculations Fix External GNSS data processing based on Internal GNSS Enabled filter option
2.3	2021-09-06	Update to support hardware v2.3 Fix Magnetometer Axis Swap for hardware v2.0
2.2	2021-06-29	Fix Magnetometer axis swap for hardware v2.1 and v2.2
2.1	2021-12-14	Bug fix for calibration issue (magnetometer scaling) Minor fixes to magnetic initialisation
1.8	2020-03-02	Support for v2 hardware
1.6	2019-11-28	Improvements to raw sensor calibration algorithm Prevents fast temperature variations from introducing digital noise into accelerometer values which was regression from firmware version 1.4
1.5	2019-	Improvements to raw sensor calibration algorithm

Version	Date	Changes
	10-03	More accurate temperature compensation of sensors leads to overall performance improvement Improvements to heading integrity monitoring algorithm to better reject erroneous measurements
1.4	2019-09-05	Improvements to raw sensor calibration algorithm Better sensor performance through rapid temperature changes leads to performance improvement in cases of fast temperature ramps
1.3	2019-05-31	Improvements to integrity monitoring allows better rejection of erroneous position and velocity data
1.1	2018-07-17	New zero angular velocity functionality Better hot start performance
1.0	2016-03-01	Initial release

Table 3: Firmware Changelog

3 Evaluation Kit

Motus is supplied in Evaluation Kits for the Rugged and OEM variants. Each kit contains everything required to get started operating the system right away. The Evaluation Kit is supplied in a rugged transport case to protect the equipment during transit.

3.1 Rugged Evaluation Kit Contents



Figure 2: Motus Rugged Evaluation Kit Contents

3.2 OEM Evaluation Kit Contents

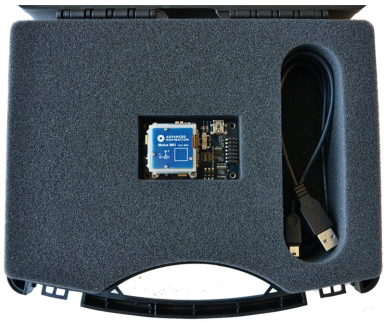


Figure 3: Motus OEM Evaluation Kit Contents

4 Getting Started

1. Plug the interface cable into Motus.
2. Plug the USB end of the interface cable into your computer.
3. Download the Motus Manager software from the Motus page of the **Advanced Navigation website**.

   Java 11 is required to run Motus Manager. Adoptium JRE 11 is the recommended Java installer to be used on all platforms. Install the latest version of Java 11 from **Adoptium**, selecting the correct operating system and architecture for your computer.
4. Click the connect button in Motus Manager.
5. The various windows in Motus Manager can be used to view the real time data.
6. To view the data logs, click disconnect in Motus Manager. In the Tools menu, select Log Converter and press Convert. The *.anpp binary log file will be converted to CSV files that can be opened with popular data processing programs such as Matlab or Microsoft Excel. The log files can be found in the same folder as the Motus Manager software.

There are additional factors that you must review before you proceed with the installation. For more information, see **7 Installing**

5 Part Numbers and Ordering Options

5.1 Evaluation Kits

Part Number	Description	Notes
MOTUS-RUG-EK	Motus Rugged Evaluation Kit (RS232)	Motus Rugged unit 2 metre USB to RS232 cable
MOTUS-OEM-EK	Motus OEM Evaluation Kit (RS232)	Motus OEM unit Motus OEM development board Mini USB cable

Table 4: Evaluation Kit Part Numbers

5.2 Standalone Units

Part Number	Description	Notes
MOTUS2-RUG	Motus Rugged Unit (RS232)	Motus Rugged unit No cables included IMU/AHRS functionality only
MOTUS2-OEM	Motus OEM Unit (RS232)	Motus OEM unit No cables included IMU/AHRS functionality only
MOTUS2-RUG-422	Motus Rugged Unit (RS422)	As above with optional RS422 interface instead of standard RS232 interface

Table 5: Standalone Unit Part Numbers

5.3 Accessories

Part Number	Description	Notes
A503-SDC20796	ODU plug to FTDI USB cable	Motus rugged ODU plug with 2 metres of cable to FTDI USB plug, see **6.7.1 Rugged Evaluation Kit USB Cable**
A503-SDC20740-2M	ODU plug with 2m cable (unterminated)	Motus rugged ODU plug with 2 metres of unterminated cable, see **6.6 Connector Pin-out**
A503-SDC20740-5M	ODU plug with 5m cable (unterminated)	Motus rugged ODU plug with 5 metres of unterminated cable, see **6.6 Connector Pin-out**
A503-SDC20740-10M	ODU plug with 10m cable (unterminated)	Motus rugged ODU plug with 10 metres of unterminated cable, see **6.6 Connector Pin-out**
A503-SDC20740-	ODU plug with 20m cable (unter-	Motus rugged ODU plug with 20 metres of unterminated cable, see **6.6 Connector Pin-out**

Part Number	Description	Notes
20M	minated)	
A503-SDC20753	ODU to D9 connectors and DC socket (RS232)	Motus rugged ODU plug with 2m of cable to industry standard D9 connectors and DC socket, see **6.7.2.1 RS232 Breakout Cable (A503-SDC20753)**
A503-SDC20755	ODU to D9 connectors and DC socket (RS422)	Motus rugged ODU plug with 2m of cable to industry standard D9 connectors and DC socket, see **6.7.2.2 RS422 Breakout Cable (A503-SDC20755)**

Table 6: Accessories Part Numbers

6 Specifications

6.1 Navigation Specifications

Motus as a standalone IMU/AHRS can achieve the following performance:

Parameters	Specification
Roll & Pitch Accuracy	0.05 °
Heading Accuracy (magnetic)	0.8 °
Orientation Range	Unlimited
Hot Start Time	500 ms
Internal Filter Rate	1000 Hz
Output Data Rate	Up to 1000 Hz

Table 7: Motus Orientation Specifications

As an INS, Motus can be interfaced to any of the leading brands of GNSS receivers to create an OEM GNSS/INS solution. When using a Trimble BD992 receiver, for example, the following specifications are achievable:

Parameters	Specification
Horizontal Position Accuracy	0.8 m
Vertical Position Accuracy	1.5 m
Horizontal Position Accuracy (RTK)	0.008 m
Vertical Position Accuracy (RTK)	0.015 m
Velocity Accuracy	0.007 m/s
Roll & Pitch Accuracy (GNSS aided)	0.03 °
Heading Accuracy (GNSS aided)	0.1 °
Heading Accuracy (dual antenna GNSS)	0.05 °

Table 8: Motus Navigation Specifications (with external GNSS)

6.2 Sensor Specifications

Parameter	Accelerometers	Gyroscopes	Magnetometers	Pressure
Range (dynamic)	±10 g	±475 °/s	±8 G	10-120 kPa
Bias Instability	8 ug	0.2 °/hr	-	10 Pa
Initial Bias	< 0.45 mg	< 3 °/hr	-	-
Initial Scaling Error	< 0.03 %	< 0.02 %	< 0.07 %	-

Parameter	Accelerometers	Gyroscopes	Magnetometers	Pressure
Scale Factor Stability	< 0.04 %	< 0.03 %	< 0.09 %	-
Non-linearity	< 0.05 %	< 0.03 %	< 0.08 %	-
Cross-axis Alignment Error	< 0.05 °	< 0.05 °	< 0.05 °	-
Noise Density	2 ug/√Hz	6 °/hr/√Hz	210 uG/√Hz	0.56 Pa/√Hz
Bandwidth	250 Hz	200 Hz	110 Hz	50 Hz

Table 9: Motus Sensor Specifications

6.2.1 Sensor Calibration

Motus's sensors are calibrated for bias, sensitivity, misalignment, cross-axis sensitivity, non-linearity and gyroscope linear acceleration sensitivity across the full operating temperature range.

6.3 Hardware Specifications

Parameter	Rugged	OEM
Operating Voltage*	5 to 36 V	4.5 to 5.5 V
Input Protection	±60 V	None
Power Consumption (typical)	1.4 W	1.4 W
Power Consumption (maximum)	1.8 W	1.8 W
Inrush Current	0.5 A	0.5 A
Hot Start Battery Capacity	> 48 hrs	> 48 hrs
Hot Start Battery Charge Time	30 mins	30 mins
Hot Start Battery Endurance	> 10 years	> 10 years
Operating Temperature	-40 °C to 85 °C	-40 °C to 85 °C
Environmental Sealing	IP67 MIL-STD-810G	None
Shock Limit	150 g, 6 ms, half sine	150 g, 6 ms, half sine
Vibration Limit	8 g RMS	8 g RMS
Dimensions	42 x 67 x 30 mm	34 x 39 x 24 mm
Weight	98 grams	26 grams

Table 10: Hardware Specifications

Note: * Devices should be designed to operate within the bounds of the operating voltage to avoid tripping the built-in protection circuitry.

6.3.1 Rugged Mechanical Drawings

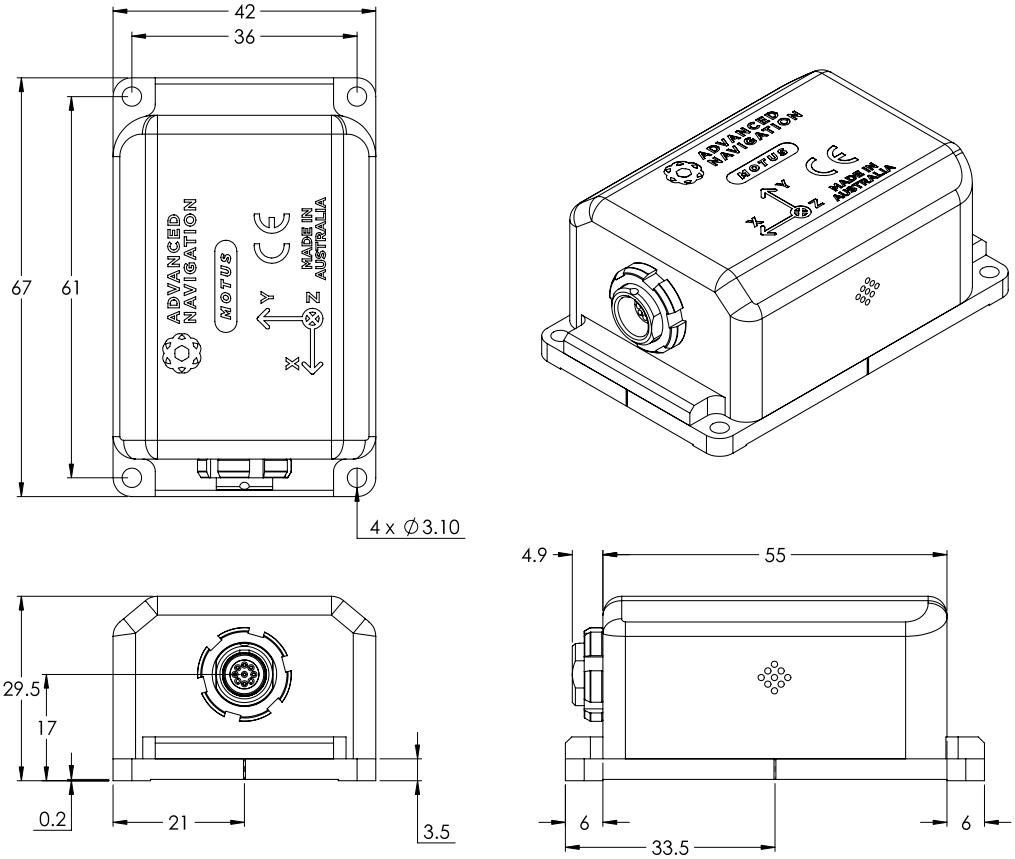


Figure 4: Mechanical drawings of Motus

6.3.2 OEM Mechanical Drawings

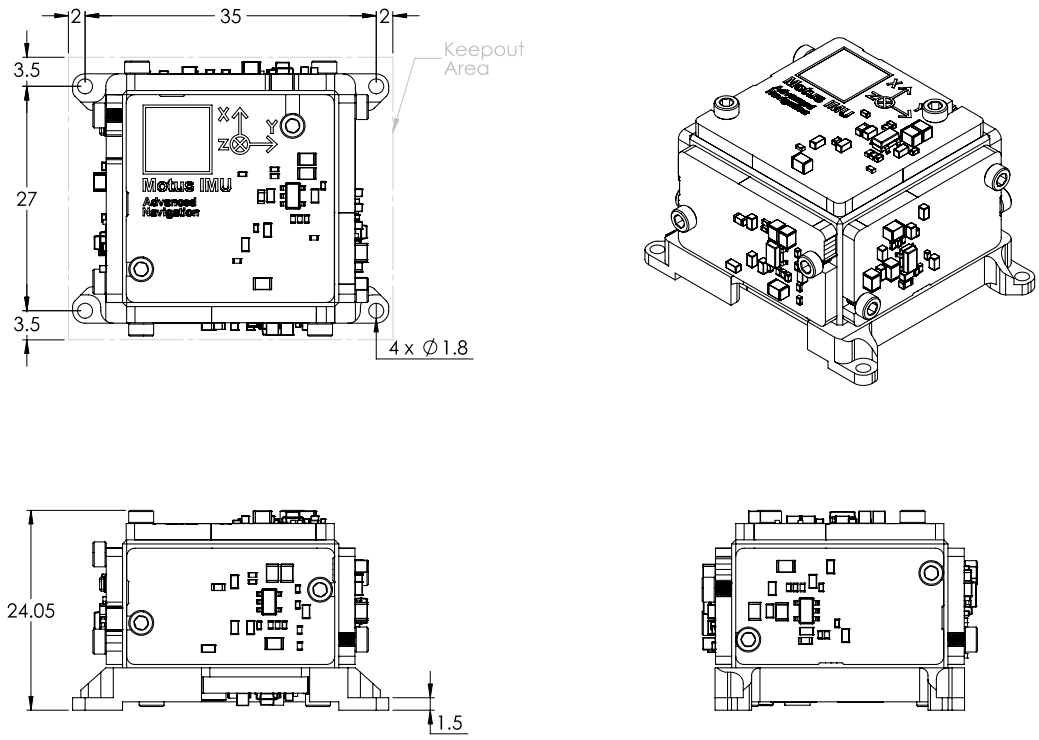


Figure 5: Mechanical drawings of Motus OEM

6.3.3 Serial Number

Units can be identified and tracked using the device serial number. This can be observed in a numbers of ways:

- Through the Device Information page in the **Device Information** page in the Manager application.
- Via the 2D data matrix bar code on the outside rear of the enclosure. Applications are available for most smart-phones that can scan the 2D data matrix barcode to display the serial number. The label also contains the hardware version and build date.



Figure 6: Motus External Serial Number

6.4 Communication Specifications

Parameter	Rugged	OEM
Interface	RS232 (RS422 optional)	3.3V TTL serial port (5V tolerant input)
Primary and Auxiliary Port Speed	2400 to 2 M baud	2400 to 2 M baud
GPIO Port Speed	2400 to 250 k baud	2400 to 250 k baud
Protocol	AN Packet Protocol NMEA TSS1 Simrad	AN Packet Protocol
Peripheral Interface	2 x GPIO and 1 x Auxiliary RS232	2 x GPIO and 1 x Auxiliary RS232
GPIO Level	5 V or RS232	3.3V TTL (5V tolerant input)

Table 11: Communication Specifications

6.5 Electrical Specifications

Parameter	Rugged			OEM		
	Minimum	Typical	Maximum	Minimum	Typical	Maximum
Power Supply						
Input Supply Voltage	5 V		36 V	4. 5 V		5.5 V
Input Protection Range	-60 V		60 V	None		None
RS232						
Tx Voltage Low		-5.7 V	-5 V	0 V		0.4 V
Tx Voltage High	5 V	6.2 V		2.4 V		3.3 V
Tx Short Circuit Current			±70 mA			10 mA
Rx Threshold Low	0.8 V	1.3 V		0 V		5.5 V
Rx Threshold High		1.7 V	2.5 V			0.8 V
GPIO				2 V		

Parameter	Rugged			OEM		
	Minimum	Typical	Maximum	Minimum	Typical	Maximum
Output Voltage Low	0 V		0.3 V	0 V		0.4 V
Output Voltage High	4.8 V		5 V	2.4 V		3.3 V
Output Current			5 mA			10 mA
Input Voltage	-20 V		20 V	0 V		5.5 V
Input Threshold Low			1.5 V			0.8 V
Input Threshold High	3.5 V			2 V		

Table 12: Electrical Specifications

Note: Devices should be designed to operate within the bounds of the operating voltage to avoid tripping the built-in protection circuitry.

6.6 Connector Pin-out

Power supply and signal connections are made through a ODU Mini-Snap Series B 9 pin connector (part number S30B0S-P09MCC0-5000). The connector provides a reliable and rugged connection to the Motus under demanding conditions and is rated to IP68 in the mated condition. Advanced Navigation stocks 2 metre unterminated ODU cables that are shielded and have external TPE insulation, see **5 Part Numbers and Ordering Options** for ordering. Each individual wire is colour coded, FEP coated 28AWG wire, labelled with the ODU pin number on the unterminated end. Custom cable lengths can be ordered by request.



Figure 7: ODU B series mating plug for Motus

See **Figure 8: ODU B series pin locations** for ODU connector pin locations and **Table 13: Motus Pin allocation table** for each pin function. **Figure 9: Pin-out view perspective** shows the perspective used for the pin location diagram.

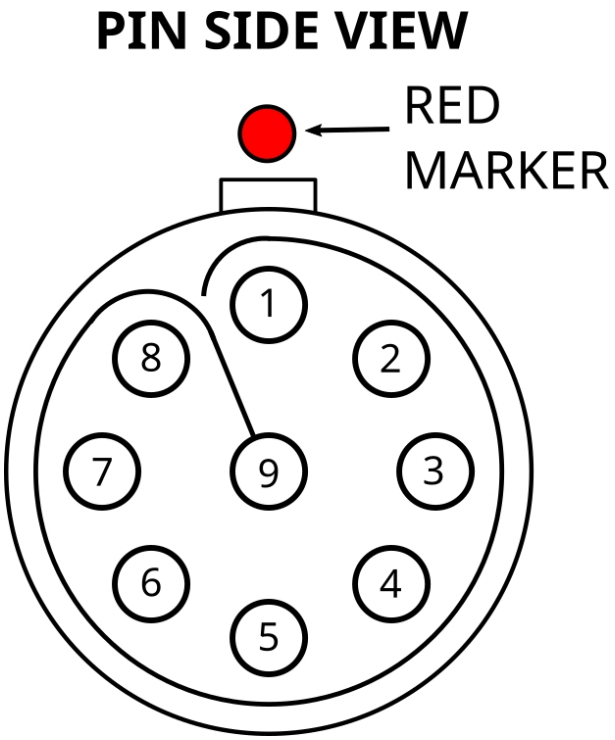


Figure 8: ODU B series pin locations



Figure 9: Pin-out view perspective

Pin	Colour	Function (RS232 version)	Function (RS422 version)
1	Black	Ground	Ground
2	Brown	Power Supply	Power Supply
3	White	GPIO 1	GPIO 1
4	Green	GPIO 2	GPIO 2
5	Red	Primary RS232 Transmit	RS422 Transmit +
6	Orange	Primary RS232 Receive	RS422 Receive +
7	Yellow	Auxiliary RS232 Transmit	RS422 Transmit −
8	Blue	Auxiliary RS232 Receive	RS422 Receive −
9	Pink	Ground	Ground

Table 13: Motus Pin allocation table

6.7 IO Connection

Advanced Navigation offers two pre-terminated IO breakout cables (one for the RS232 version and one for the RS422 version) for quick connection to Motus. The evaluation kit USB Cable is supplied as a part of the Motus Evaluation Kit. See **3 Evaluation Kit**.

6.7.1 Rugged Evaluation Kit USB Cable

The evaluation kit is supplied with a 2 metre cable with a Motus ODU plug on one end and an overmoulded RS232 to USB converter on the other end, please see **Figure 10: Evaluation kit USB cable**. The cable is the same 9 wire cable supplied with unterminated ODU cables and all 9 wires are connected on the ODU end. If required, it is possible to cut off the overmoulded RS232 converter to get access to all the individual wires. The chip used in the overmoulded RS232 converter is an FTDI FT232R and supports baud rates of up to 2,000,000 baud.

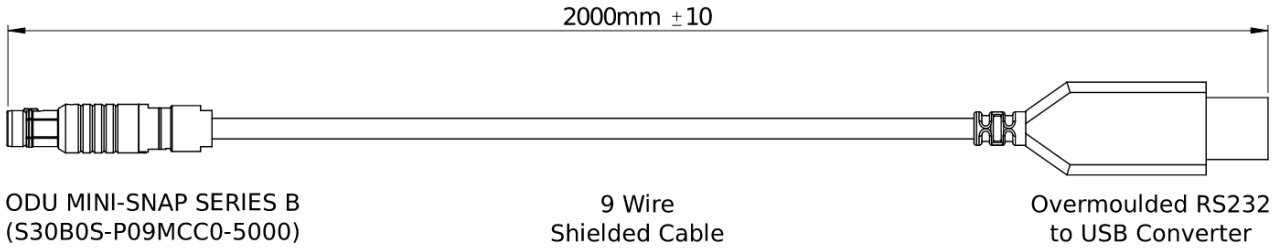


Figure 10: Evaluation kit USB cable

Pin	Function	RS232 Converter Connection
1	Ground	USB Ground
2	Power	USB 5 volt supply
3	GPIO 1	-
4	GPIO 2	-
5	Primary RS232 Tx	FTDI RS232 Rx

Pin	Function	RS232 Converter Connection
6	Primary RS232 Rx	FTDI RS232 Tx
7	Auxiliary RS232 Tx	-
8	Auxiliary RS232 Rx	-
9	Ground	-

Table 14: Evaluation kit USB cable pin-out

6.7.2 Optional Breakout Cables

Advanced Navigation offers pre-terminated breakout cables for access to all of Motus' pins. There are two breakout cables — one for RS232 communication and another for RS422 communication. All external signal and power connections are supported with 2 m of cable. The breakout cables are an optional extra and are not supplied as standard with the evaluation kit.

6.7.2.1 RS232 Breakout Cable (A503-SDC20753)

The RS232 interface cable is provided with industry standard 9 pin DSUB connectors for each of the two RS232 communication channels and GPIO pins.

Please note that this cable is only applicable to the RS232 version of Motus (part number MOTUS2-RUG)

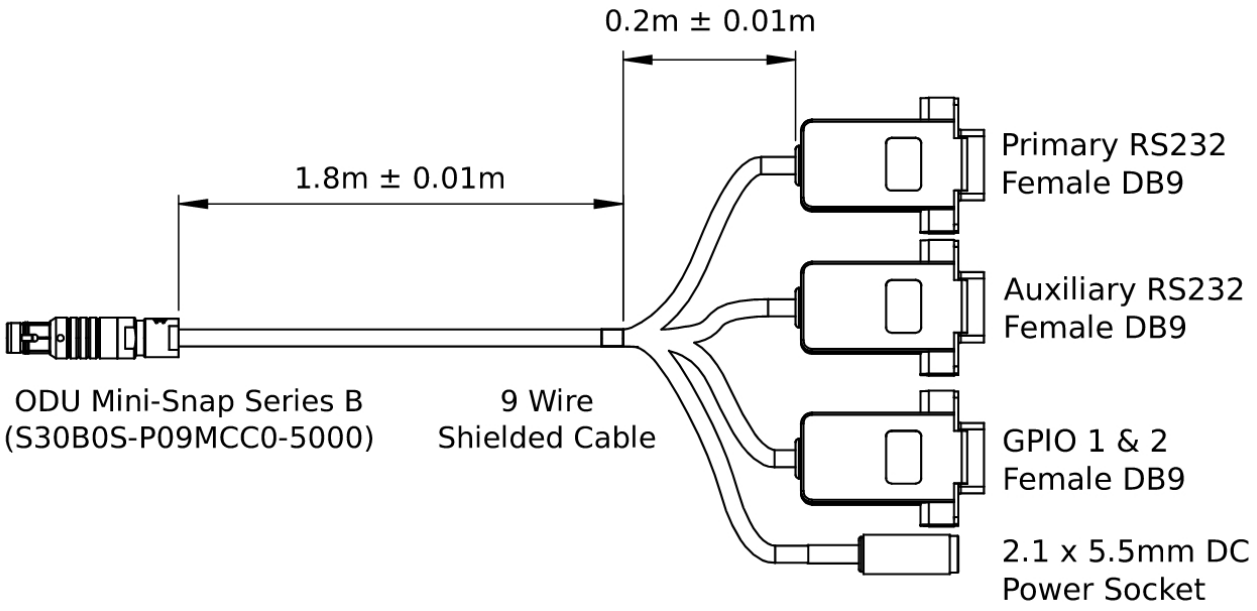


Figure 11: Motus Optional RS232 breakout cable

Pin	Function	Primary	Auxiliary	GPIO	Power
1	Ground	5	5	5	-
2	Power	-	-	-	Tip
3	GPIO 1	-	-	2	-
4	GPIO 2	-	-	3	-

Pin	Function	Primary	Auxiliary	GPIO	Power
5	Primary RS232 Tx	2	-	-	-
6	Primary RS232 Rx	3	-	-	-
7	Auxiliary RS232 Tx	-	2	-	-
8	Auxiliary RS232 Rx	-	3	-	-
9	Ground	-	-	-	Ring

Table 15: Motus optional RS232 breakout cable connector pin-out

6.7.2.2 RS422 Breakout Cable (A503-SDC20755)

The RS422 interface cable is provided with an industry standard 9 pin DSUB connector for the RS422 communication channel and GPIO pins.

Please note that this cable is only applicable to the RS422 version of Motus (part number MOTUS2-RUG-422)

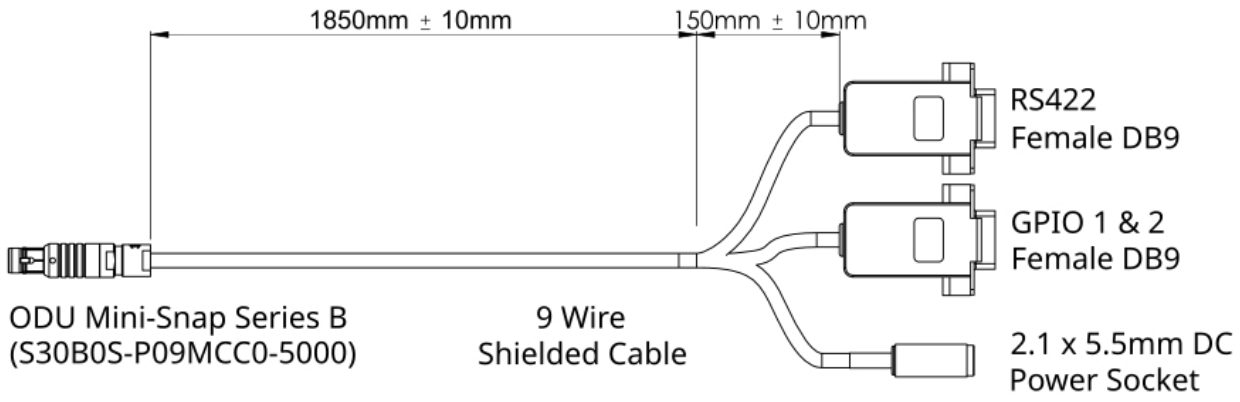


Figure 12: Motus Optional RS422 breakout cable

Pin	Function	Primary	Auxiliary	GPIO
1	Ground	5	5	-
2	Power	-	-	Tip
3	GPIO 1	-	2	-
4	GPIO 2	-	3	-
5	RS422 Tx +	2	-	-
6	RS422 Rx +	3	-	-
7	RS422 Tx −	8	-	-
8	RS422 Rx −	7	-	-
9	Ground	-	-	Ring

Table 16: Motus optional RS422 breakout cable connector pin-out

7 Installing

Refer to **4 Getting Started**, for a summary of steps to get started.

Note: For ordering an evaluation kit, a standalone kit or accessories for Motus, see **5 Part Numbers and Ordering Options**.

7.1 Mounting Motus

When installing Motus into a vehicle, the mounting position and alignment of the unit are essential for achieving optimal performance. Position refers to the physical location within the vehicle where the unit is installed while alignment refers to the orientation of the unit with respect to the orientation of the vehicle.

7.1.1 Mounting Considerations

Before you install Motus, review the mounting location considerations as follows:

- Mount the unit in an area that will not exceed the rated temperature range, and with minimal fluctuations in temperature.
- Mount the unit away from high levels of vibration where possible.
- Mount the unit within several metres of the GNSS antennas where possible.
- Mount close to the centre of gravity of the vehicle where possible.
- Mount the unit at least 10 cm away (and preferably as far as possible) from sources of dynamic magnetic interference such as high current wiring or large motors for best performance.
- Do not obstruct the two vents on either the side of the enclosure unless **8.9 Atmospheric Altitude** is a feature that is not desired and is to be disabled.

7.1.2 Mounting Plate Layout

The Motus mounting plate and hole guide are shown in the following diagram. The holes are designed for M3 screws.

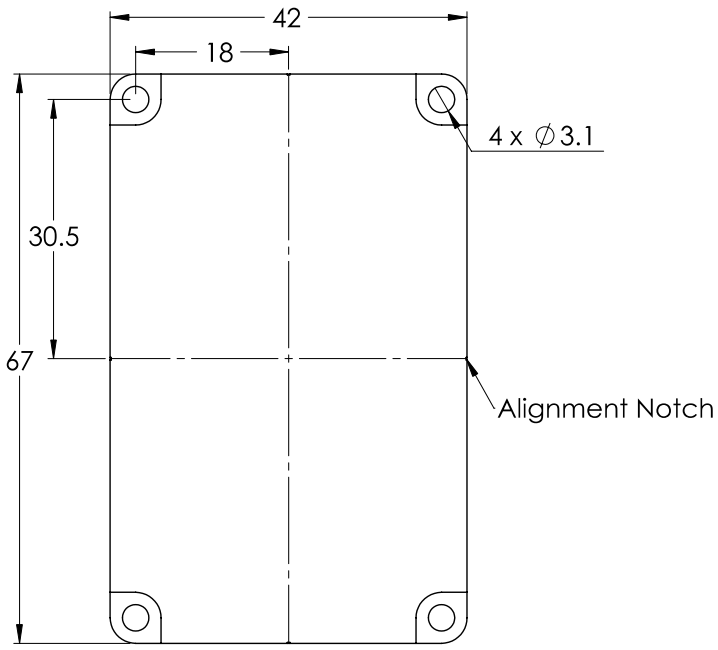


Figure 13: Mechanical drawings of Motus mounting

7.2 Mounting Alignment

The easiest way to align Motus is by installing the unit with its sensor axes aligned with the vehicle axes. There are notches marked on the X and Y axis to assist with alignment. Install the unit such that the X+ axis points forward towards the front of the vehicle and the Z+ axis points straight down towards the ground as shown below.

If aligning Motus with the vehicle axes is not possible within vehicle size or shape constraints, it may be mounted in a different alignment and the alignment offset can be configured in the **Alignment** dialogue in the INS Manager.. For easy roll and pitch alignment, use the **Set Zero Orientation** button in the **Alignment** page to set the current orientation as the level alignment.

Note: This will only correct for roll and pitch offsets. Any heading offset will need to be entered manually and saved after using this function.

7.3 Power Supply

Power supply filtering has been built into the unit.

However, a power supply with low noise and ripple levels, and adequate transient response is still recommended.

The unit contains a non-isolated power-supply.

The power supply must be able to reliably provide the maximum current at the chosen voltage, especially in systems with substantial load variations. Make sure that all system components are rated (including safety factors) for the anticipated environmental operating conditions and stressors of the system. Refer to the Hardware Specifications for further information.

The unit contains an active protection circuit on the power supply input that protects the unit from:

- Under Voltage
- Over Voltage
- Reverse Polarity

It is recommended to always operate the unit at a voltage higher than the minimum mentioned in the Electrical Specifications. Take caution when using a supply voltage close to the minimum voltage as minor voltage fluctuations could trigger under-voltage protection, leading to unexpected oscillations between the on and off states.

7.4 Odometer

Note: Motus AHRS variants can be used as an INS with velocity and position. It can also be interfaced to any leading brand of GNSS receivers to create an OEM GNSS/INS solution.

On ground vehicles, the use of an odometer input will greatly improve the Motus navigation and orientation solution.

There are several different options for odometer that are explained as follows.

7.4.1 Factory VSS Signal

Most mass-market vehicles since 1980 contain a VSS (vehicle speed sensor) signal that can be wired directly into one of the Motus GPIO pins. Any voltage limits on this pin that must be taken into consideration.

To set up the odometer, the appropriate GPIO pin should be set to Odometer Input using INS Manager. The odometer pulse length must then be set either manually or automatically, see **8.10 Odometer Pulse Length** for more information.

For more information on the GPIO signals and their requirements see **12.3 GPIO Pins and the Auxiliary Port**.

7.4.2 OBDII Odometer Interface

For applications where it is undesirable to modify the vehicle or the system needs to be used with multiple vehicles, the OBDII odometer interface may be a better solution. OBDII is a vehicle diagnostic port standard and most vehicles from the mid 1990s onwards contain an OBDII port in the drivers side foot well. Advanced Navigation produces an inexpensive OBDII Odometer Interface that plugs into this OBDII port and feeds Motus with odometer data over the Auxiliary RS232 port. See the figure below. Contact the Advanced Navigation sales team at **sales@advancednavigation.com** for more information.



Figure 14: Advanced Navigation OBDII Odometer

7.4.3 Aftermarket Wheel Speed Sensor

Applications requiring very high performance without GNSS for any length of time are recommended to use a high precision aftermarket wheel speed sensor. Advanced Navigation recommends aftermarket wheel speed sensors from Kistler, Pegasem or GMH Engineering.



Figure 15: Aftermarket Wheel Speed Sensor

7.4.4 Radar Speed Sensor

For applications requiring high performance in harsh conditions where aftermarket wheel speed sensors are not feasible, a radar speed sensor is recommended. Advanced Navigation recommends radar speed sensors from Stalker or GMH Engineering.



Figure 16: Radar Speed Sensor

7.5 Magnetics

Motus contains magnetometers which it can use to measure the Earth's magnetic field in order to determine its heading. The principle is the same as that of a compass. When operating Motus with magnetic heading enabled, sources of magnetic interference can degrade the Motus solution if not compensated for.

Note: Magnetic interference affects the performance of gyroscopes by causing rotation errors; it is recommended that you mount Motus away from magnetic interference.

There are two types of magnetic interference: **Static** and **Dynamic**.

Sources of static magnetic interference are predominantly ferrous materials (e.g. steel, iron, nickel, cobalt), high- current cables and permanent magnets located near the Motus. Static disturbances are easily compensated for by running a magnetic calibration. A magnetic calibration should always be performed after installation into a vehicle if accurate magnetometer heading is required. See **8.6 Magnetic Calibration** for instructions on how to calibrate your device.

Dynamic magnetic interference is a much more complex and problematic issue due to its many variable factors and unpredictability. Sources of dynamic magnetic interference include high current wiring, electric motors, servos, solenoids and nearby large masses of metal that don't move with Motus. Motus should be mounted as far as possible from these interference sources.

Motus contains a special algorithm to reduce the effects of dynamic magnetic interference. This is able to compensate for most typical interference sources encountered, however certain types of prolonged dynamic interference cannot be compensated for. The best way to check for dynamic magnetic interference is to use the Raw Sensors view in INS Manager and watch the magnetometer outputs whilst the vehicle is operating but stationary. See **Using Motus Manager**. The values should be constant: if the values are fluctuating there is dynamic magnetic interference present.

If dynamic magnetic interference is causing performance problems and there is no way to mount Motus away or, where practical, magnetically shield it from these interference sources, the magnetometers should be disabled and an alternative heading source should be used, see **8.6.4 Disabling Magnetometers**.

7.6 Vibration

High vibration levels can degrade the performance of an Inertial Navigation System (INS). While Motus is designed to tolerate high vibration levels, we recommend reviewing our **Best Practices for INS Damping and Isolation** knowledge base article for optimal setup.

Monitor the *INS Manager > Raw Sensors* view to look for over-range events, and address in your design as required.

7.7 PCB Design Tips

When incorporating the Motus OEM in a PCB (Printed Circuit Board) address the following challenges in order to achieve optimal performance.

7.7.1 Heat sources

Place the Motus OEM away from heat sources, such as CPUs.

Reason: IMU (Inertial Measurement Unit) accuracy is impacted by temperature fluctuations.

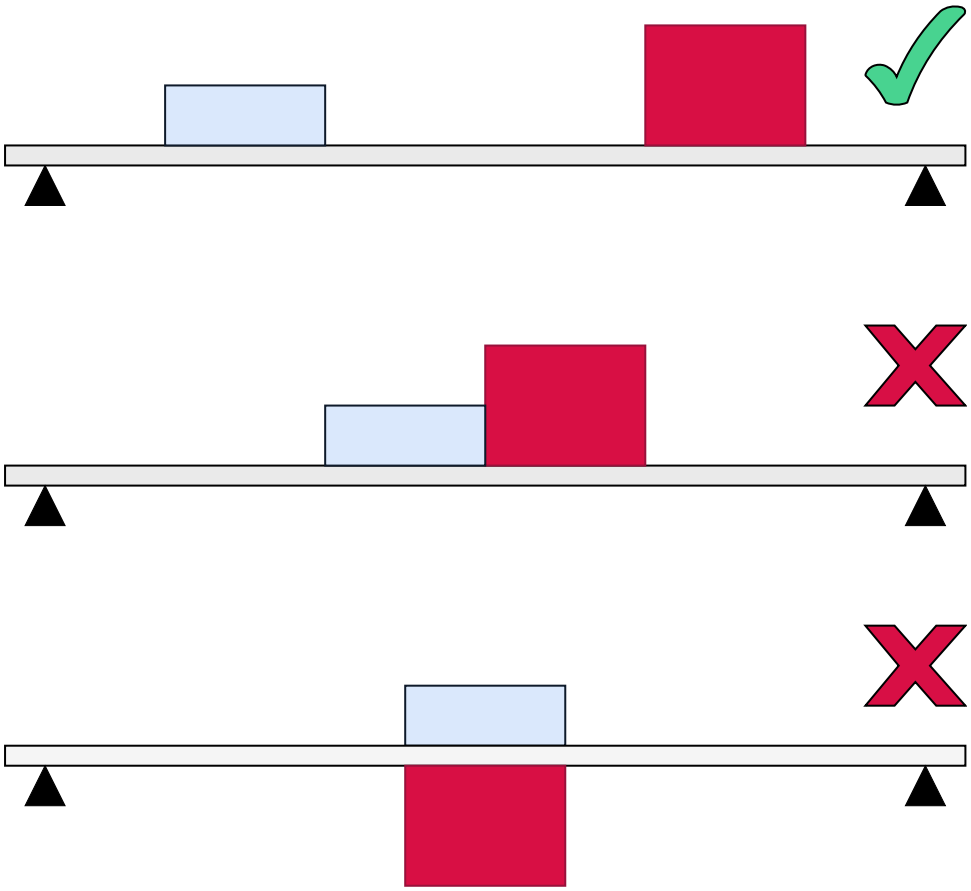


Figure 17: Diagram: Position Motus OEM away from heat sources.

7.7.2 Vibration and resonance

Reduce vibration and resonance.

Reason: Vibration from resonance will introduce noise into IMU readings, which will impact accuracy.

7.7.2.1 Low harmonic location

Place the Motus OEM on an area of the PCB which is less prone to vibration, typically near the board's center.

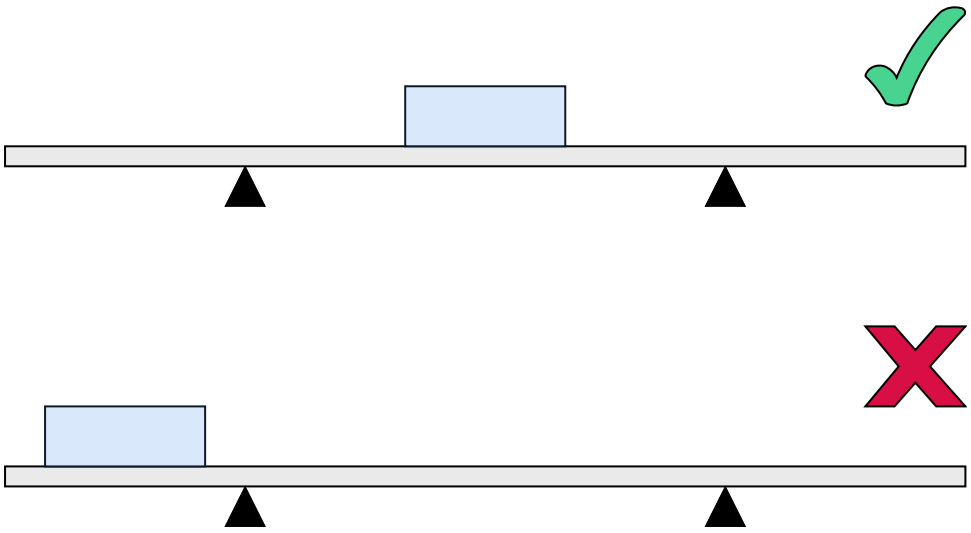


Figure 18: Diagram: Position the Motus in the center of the PCB.

7.7.2.2 Avoid overhangs

Secure the PCB to reduce resonance, such as:

- Attach to the platform using screws at the corners of the PCB.
- Avoid overhanging sections of the PCB.

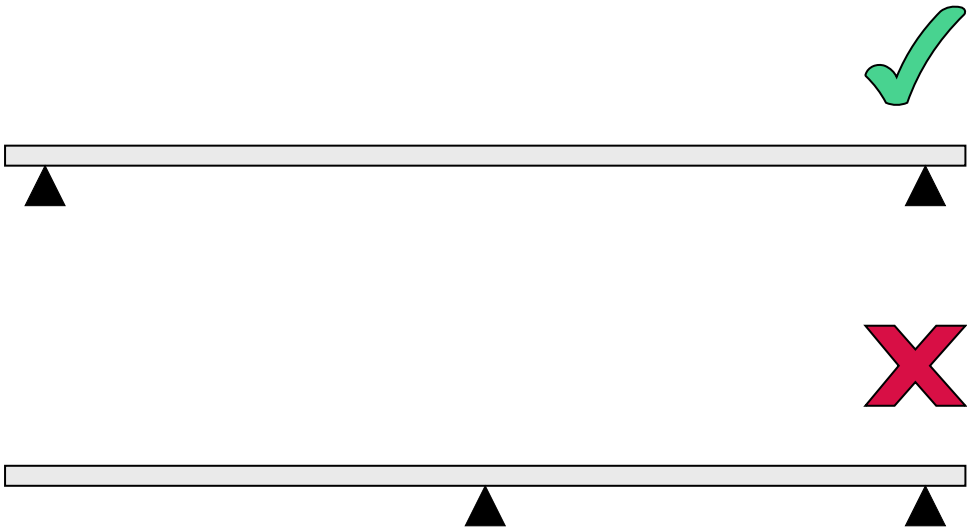


Figure 19: Diagram: Don't have overhanging sections of PCB which are prone to vibration.

7.7.2.3 Avoid contact with foreign structures

Avoid contact with foreign structures, which might move and nudge the INS, such as a heat sink.

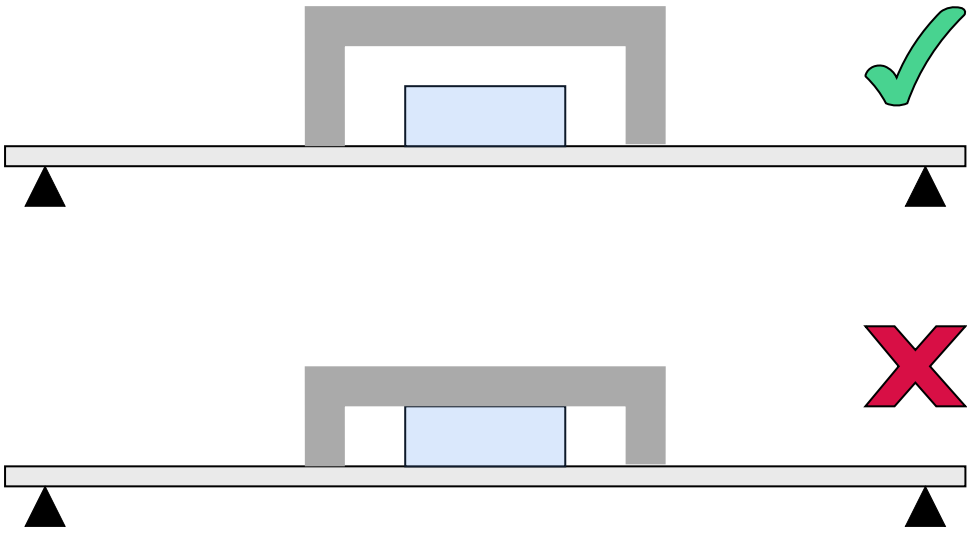


Figure 20: Diagram: Ensure a heat sink is not touching the Motus.

Avoid resin

Avoid resins or other materials over the sensor as it can change performance and calibration.

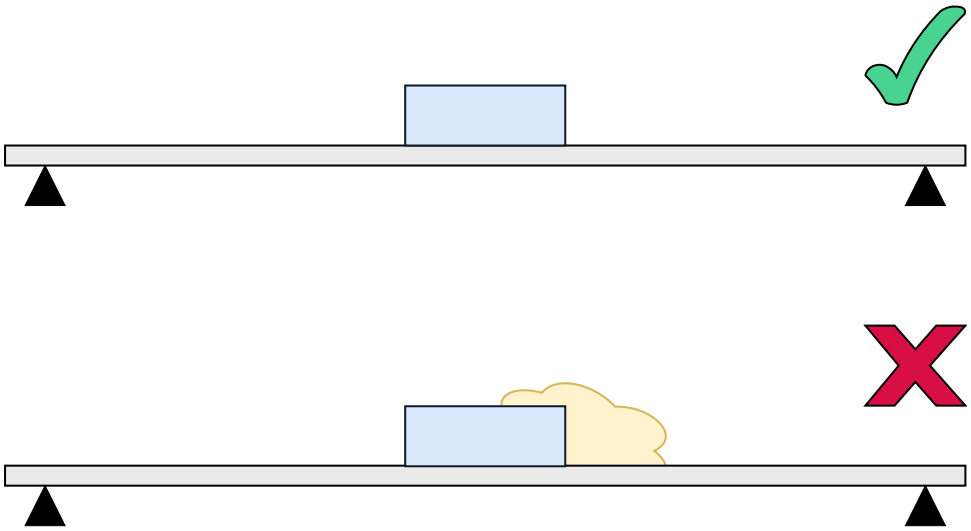


Figure 21: Diagram: Ensure the Motus is not touching epoxy resin.

7.7.2.4 Rigid board

Use a rigid rather than flexible PCB board.

7.7.3 Avoid board flex from screws

Avoid placing the Motus OEM next to a screw, which can cause board flexing near the screw.

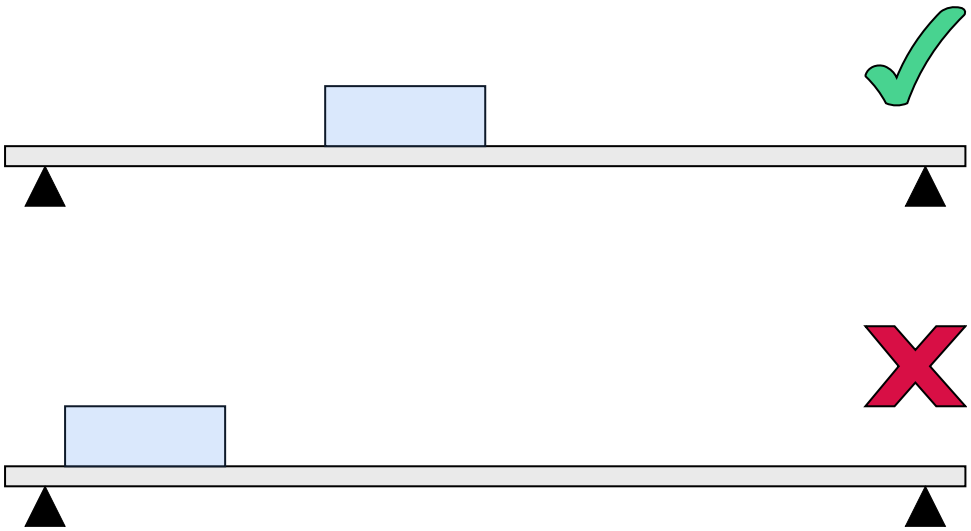


Figure 22: Diagram: Place Motus away from screw holes.

Reason: Flexing of the Motus OEM can introduce bias and noise into IMU readings.

8 Using Motus

This section covers the following topics:

8.1 Initialization for Optimal Performance

The performance of an Inertial Navigation System (INS) is dependent on the quality of its initialization.

For advice, refer to our Knowledge Base article, **How to Initialize your INS for Optimal Performance.**

8.2 Filter Initialisation

There are four different levels of filter initialisation on Motus. These are orientation, navigation, heading and time.

The initialisation of each can be monitored by inspecting the **9.6.2 View Menu** in the INS Manager.

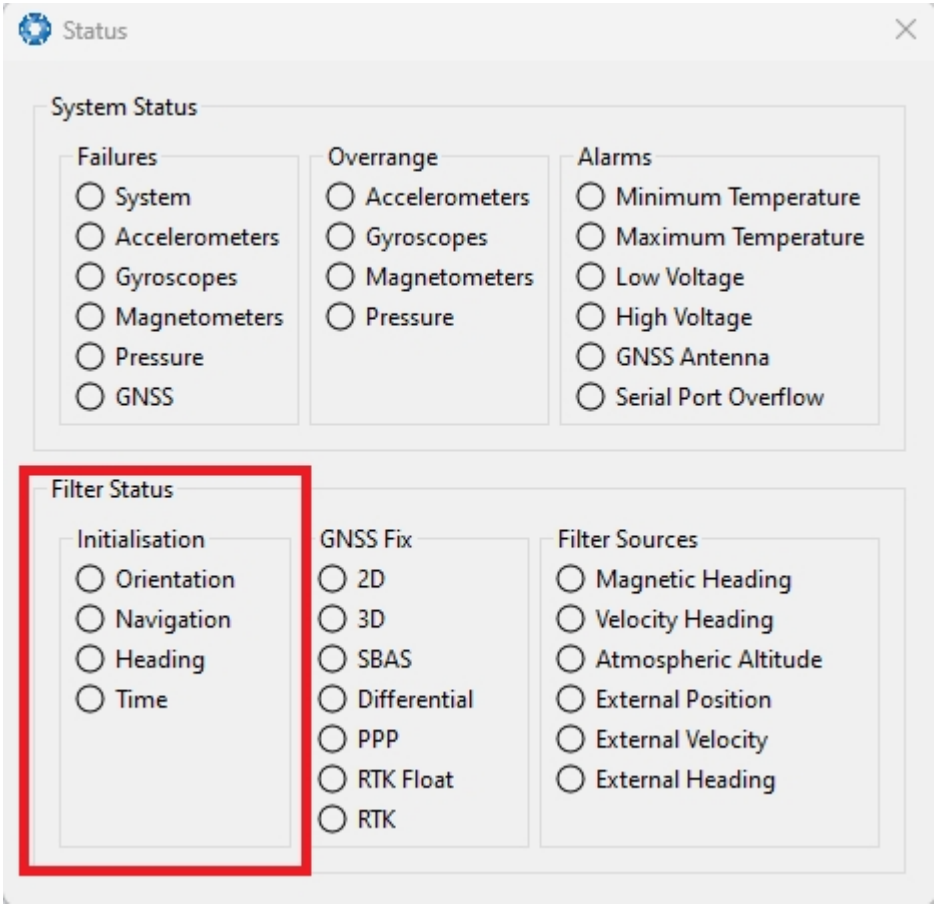


Figure 23: Initialisation Levels

After all four levels of initialisation, the Motus INS filter takes several minutes to achieve its full accuracy. It is recommended to wait two minutes after initialisation for applications requiring high accuracy.

8.2.1 Orientation Initialisation

Orientation initialisation occurs automatically upon power on and typically completes within seconds. To determine its orientation it uses the accelerometers to measure the gravity vector. Random accelerations from environmental sources can disturb this process, which can introduce minor orientation errors that will be corrected within several seconds. To prevent this, Motus monitors the accelerometers and gyroscopes and restarts the orientation detection if there are sudden movements. It is however still possible under some circumstances for it to miss minor movements and start with a small orientation error. Motus will progressively correct the orientation error over a period of several seconds.

Once orientation initialisation is complete, the roll, pitch and angular velocity values will be valid.

8.2.2 Navigation Initialisation

Note: Motus AHRS variants can be used as an INS with velocity and position. It can also be interfaced to any leading brand of GNSS receivers to create an OEM GNSS/INS solution.

Navigation initialisation completes once the system has determined a starting absolute position. If the system is hot-starting it will remember its position at the time it was switched off and use this as the starting position. The other possibility for navigation initialisation is an external position source to be used, see **13.4.26 External Position Packet** for more information. In a situation where an external position input is not available, it can be initialised manually by entering a position into the **9.6.3 Configuration Menu** dialogue in the INS Manager.

Once navigation initialisation is complete, the position, velocity and acceleration values will be valid.

8.2.3 Heading Initialisation

Heading initialisation completes once the system has determined a heading using one of its several methods. The conditions required to determine a heading depend upon the heading source being used, see **8.5 Heading Sources**. If velocity heading is enabled and the vehicle is moving, this will be very fast. If the system hot-starts, the last known heading value will be used. However, in order for the Heading Filter to initialise, the system must have a valid heading source.

Until the heading has been initialised, the system will not be able to navigate without a GNSS fix, and the roll and pitch values will not reach full accuracy.

8.2.4 Time Initialisation

Time initialisation completes once the system has determined time accurately. This requires:

- A valid timestamp from a GNSS, which occurs once the GNSS obtains its first fix, and
- A 1PPS signal.

These signals can either be sourced internal to the device if it has an internal GNSS, or input from an external source.

External timestamps can be delivered via 11.4.32 External Time Packet. This may be used if:

- Time is required prior to the GNSS obtaining a GNSS fix, or
- The GNSS antenna doesn't have visibility of satellites, or
- The GNSS is disabled.

An external 1PPS signal can be input via a user GPIO port configured as 1PPS input. If the device has an internal GNSS then `Filter Options > Internal GNSS` must be set to disabled.

After time initialisation completes:

- The `Filter Status > Time Valid` flag will be set to true, and
- The timestamps in all output messages (eg: `ANPP Unix time`, `ANPP Formatted time`, `NMEA`, etc) will contain correct times, and
- The 1PPS output will be accurately synchronised to UTC seconds.

8.3 Hot Start

Hot-start functionality allows the unit to sustain navigational state parameters and accurate time for many hours after losing power.

When the unit hot-starts, it assumes that it is in the same state as it was when it lost power and begins navigating from that position.

For units with GNSS capability, retained ephemeris, almanac and time information is provided to the GNSS receiver. This enables the GNSS receiver to achieve a GNSS fix faster than it otherwise could.

8.4 Time

Motus is designed to provide an accurate time reference via an internal oscillator.

For a device with a GNSS receiver, when a GNSS fix is available, the internal oscillator uses GNSS time to precisely correct itself to within the accuracy specified for the GNSS receiver.

When a GNSS fix is lost, time accuracy will slowly drift.

The time can be accessed via the following means:

- A 1PPS output over GPIO or the Auxiliary port, see **12.3.1.2 1PPS Output** of the Dynamic Pin Functions.

Note: The time filter will need to be initialised before the device will start sending NTP or PTP messages.

8.5 Heading Sources

There are several different heading sources available for Motus. The heading sources can be configured using the **Filter Options** page in the Manager application (see **Motus Manager**). It is possible to use multiple heading sources and this can provide performance benefits.

8.5.1 Velocity Heading

For devices that support either single or dual GNSS input, velocity heading works by deriving heading from the direction of velocity and acceleration. Velocity heading works well with cars, fixed wing aircraft and other vehicles that don't move sideways. Velocity heading cannot work for helicopters and other vehicles where the vehicle is pointed somewhere other than the direction of travel. Velocity heading requires a horizontal speed of over 1.15 m/s with a GNSS fix. Velocity heading is disabled by default.

8.5.2 Magnetic Heading

This is the default heading source and works well in the majority of cases. When using magnetic heading, magnetic calibration is required every time the static magnetic field around the Motus changes. This can occur, for example, when moving the mounting location of the unit, or making significant structural changes to the vehicle that alter the distribution of ferrous materials. The downside of magnetic heading is that prolonged exposure to dynamic magnetic interference can cause significant heading errors. See **7.5 Magnetics** and **8.6 Magnetic Calibration** for further information.

8.5.3 External Heading

This can be used if there is some other way to derive heading that is external to Motus. The heading must be supplied to Motus using the **13.4.29 External Heading Packet** or an NMEA message, see **12.3.1.29 NMEA Input**.

8.5.4 External Magnetometers Packet

Heading can derived using the magnetometers of an additional external Advanced Navigation product to input external magnetometer values. See **13.4.43 External Magnetometers Packet** for details.

8.6 Magnetic Calibration

Static magnetic interference is resolved through magnetic calibration. Dynamic magnetic interference is compensated by a filter algorithm, but should be minimised where possible by selecting the best possible mounting location or implementing magnetic shielding — see **7.5 Magnetics** for more information. To compensate for static magnetic interference, magnetic calibration should be performed whenever the Motus mounting location or operation environment changes.

Motus contains a dynamic magnetic compensation filter that is able to mitigate the effects of short term magnetic interference sources while in operation. For example, if Motus is installed in a car and the car drives over a large piece of magnetised steel, this will be compensated for. Another example is driving through a tunnel which is built from reinforced concrete, which can distort the local magnetic field substantially. It is important to note that for the Motus dynamic magnetic compensation filter to operate correctly, Motus needs to get a GNSS fix at least once every time it is moved more than 50 km. Each time Motus moves more than 50 km the new position is stored permanently and allows Motus to update its world magnetic model values.

There are three types of magnetic calibration available: 2D calibration, 3D calibration and automatic calibration. The three magnetic calibration types and their respective typical applications are described as follows.

- 2D calibration involves at least one full rotation of the vehicle about the Z-axis and is designed for vehicles that are not intended to travel whilst turned upside down, such as full-sized cars, passenger planes and boats.
- 3D calibration involves at least two full rotations of each axis (X, Y and Z) and is designed for vehicles that are intended to be manoeuvred and travel upside down, such as underwater vehicles. 3D calibration offers slightly better performance, and is recommended where possible. Be mindful that the support mechanism used to support the vehicle during calibration can influence the results.
- Automatic magnetic calibration continuously and automatically calibrates for static magnetic interference while the unit is in operation. While it is not as accurate as 2D or 3D calibration, and can take significantly

longer to calibrate, automatic magnetic calibration is very convenient for applications where it is impractical to employ those methods.

Note: If Motus is going to be used in a vehicle, the calibration should be performed while Motus is mounted in and fixed to that vehicle. This means that the whole vehicle must be moved to perform the calibration. The calibration needs to be performed in an area away from other sources of magnetic interference. For example if Motus is installed in a car, the calibration should not involve driving over steel drains or reinforced concrete etc. If Motus is being calibrated to operate standalone, the calibration should not be done on a desk with a steel frame.

8.6.1 2D Magnetic Calibration

The following procedure should be used to perform a 2D magnetic calibration.

8.6.1.1 2D Magnetic Calibration Using Motus Manager Software

1. The unit should be powered on while the vehicle is in a level orientation and kept stationary. It is recommended this be done in an outdoor area away from metal or obstructions.
2. After power on wait 5 minutes for the temperature and filter to stabilise.
3. Open Motus Manager and connect to the device.
4. In the **Tools** menu, open **Magnetic Calibration**. Click the **2D Calibration** button.
5. Whilst keeping as level as possible, rotate the vehicle in either direction through one and a quarter full rotations. In a car or boat his can be achieved by driving in one and a quarter circles.
6. Check the status in the **Magnetic Calibration** window to ensure that the calibration completed successfully. If not successful, click **Cancel**, wait 2 minutes and repeat from step 4.

Note: Do **not** press the Save button in the lower half of the page, this will overwrite the new calibration values. The auto-generated calibration values are automatically saved internally by Motus.

8.6.1.2 2D Magnetic Calibration Using the Packet Protocol

1. The unit should be powered on while the vehicle is in a level orientation and kept stationary.
2. After power on wait 5 minutes for the temperature and filter to stabilise.
3. Ensure that the device has a position fix, either from a GNSS receiver, or using the External Position and Velocity packet #44 or the External Position packet #45, before proceeding.
4. Send the **13.5.8 Magnetic Calibration Configuration Packet** with the action **Start 2D Magnetic Calibration**.
5. Whilst keeping as level as possible, rotate the vehicle in either direction through one and a quarter full rotations. In a car this can be achieved by driving in one and a quarter circles.
6. Read the **13.5.9 Magnetic Calibration Status Packet** to ensure that the calibration completed successfully. If not successful, send the **13.5.8 Magnetic Calibration Configuration Packet** with the action **Cancel**, wait 2 minutes and repeat from step 4.

8.6.2 3D Magnetic Calibration

The following procedure should be used to perform a 3D magnetic calibration.

8.6.2.1 3D Magnetic Calibration Using the Motus Manager Software

1. The unit should be powered on and the vehicle kept stationary. It is recommended this be done in an outdoor area away from metal or obstructions.
2. After power on wait 5 minutes for the temperature and filter to stabilise.
3. Open Motus Manager and connect to the device. Ensure that the device has a GNSS fix before proceeding.
4. In the **Tools** menu, open **Magnetic Calibration**. Click the **3D Calibration** button. The following steps 5, 6 and 7 can be performed in any order.
5. Slowly rotate the vehicle twice around the X-axis (roll).
6. Slowly rotate the vehicle twice around the Y-axis (pitch).
7. Slowly rotate the vehicle twice around the Z-axis (heading).
8. Check the status in the **Magnetic Calibration** window to ensure that the calibration completed successfully. If not successful click **Cancel**, wait 2 minutes and repeat from step 4. Do **not** press the save button in the lower half of the page, this will over write the new calibration values. The auto-generated calibration values are automatically saved internally by Motus.

8.6.2.2 3D Magnetic Calibration Using the Packet Protocol

1. The unit should be powered on and the vehicle kept stationary.
2. After power on wait 5 minutes for the temperature and filter to stabilise.
3. Ensure that the device has a position fix, either from a GNSS receiver, or using the External Position and Velocity packet #44 or the External Position packet #45, before proceeding.
4. Send the **13.5.8 Magnetic Calibration Configuration Packet** with the action **Start 3D Magnetic Calibration**. The following steps 5, 6 and 7 can be performed in any order.
5. Slowly rotate the vehicle twice around the X-axis (roll).
6. Slowly rotate the vehicle twice around the Y-axis (pitch).
7. Slowly rotate the vehicle twice around the Z-axis (heading)
8. Read the **13.5.9 Magnetic Calibration Status Packet** to ensure that the calibration completed successfully. If not successful, send the **13.5.8 Magnetic Calibration Configuration Packet** with the action **Cancel**, wait 2 minutes and repeat from step 4.

8.6.3 Automatic Magnetic Calibration

Automatic magnetic calibration is an algorithm that continuously and automatically calibrates for static magnetic interference. Automatic magnetic calibration is enabled by default and it is recommended to keep it enabled for all vehicle types. When used on top of a 2D or 3D calibration it can give the best results from magnetic heading.

The automatic calibration for 2D vehicles differs from that for 3D vehicles. To ensure the correct calibration is being applied, ensure the correct vehicle profile is selected for your application in the **9.6.3 Configuration Menu** dialogue in the Motus Manager.

8.6.3.1 2D Vehicle Profiles

2D vehicles are those which cannot easily or safely be turned upside down, and typically include vehicles such as cars and bicycles.

Each time Motus is powered on in a cold start (48 hours without power) the automatic calibration is reset. The automatic magnetic calibration starts operating when Motus is travelling at a speed over 5 m/s with a GNSS fix and either dual antenna is the heading source or velocity heading is turned on.

Automatic magnetic calibration is not as accurate as 2D or 3D calibration, however, it is sufficient for most vehicles and allows Motus to be installed into vehicles and operated without user intervention.

Automatic magnetic calibration can be turned off using the **9.6.3 Configuration Menu** dialogue in the Motus Manager.

8.6.3.2 3D Vehicle Profiles

3D vehicles are those which can easily or safely be turned upside down, and typically include vehicles such as humans, underwater vehicles and fixed wing planes.

Each time Motus is powered on in a cold start (48 hours without power) the automatic calibration is reset, and a new calibration will begin if a significant change in calibration values is detected. The automatic calibration for 3D vehicle profiles requires no prior calibration or minimum velocity to begin.

The calibration time is dependent upon the rotation about each axis; approximately, 1 degree of uniform motion in each axis will take up to an hour to converge, motion as low as 0.2-0.5 degrees will take upwards of 12 hours to converge, and 10 degrees of motion will take around 40 seconds to converge. These time estimates are dependent on the environment and nature of the motion.

Automatic magnetic calibration can be turned off using the **9.6.3 Configuration Menu** dialogue in the Motus Manager.

8.6.4 Disabling Magnetometers

In situations where there are very strong dynamic magnetic disturbances present, it is recommended to disable the magnetometers. When the magnetometers are disabled, an alternative heading source is required or else Motus may become inaccurate. Dual antenna heading is the recommended source. See **8.5 Heading Sources** for information. The magnetometers can be disabled using the Magnetic Heading setting on Filter Options page in the Motus Manager. See **Using Motus Manager**.

8.7 Data Anti-aliasing

Internally the Motus filters update at 1,000 Hz. When Motus outputs data, most applications require the data at a much lower rate (typically less than 100 Hz). This causes a problem for time-based data such as velocities and accelerations where aliasing will occur at the lower rate. To prevent this problem, if the output rate is lower than 1000 Hz, Motus will low-pass filter the values of the time dependent data between packets to prevent aliasing. This is only the case when a packet is set up to output at a certain rate. If the packet is simply requested, no anti-aliasing will occur. Additionally there is no anti-aliasing for non time-dependent fields such as position.

8.8 Vehicle Profiles

Motus supports a number of different pre-configured vehicle profiles. These vehicle profiles impose constraints upon the filter to increase performance. If your application matches one of the available vehicle profiles, it is required that you select it for use in the **Filter Options** page in the INS Manager. For a list of the different vehicle profiles, see **13.5.5.1 Vehicle Types**. If the wrong vehicle profile is selected, it can cause a significant decrease in performance.

8.9 Atmospheric Altitude

Motus contains an internal atmospheric pressure sensor. This is used as an altitude aid in the filter. If Motus is mounted in an area subject to unusual pressure waves, for example helicopters, or is mounted inside an enclosure that is not exposed to atmospheric pressure, the pressure sensor input should be disabled. This can be done by disabling Atmospheric Altitude in the **9.6.3 Configuration Menu** dialogue in the INS Manager.

8.10 Odometer Pulse Length

For Motus to use a wheel speed sensor or odometer input, it must know the pulse length of the signal. The pulse length is the distance in metres between low to high transitions of the signal. By default, the odometer will automatically calibrate itself. To automatically calibrate the odometer pulse length, see **8.10.1 Calibrating Odometer Automatic Pulse Length**.

8.10.1 Calibrating Odometer Automatic Pulse Length

1. Ensure that the signal is connected correctly and that the GPIO pin is configured as an odometer input using the **9.6.3.7 GPIO** dialogue in the INS Manager.
2. Wait until Motus has a continuous GNSS fix from a GNSS receiver and then conduct the test in an area with no obstructions over the test circuit.
3. Open INS Manager, connect to Motus and open the Odometer Configuration dialogue. In the **Odometer Configuration** dialogue select the **Automatic pulse length calibration** check box and click **Save**.
4. If Motus loses a GNSS fix for any extended period of time during the calibration, the distance travelled will be reset. The distance travelled can be checked in the Odometer Configuration dialogue to ensure that it has passed 1,000 metres.
5. Once 1000 metres has been driven, check that the automatic pulse length check box has become un-ticked and the Pulse Length value has changed. This indicates a successfully completed calibration.

Note: To enter the pulse length manually, use the **Odometer** page in the Configuration menu in the **INS Manager**.

Note: Maximum speed that can be processed by Motus is 216 km/h. If the maximum speed is exceeded, the Odometer Automatic Pulse Length Calibration will reset. Calibration will restart once speed drops below the maximum speed.

8.11 Reversing Detection

Reversing detection is an algorithm that can detect when the vehicle is travelling backwards. Knowledge of reverse motion is important when using velocity heading or odometer input to provide correct results. If Motus is fitted to a vehicle that does not reverse or doesn't use velocity heading or an odometer, this function should be disabled. Reversing detection is enabled by default and it can be disabled using the Filter Options page in the web UI or INS Manager. See or **Using Motus Manager**.

8.12 DVL Integration

To understand how to integrate a Doppler Velocity Log (DVL) with your INS, refer to **Advanced Navigation INS & DVL Integration.**

8.13 Environmental Exposure

Whilst the rugged unit is environmentally protected, there are limits to this protection that must be observed to avoid damaging the device.

Motus is only protected when its connectors are mated (or sealed with the blanking caps). If any of these connections are not tightened to the recommended torque, the unit will not achieve its rated protection.

Spanners or tools should never be used to tighten the connectors. Use your fingers to firmly tighten the connectors.

OEM units offers limited environmental protection. Care should be taken in choosing a suitable mounting location.

8.13.1 Temperature

The unit should not be subjected to temperatures outside of its operating range. Subjecting the unit to temperatures outside of the specified operating range can affect the factory sensor calibration, potentially causing a permanent reduction in performance or premature failure.

8.13.2 Water

The rugged unit complies with IP67 standard, which means that it can be submersed in water to a depth of up to 1 m deep for a maximum of 30 minutes. Submersion to depths beyond 1 metre can cause water entry and destruction of the internal electronics, and the sealing is not designed for prolonged or repetitive immersion. Do not direct streams of water directly at the unit even at low pressure. For cleaning the unit, ensure that no pressure is used.

Note: Connectors must be installed or blanking caps must be used if the unit is likely to be exposed to water.

8.13.3 Salt

The rugged unit housing is made from hard-anodized marine-grade aluminium and stainless fittings to provide good corrosion resistance, however cannot tolerate extended periods of time in salt water environments. After any contact with salt water or salt spray, Motus should be rinsed with distilled or RO water.

8.13.4 Dirt and Dust

The rugged unit is completely sealed against dirt and dust entry whenever all connectors are mated. When disconnecting the connectors, any dirt or dust should be rinsed or wiped off with fresh water and then dried to minimise the chance of contaminants entering the connectors or plugs.

8.13.5 pH Level

Environments with acidic or alkaline water or atmospheres can cause the Motus enclosure to corrode or discolor. If Motus comes into contact with these environments it should be rinsed in fresh water as soon as

possible. It is not recommended to operate Motus in non-neutral pH environments.

8.13.6 Shocks

Motus is a rugged piece of equipment that can withstand reasonable amounts of shock. However, continuous shocks may cause premature failure as a result of materials fatigue, and may also affect the sensor calibration resulting in degraded performance.

Shocks applied directly to the Motus enclosure can affect the unit. Therefore, care should be taken when handling the unit prior to mounting. When mounted in a vehicle, the effect of most operating shocks is negligible.

8.13.7 Atmospheres

Hydrogen-rich atmospheres may damage the MEMS. Motus is not rated for use where explosive gases may be present.

8.14 Upgrading the Motus Firmware

Advanced Navigation regularly provides firmware updates to improve performance and add features. These are described on the Advanced Navigation website. Advanced Navigation recommends that you upgrade to the latest version whenever available.

The firmware update function is robust, such that power loss, file corruption or other issues cannot brick the device.

See **9.6.4.2 Firmware Update** for instructions on how to upgrade the firmware using the INS Manager.

9 INS Manager (Web and Desktop)

Configuration, monitoring and log analysis of Advanced Navigation Inertial Navigation System (INS) products is done using one of our INS Manager products, or directly via API calls:

- **INS Manager - Web** provides a web user interface, accessed via a web browser, which connects via ethernet to a web server running on the INS.
- **INS Manager - Desktop** is a java based desktop application, which connects to the INS via a serial or ethernet cable.
- Direct API calls using Advanced Navigation Packet Protocol (**9.7 Using ANPP**) .

Note
Motus doesn't support the *INS Manager - Web* interface.

9.1 What's the difference between Desktop and Web Interfaces?

INS Manager - Web and *INS Manager - Desktop* provide a very similar look-and-feel and offer most of the same features, but they are technically two different products. The following table summarizes the differences.

Feature	INS Manager - Desktop	INS Manager - Web
Availability	All INSs (No Subsea)	Most INSs with ethernet (Subsea)
Ease of installation	Java desktop installation	Zero installation, browser based
Connection	Serial / Ethernet	Ethernet (Wifi for Hydrus)
Majority of functions	Good	Good
High bandwidth data	Good	Network constrained
Download logs	No	Yes
Convert logs	Yes	No

Table 17: Summary of differences between INS Manager and web interfaces.

9.2 Which products have hidden pages?

The interface dynamically hides pages that are not relevant to the connected INS.

Notably, products without an internal or externally connected GNSS receiver will not have access to GNSS specific pages.

9.3 Which products support Desktop and Web interfaces?

The following table describes which Advanced Navigation products support the INS Manager Desktop and Web interfaces.

Product	INS Manager - Desktop?	INS Manager - Web?	Internal Logging?
Boreas 50/70/90	Yes	Yes	Yes
Certus / Certus Evo	Yes	Yes	Yes

Product	INS Manager - Desktop?	INS Manager - Web?	Internal Logging?
Certus Mini	Yes	No	No
Motus	Yes	No	No
GNSS Compass	Yes	Yes	No
Hydrus	No	Yes (via WiFi)	No
Subsonus	No	Yes	No
Subsonus Tag	No	No (Uses mobile app + bluetooth)	No
Legacy Products			
Spatial	Yes	No	No
Spatial Dual	Yes	No	No
Orientus	Yes	No	No
Spatial FOG	Yes	No	No
Spatial FOG Dual	Yes	No	No

Table 18: Products which support INS Manager and web interfaces.

9.4 Installing INS Manager - Desktop

This section is applicable to the desktop variant of INS Manager. Check if the web interface is easier for you.

Note: The *INS Manager Desktop* is only available for systems with x64 and ARM architectures.

Note: The older Java version 11 is required by *INS Manager Desktop*.

- Install the latest version of Java 11 from **Adoptium**, selecting the correct operating system and architecture for your computer. Adoptium JRE 11 is the recommended Java installer to be used on all platforms.
- Download *INS Manager Desktop* from the Advanced Navigation website.
- Double click the *INS Manager Desktop* icon on your computer to start the application.
- Connect to the INS, following instructions in the following section: *Using INS Manager*.

9.4.0.1 Troubleshooting

Symptom	Likely Cause	Resolution
The *INS Manager Desktop* application fails to launch or start.	A 32-bit version of the Java Runtime Environment (JRE) is installed on a 64-bit operating system, causing an architecture mismatch execution failure.	Uninstall the existing 32-bit Java version from your system and install a compatible 64-bit version.
Cannot view *INS Manager - Web* page (such as http://boreas.local) for ethernet enabled units	Active internet connectivity is unavailable.	Verify the local network internet connection status and ensure routing endpoints are clear.

Symptom	Likely Cause	Resolution
The *INS Manager - Java Desktop* fails to discover or list any connected network devices.	Active internet connectivity is unavailable.	Verify the local network internet connection status and ensure routing endpoints are clear.
	An explicit Inbound block configuration rule exists inside Windows Defender Firewall against *INS Manager Desktop*, or AdNav *INS Studio* or java. Because Windows firewall priority rules dictate that an active "Block" action immediately supersedes an active "Allow" rule, all network discovery traffic (such as UDP multicast/broadcasts or TCP polling) is blocked.	1. Open **Windows Defender Firewall with Advanced Security** on your laptop. 2. Navigate to the **Inbound Rules** pane in the left-hand column. 3. Locate the offending rule and change from "Block" to "Allow". 4. Relaunch the application.
The *INS Manager - Java Desktop* cannot see the INS unit via a serial connection.	Serial or USB communication settings (COM interface, wiring protocol, or data rate) are misconfigured or experiencing connectivity dropouts.	1. Ensure the physical **RS232 / RS422** hardware switch toggle matches the required network configuration mode. The default hardware mode is RS422. 2. Verify the targeted **COM port** assignment. If ambiguous, cycle connection stability by unplugging and re-inserting the physical USB connector to re-initialize hardware handshake rules. 3. Confirm that the application data transmission **baud rate** configuration matches the source device transmission specification. The factory default is 115200 bps.
The serial port does not show up when you plug in the FTDI USB to RS232/RS422 adaptor cable.	The FTDI Virtual COM Port (VCP) driver is not installed. Windows usually installs it automatically, but it can be missing on some systems.	1. Download and install the FTDI VCP drivers from **ftdichip.com/drivers/vcp-drivers**. 2. Unplug and reconnect the USB cable. 3. Confirm a COM port appears under **Ports (COM & LPT)** in Windows Device Manager.
On Linux, the serial port does not show up, or cannot be opened, when the FTDI USB to RS232/RS422 adaptor cable is plugged in.	The user account does not have permission to access serial ports (not a member of the `dialout` group).	Add the user to the `dialout` group, then log out and log back in: sudo adduser username dialout
	ModemManager has claimed the FTDI USB-serial device.	If you are not using a modem, remove ModemManager: sudo apt-get remove modemmanager
After connecting the	Windows serial enumerator	A computer restart often resolves the problem.

Symptom	Likely Cause	Resolution
FTDI USB-to-serial cable on Windows, the mouse pointer jumps around the screen or clicks randomly, like a “crazy mouse”.	(`serenum.sys`) misidentifies serial data from the INS as a legacy serial mouse (Microsoft Serial Ballpoint) and loads a mouse driver for the COM port.	Alternatively: 1. Unplug the FTDI USB cable (or power off the INS) to regain control of the mouse. 2. Reconnect the cable. In Windows Device Manager, expand **Ports (COM & LPT)** and open the USB Serial Port. 3. Select **Properties** > **Port Settings** > **Advanced**, then clear the **Serial Enumerator** check box. 4. If a **Microsoft Serial Ballpoint** (or Microsoft serial mouse) appears under **Mice and other pointing devices**, disable it.
Data packet loss occurs when running the INS at very high data rates.	High driver latency within the FTDI USB-to-serial communication pipeline leads to buffer overruns and dropped telemetry frames.	In the *INS Manager's* Advanced Settings, reduce the **Latency Timer** to **1 ms**.

9.5 INS Manager Changelog

Version	Date	Changes
8.2	2026-09-22	**Aiding Source Configuration:** Added Aiding Source Configuration support to Motus. Dual Antenna Configuration window has been deprecated and dual antenna offsets are now configured in the Aiding Source Configuration window. Dual Antenna offset Auto mode has been deprecated due to incompatibility with the latest Fusion updates. Dual Antenna offset Manual mode has been updated with an option for pre-defined offsets or manual input. **Aiding Source Calibration:** Added Aiding Source Calibration window for odometer, body velocity, and DVL scale factors. **Internal Depth Sensor Offset Units:** Fixed Internal Depth Sensor position offset reporting incorrect units.
8.1	2026-06-26	**Improved External Aiding Configuration:** Added Aiding Sources Configuration window. Improved external aiding source lever-arm and alignment configurations and status reporting across aiding sources. **Improved Diagnostics and Monitoring:** Device Information now displays navigation filter version when available; Velocity and Acceleration window adds velocity standard deviation plotting; Satellites window indicates stale data. **Improved Reliability:** Resolved excessive memory use over long sessions, map rendering instability when position or orientation data is invalid, and NTRIP mountpoint download failures on some servers.

Version	Date	Changes
8.0	2025-11-12	Key Features: This INSManager connects to all Advanced Navigation's INS/AHRS products. Previously there was a different Manager for each product. Added support for Boreas 50 series. Added support for Mosaic X5 receiver. Added support for X20P receiver. Added support for Certus Mini receiver. Added support for GNSS Jamming and Spoofing status in new packets 92 GNSS PVT and 93 GNSS Orientation. Extra changes: Added baud rates check before being sent to the device. Added log conversion of IP configuration packet. Added drop-down to set plots timescale. Added plot units. Added Follow function to the map view. Added units to attitude indicator. Added indication of datarate in NTRIP window. Added warning when packet period is not multiple of a timestamped packet period. Added menu item pointing to the online device documentation. Added device type to window title when a device is connected. Improved Trimble receiver firmware update. Improved alignment window antenna position. Added plot lines can be hidden by clicking their legend label. Increased maximum number of packet rates that can be configured. Increased maximum number of satellites displayed in table. Added automatic disconnection if USB-to-serial cable is unplugged. Updated Omnistar L-Band frequencies. Fixed serial baud rate restore after failed firmware update. Fixed alignment parameters input on Linux. Fixed error caused by empty packet rates period. Fixed UDP Server connection mode. Fixed NTRIP disconnection error. Fixed leap seconds download error. Fixed Zero Current Orientation. Fixed multi-part ANPP packets log conversion. Fixed support for Mondo Pin NTRIP mountpoints.
7.4	2024-08-19	Release to support Certus Mini hardware
7.3	2024-04-11	Added support for Boreas A variants Added vessel motion packet, updated and expanded Heave window to Vessel Motion, including surge and sway Added Train profile as Filter Option vehicle type Improved Trimble license options decoding Added support for NavIC configuration on Trimble receiversImproved usage and syntax information in the command line help Added support for more GPIO functions during log conversion Added command line argument to connect to the device on launch Added argument to open the Manager window maximised

Version	Date	Changes
		Updated device information window to show "BOOTLOADER" when applicable to software version
7.2	2023-11-10	Clear map history on reconnection Fix incorrect timezone text displayed in terminal.csv Updated leap-seconds list URL Add automatically log configuration at the start of a new log Added support for NavIC satellite system Fixed indicators behaviour on disconnect Updated support for Mac silicon with latest macOS Added ability to Factory Restore via Tools > Device Discovery Corrected list of available auxiliary functions Fix unable to change offsets in alignment window Added support for extra NMEA 0183 messages Removed GNSS status indicator Fixed issue with opening windows containing charts Renamed External GNSS Enabled to Internal GNSS Enabled Increased NTRIP client timeout Update support for Trimble BD9xx RTX frequencies as of 30 Sep 2022 Show COCOM status of Trimble BD9xx receivers in the licence field In satellites window, only populate azimuth and elevation fields in the table view when the values have been determined Added elevation to the gimbal state packet Add support for Boreas subtypes Add ability to set Packet Timer period
7.0	2023-02-13	Updated from JDK8 to JDK11. **JRE11/JDK11 required to run this release** Refresh Port Mode information on reconnection Refresh Packet Rates information on reconnection Fixed Ethernet Settings Static IP setting Improved reliability of firmware update Changed Low Voltage label to Internal Data Logging Error to support newer firmware Fixed issue with NTRIP client streaming from some servers When the NTRIP sourcetable says authentication type is any of '', 'B', 'N', then use Basic Authentication. This is to workaround NTRIP servers that incorrectly populate the authentication field Overwrite/clear existing packet rates on import of configuration Added Disable Dual Antenna Heading option Updated Seastar frequencies for Trimble receivers Updated RTX frequencies for Trimble receivers Added support for ARM (aarch64) processors with 64-bit OS Added support for Apple M1 processors and improved MacOS support Various UI improvements
6.2	2022-10-10	Fixed Heading indicator reporting as invalid after a successful Coarse Heading Alignment Added Moving Base output to GPIO configuration Fixed Packet Rates not being refreshed after a save Removed unavailable GPIO options

Version	Date	Changes
		Fixed status indicator message in case of a logging error Fixed configuration import via network Added allow firmware update over ANPP using Auxiliary, GPIO and Ethernet ports Added GNSS Firmware Update for BD992 variants Added CAN settings configuration Added ANPP14 Subcomponent information to Log Converter Added support for NTRIP Casters that request Username & Password to retrieve source table Added copying version numbers etc from Device Information and GNSS Receiver Information dialog boxes Renamed Log Conversion menu item Added Settings menu item BaseStation.csv output "Fix Type" column now split into "Base Fix Type" and "Rover Fix Type" Added GPIO voltage selection Removed of SBAS corrections option when SBAS is not supported on the device Fixed configuration export for POE variant Improved reliability of GNSS Firmware update Various bug fixes Renamed position error to position standard deviation in position page of view menu
6.1.1	2022-02-18	Device information updated automatically after a firmware update Fixed configuration export Add ability to control log file location Included software version number in title Improved NTRIP server reconnection Improve export configuration reliability Added external GNSS Filter option Improved Log files saved by default in the directory from where the application is launched Improved NTRIP mountpoint selection. Added handling of nonstandard NTRIP server responses Force the device to exit bootloader mode if a firmware update is not in progress
6.1	2022-02-18	Initial release of Boreas Manager Improved NTRIP mountpoint selection Added handling of non-standard NTRIP server responses Changed "Low Voltage" Status Flag to "Internal Data Logging Error" Added EMSAT, OCSAT, IOSAT, etc and corresponding default frequencies to the Omnistar GNSS Configuration list Added ability to set custom L Band satellite frequency and baud GNSS receiver information is updated after applying a receiver option Updated source of map tiles Improved NTRIP server reconnection Improved export configuration reliability Added Device Discovery tool to find Advanced Navigation devices on the network Improve display of Trimble receiver options Added the ability to configure the Manager logging folder
6.0	2021-	Major update to support v2.0 hardware and v2.0+ firmware

Version	Date	Changes
	10-29	
5.9	2021-06-28	Add ability to disable logging Add ability to control log file location Add GNSS configuration page, support GNSS auto detect Log Converter now writes program version/name to output file Extract Magnetometers Enabled and Automatic Magnetic Calibration Enabled to Configuration.txt Add sensor temperatures packet to log converter Reset running time display upon connection Support new style GPIO output packet Fix GPIO output configuration window not working Fix NTRIP NMEA Message Output at Irregular Periods Various bug fixes
5.7	2020-09-23	Added support for Certus Evo Configuration Export and Import
5.6	2020-09-09	Updated raw sensors dialogue to include all temperatures Fixed bug with communications statistics dialogue cutting off numbers Added zero angular velocity update GPIO function Added Mavlink output GPIO function
5.3	2020-06-30	Add support for serial firmware update via Certus application for ANFW >2MB Fix bug in Extended Satellites display where it could show stale data. Add ability to set port modes (eg RS232 / RS422)
5.2	2020-05-20	Initial Release

Table 19: INS Manager Software Changelog

9.6 Using INS Manager

The following sections describe how to use the Desktop and Web versions of the INS Manager.

Note: The web and desktop variants have slightly different features.

Note: The features available varies slightly by INS product type, and a feature described here might not be available on your product.

9.6.1 Main View

The Main View is accessible via `View > Map`.

It provides an immediate overview of the system's status, attitude, GNSS connectivity, and current position on the map.

The desktop version additionally includes controls for serial connection, while the web version includes the current time and position readout .

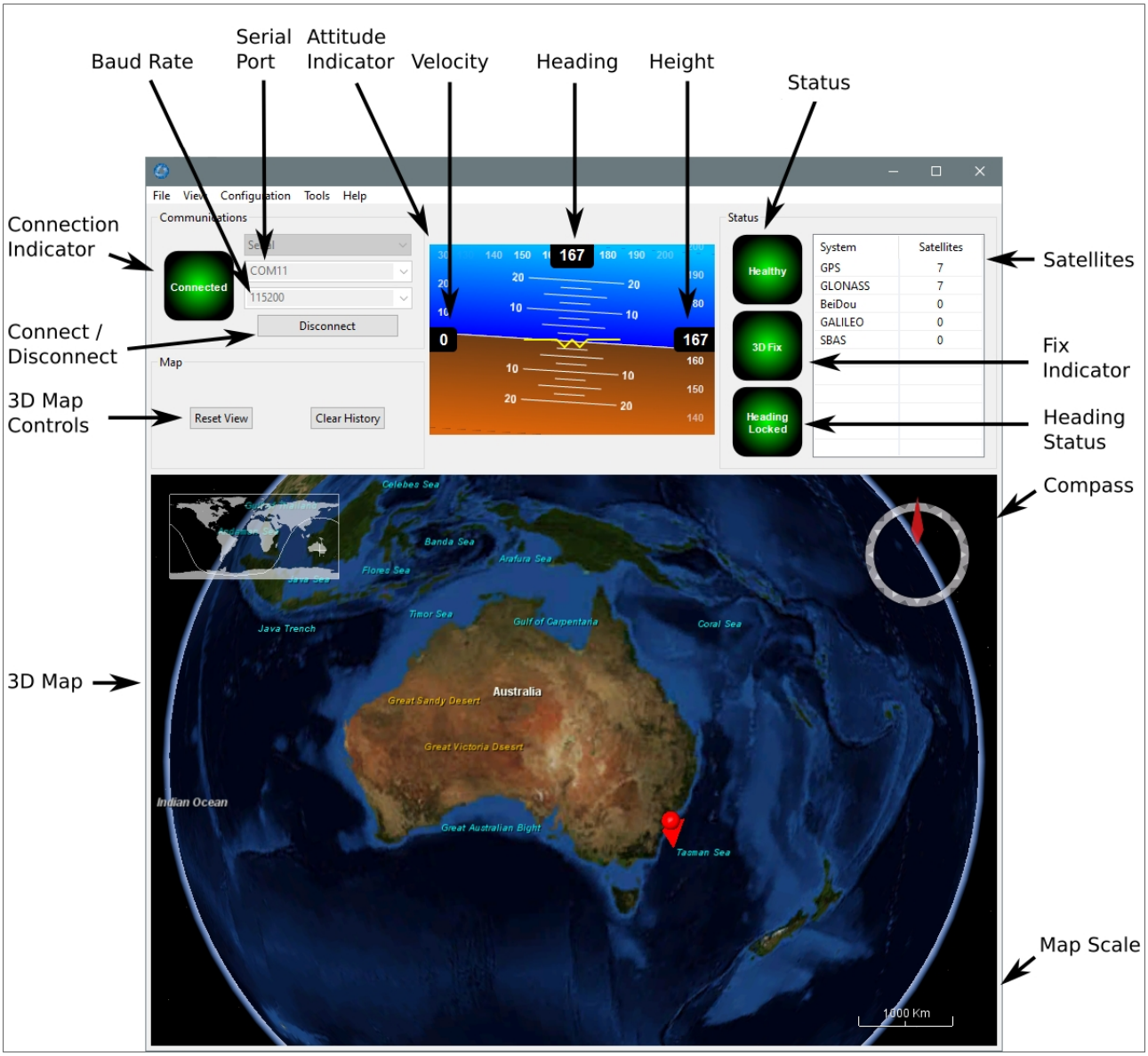


Figure 24: INS Manager Desktop Main View

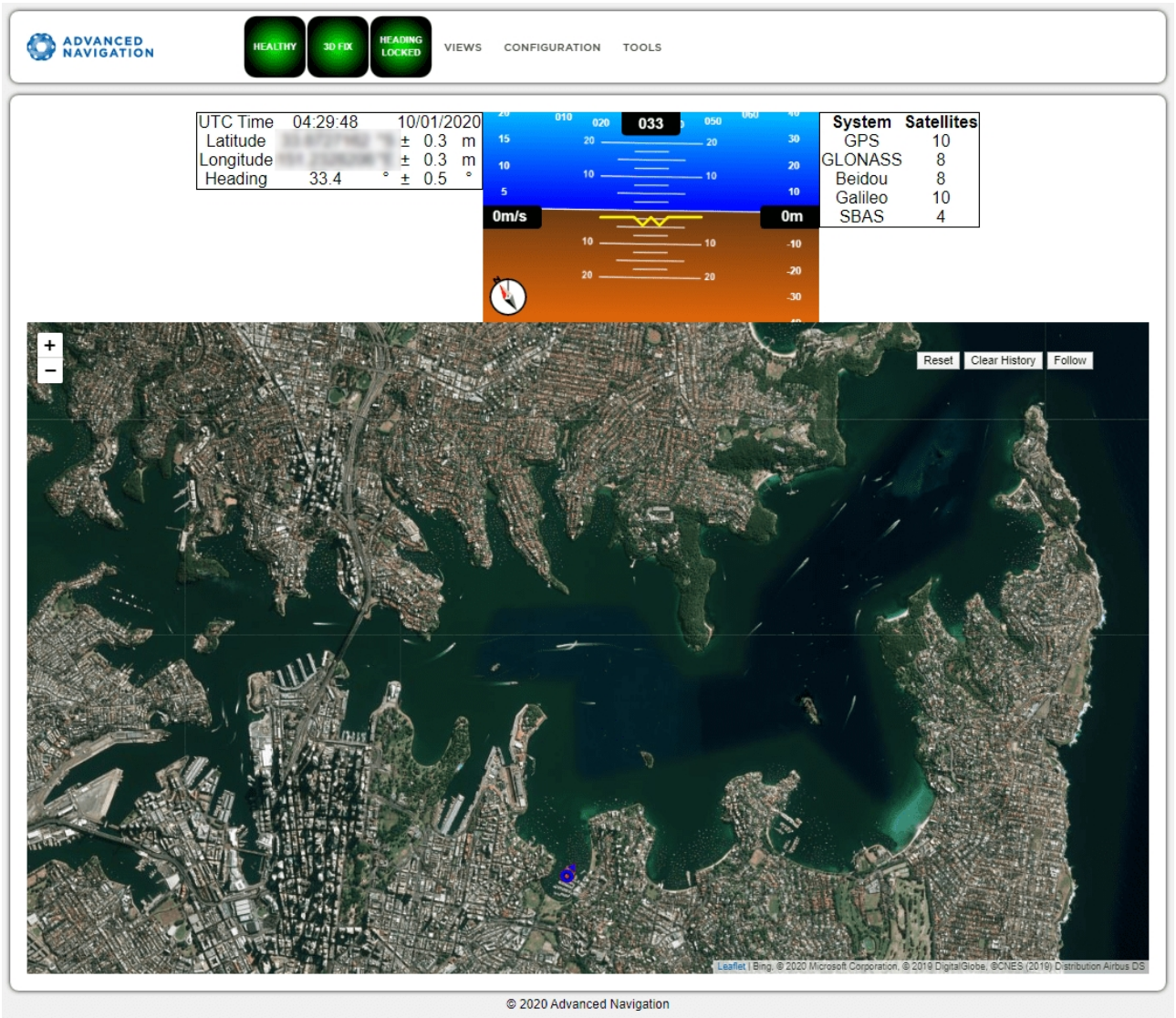


Figure 25: INS Manager Web Main View

Tip

1. *INS Manager* is best viewed on a screen size of 1280 x 1024 or larger.
2. When viewing *INS Manager - Web* from a small screen, you can change your browser zoom level better use screen real estate.

9.6.1.1 Communications Section

[INS Manager Desktop only]

Within the Communications Pane, you can select to connect to a device using a `Serial` or `Network` connection.

For serial communications:

- Select `Serial`
- Select `COM Port`
- Select `Baud Rate` (default is 115200)

For network communications. In most cases, you can find your device from `Tools > Device Discovery`

- Select `Tools > Device Discovery`
- Select your device from the list.
- Select `Connect` (default is 115200)
- Close the `Device Discovery` Popup

Alternatively you can select parameters directly

- Select `Network`
- Select `IP Address`
- Select `Port` (Default 16718. Can use up to 4 ports.)

Select `Connect / Disconnect` to connect the Desktop application to the INS unit.

9.6.1.2 Current Time, Position and Heading

[INS Manager Web only]

This section displays the current UTC time, position, and heading. The data will only be valid if the INS has a HEALTHY status, there is a 3D GNSS fix or better, and the Heading is initialised.

9.6.1.3 Attitude Indicator

The Attitude Indicator visualises roll and pitch using a standard artificial horizon, similar to a traditional aircraft instrument. Heading, velocity, and height are displayed adjacent to the horizon. All displayed units are in SI (metric) and degrees.

9.6.1.4 Satellites Table

The Satellites page provides the satellites visible to the primary antenna. More detailed information is available in `View > Satellites`.

Note: The satellites table will only display satellite data if your device has an internal GNSS receiver, or is interfaced to an external GNSS receiver.

9.6.1.5 Status Indicator

The Status Indicator section provides immediate visual feedback on the operational state of the INS unit. These indicators are visible on every page of the interface.

- Status Indicator: This indicator reports any critical problems. Before initialisation, the status may show “Filter not initialised” or similar. Once the filter has initialised, it will show HEALTHY.
- GNSS Fix / Fix Indicator: This indicates the quality of the GNSS position fix. Under normal conditions, this will show 3D FIX or SBAS FIX. Visibility issues may result in 2D FIX or NO FIX.
- Heading Status: This indicator shows the current status of the heading filter. When a valid source of heading is active, it will show HEADING LOCKED.
- Satellites Table: This table shows the number of active satellites contributing to the current GNSS solution.

9.6.1.6 Map

[Option in INS Manager Web. Available from main page in INS Manager Desktop.]

The map provides a visual representation of the unit's current position on Earth and displays a red trail showing the position history since the last reset. Upon filter initialisation, the map automatically centers the view on the unit's current position. Basic map controls include:

- Movement: Click and drag on the map to move the camera view.
- Zoom: Use the mouse scroll wheel to zoom in and out.
- 3D Camera View: *[INS Manager Desktop only]* To manipulate the camera angle in 3D, use right click and drag or shift click and drag.

9.6.1.7 Map Controls

The map controls are functionally identical across both the Web and Desktop interfaces:

Reset

This resets the map view to center on the unit's current position.

Clear History

This clears the current red position history trail, immediately restarting the trail from the current location. This action does not impact active data logging.

Follow

This button automatically keeps the map view centered and tracking the device's current position.

9.6.2 View Menu

From the `View` menu, you can select pages to view real-time data from the INS.

9.6.2.1 Map

[INS Manager - Desktop only]

`View > Map` takes the user back to the home which features the map display.

9.6.2.2 Device Information

`View > Device Information` displays the unit's:

- Serial number.
- Device ID.
- Firmware version.
- Hardware version.

This page is useful when requesting technical support.

9.6.2.3 Subcomponent and GNSS Receiver Information

The `INS Manager Web > View > Subcomponent Information` and `INS Manager Desktop > View > GNSSS Receiver Information` displays detailed information about the GNSS receiver fitted to the INS.

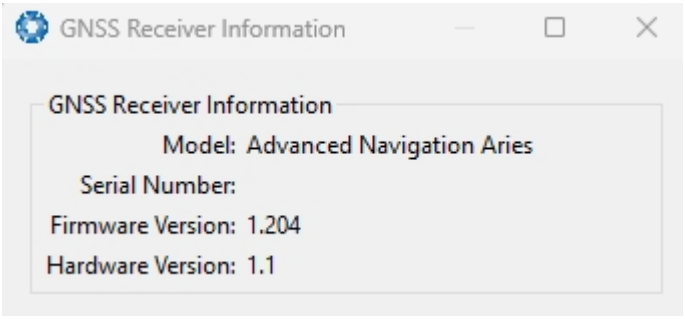


Figure 26: INS Manager Desktop GNSS Receiver Information Dialogue

If your device is fitted with a Trimble GNSS receiver, the Desktop dialogue features a `License Update` text field. To upgrade your licence, copy and paste the license key into the `License Update` field and click `Send`.

9.6.2.4 Status

`View > Status` shows the current system status, filter status, and GNSS status flags, as reported by **13.4.1 System State Packet** (ID#20) and **13.4.10 Raw GNSS Packet**. (ID#29)

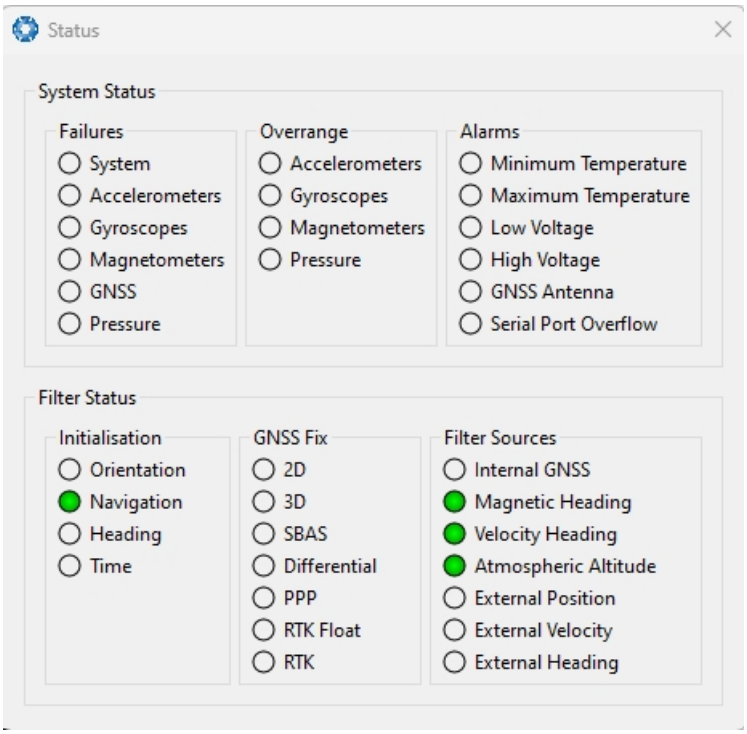
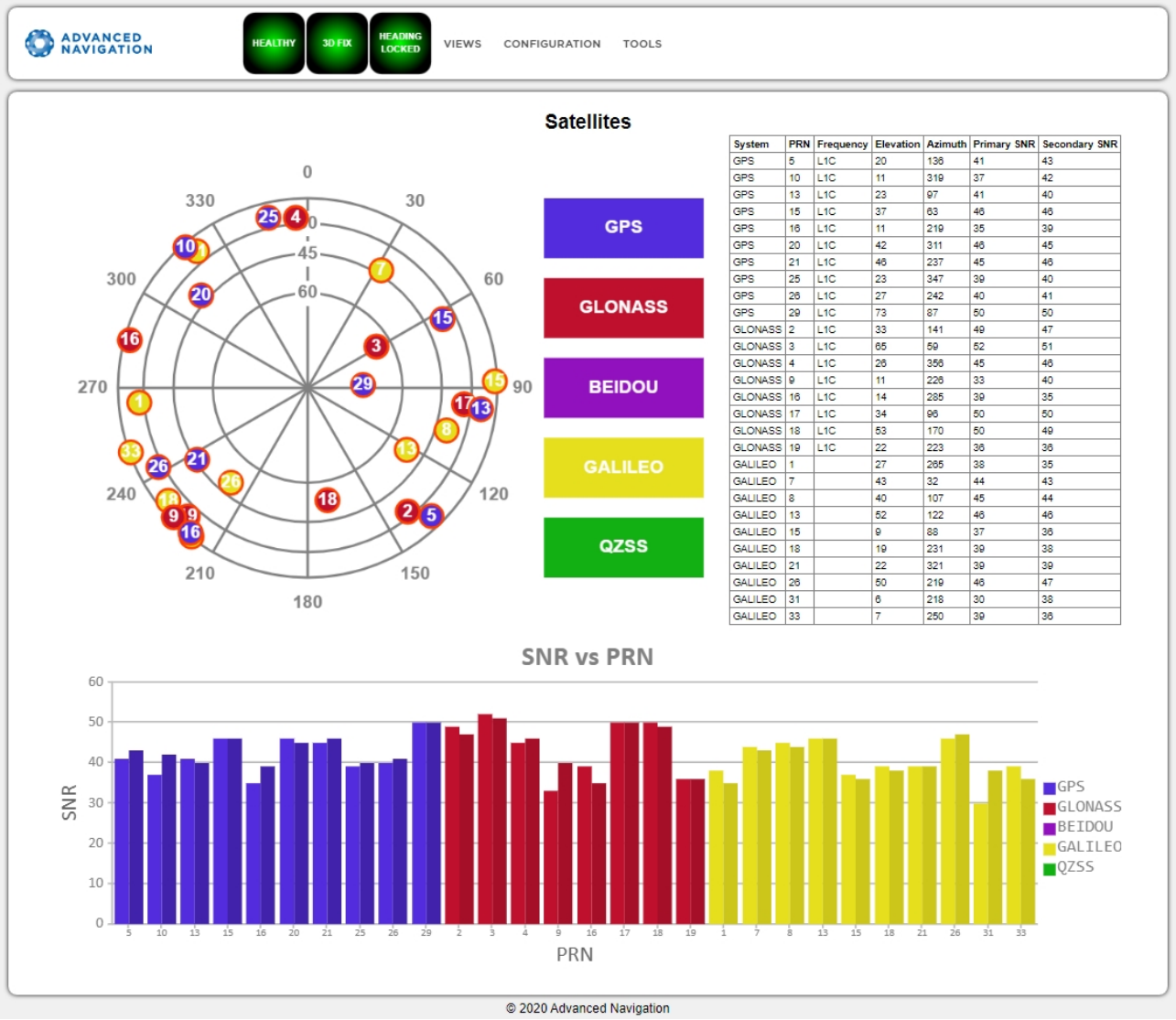


Figure 27: INS Manager Desktop Status Dialogue

9.6.2.5 Satellites

`View > Satellites` shows detailed information on the satellites visible to the INS GNSS receiver. The display includes a satellite skyplot, a table, and an SNR (Signal-to-Noise Ratio) graph.

The plot shows the SNR of each satellite or each satellite at the primary and secondary antenna. As a guideline, SNRs should be within 5-10 dB of each other in the absence of cable or antenna faults. Elevation and azimuth are displayed in units of degrees.



System	PRN	Frequency	Elevation	Azimuth	Primary SNR	Secondary SNR
GPS	5	L1C	20	136	41	43
GPS	10	L1C	11	319	37	42
GPS	13	L1C	23	97	41	40
GPS	15	L1C	37	63	46	46
GPS	16	L1C	11	219	35	39
GPS	20	L1C	42	311	46	45
GPS	21	L1C	46	237	45	46
GPS	25	L1C	23	347	39	40
GPS	26	L1C	27	242	40	41
GPS	29	L1C	73	87	50	50
GLONASS	2	L1C	33	141	49	47
GLONASS	3	L1C	65	59	52	51
GLONASS	4	L1C	26	356	45	46
GLONASS	9	L1C	11	226	33	40
GLONASS	16	L1C	14	285	39	35
GLONASS	17	L1C	34	96	50	50
GLONASS	18	L1C	53	170	50	49
GLONASS	19	L1C	22	223	36	36
GALILEO	1		27	265	38	35
GALILEO	7		43	32	44	43
GALILEO	8		40	107	45	44
GALILEO	13		52	122	46	46
GALILEO	15		9	88	37	36
GALILEO	18		19	231	39	38
GALILEO	21		22	321	39	39
GALILEO	26		50	219	46	47
GALILEO	31		6	218	30	38
GALILEO	33		7	250	39	36

Figure 28: INS Manager Web Satellites Table

9.6.2.6 Raw Sensors

`View > Raw Sensors` displays the raw IMU sensor values.

The plots include accelerometer, gyroscope, and magnetometer readings. Temperature readings are provided for a range of sensors, including T(Ax), T(Ay), T(Az) for the accelerometer axes; T(Gx), T(Gy), T(Gz) for the gyroscope axes; T(M) for the magnetometer; and T(P) for the pressure sensor.

The lower plot visualizes air pressure. Internal temperature sensors are also reported but do not reflect the general environmental temperature.

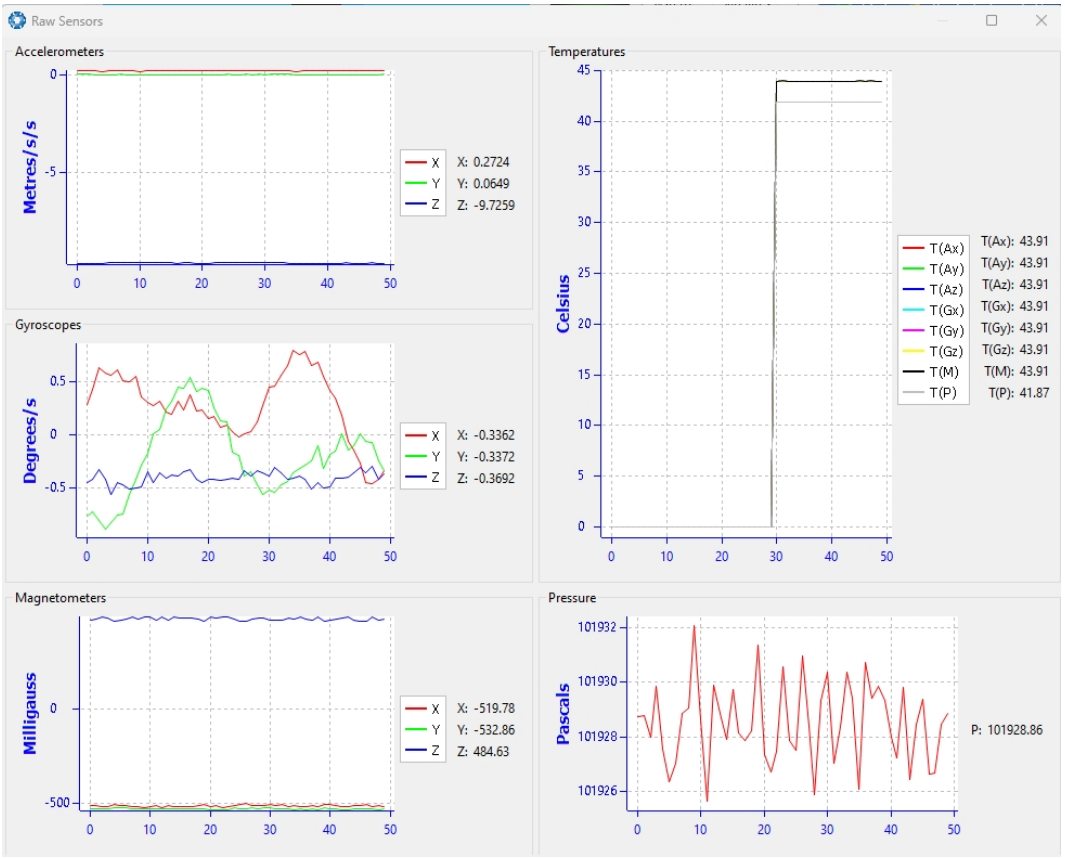


Figure 29: INS Manager Desktop Raw Sensor Outputs

9.6.2.7 Orientation

`View > Orientation` shows the INS orientation, angular velocity, and orientation error.

- `Course` is the path determined by measuring sequential positions.

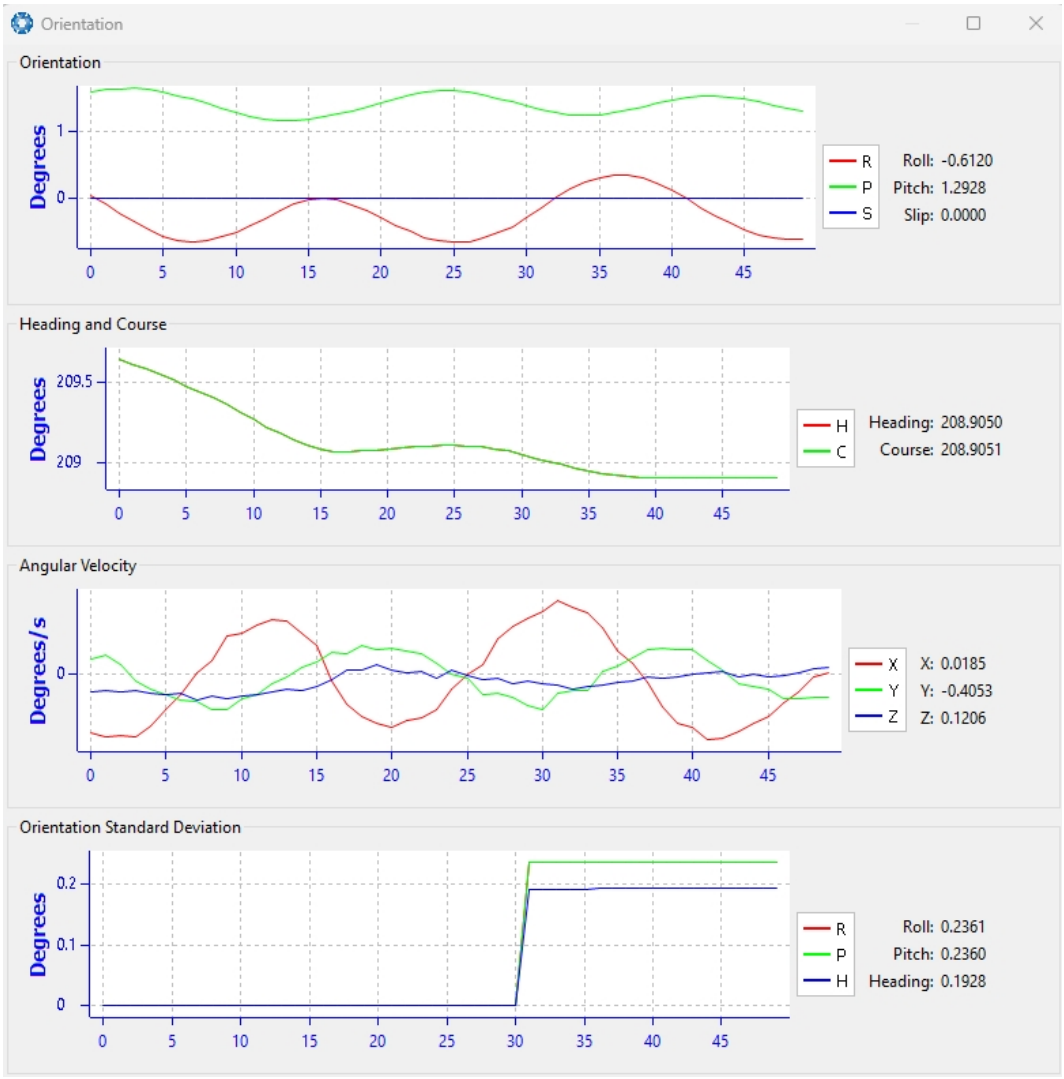


Figure 30: INS Manager Desktop Orientation Outputs

9.6.2.8 Position

`View > Position` shows the INS position and position error (standard deviation). Latitude and longitude are converted to North and East metres from a reference point.

The position reference point can be reset in `INS Manager - Desktop` or by refreshing the web page in your browser with `INS Manager - Web`.

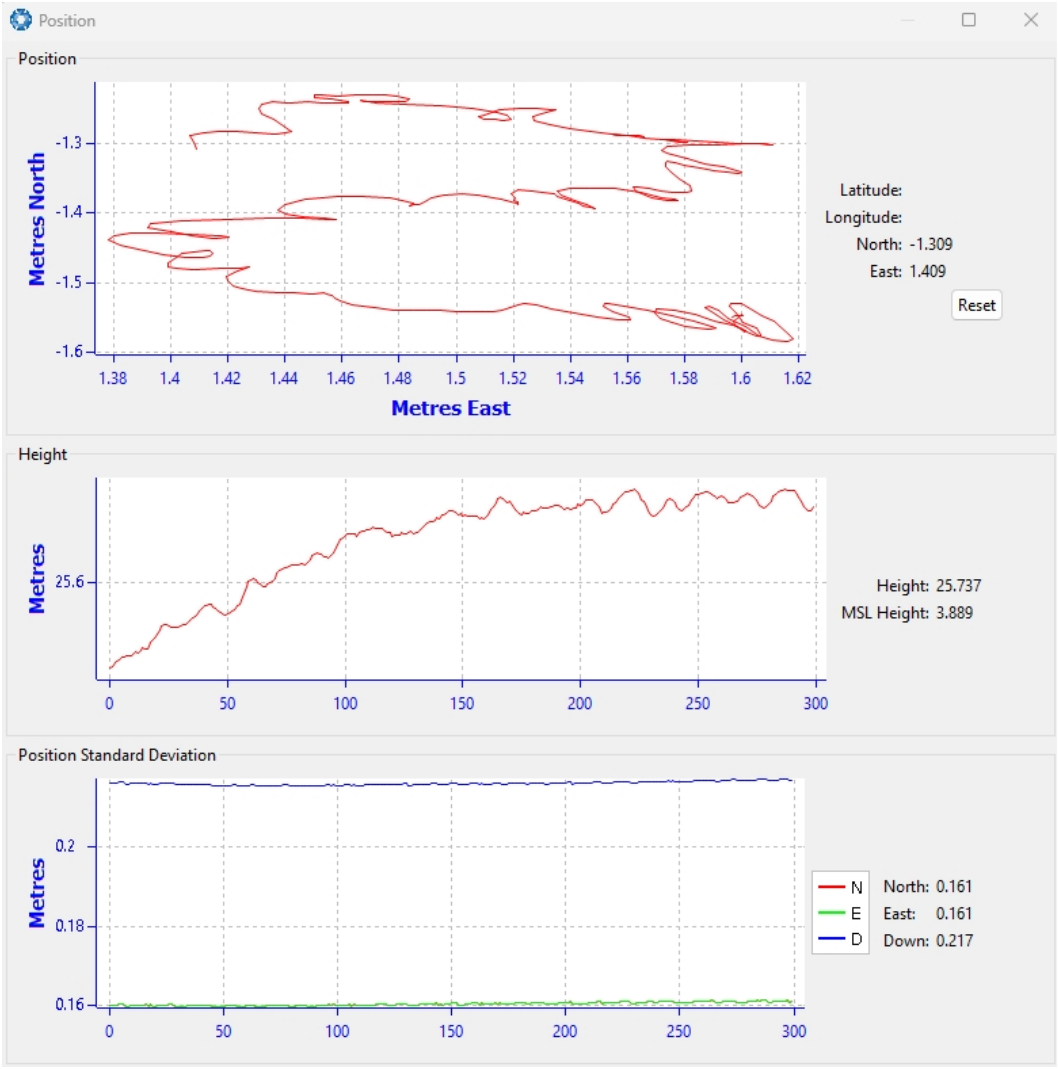


Figure 31: INS Manager Desktop Position Outputs

9.6.2.9 Velocity and Acceleration

`View > Velocity and Acceleration` shows the INS velocity, acceleration, and g-force over time.

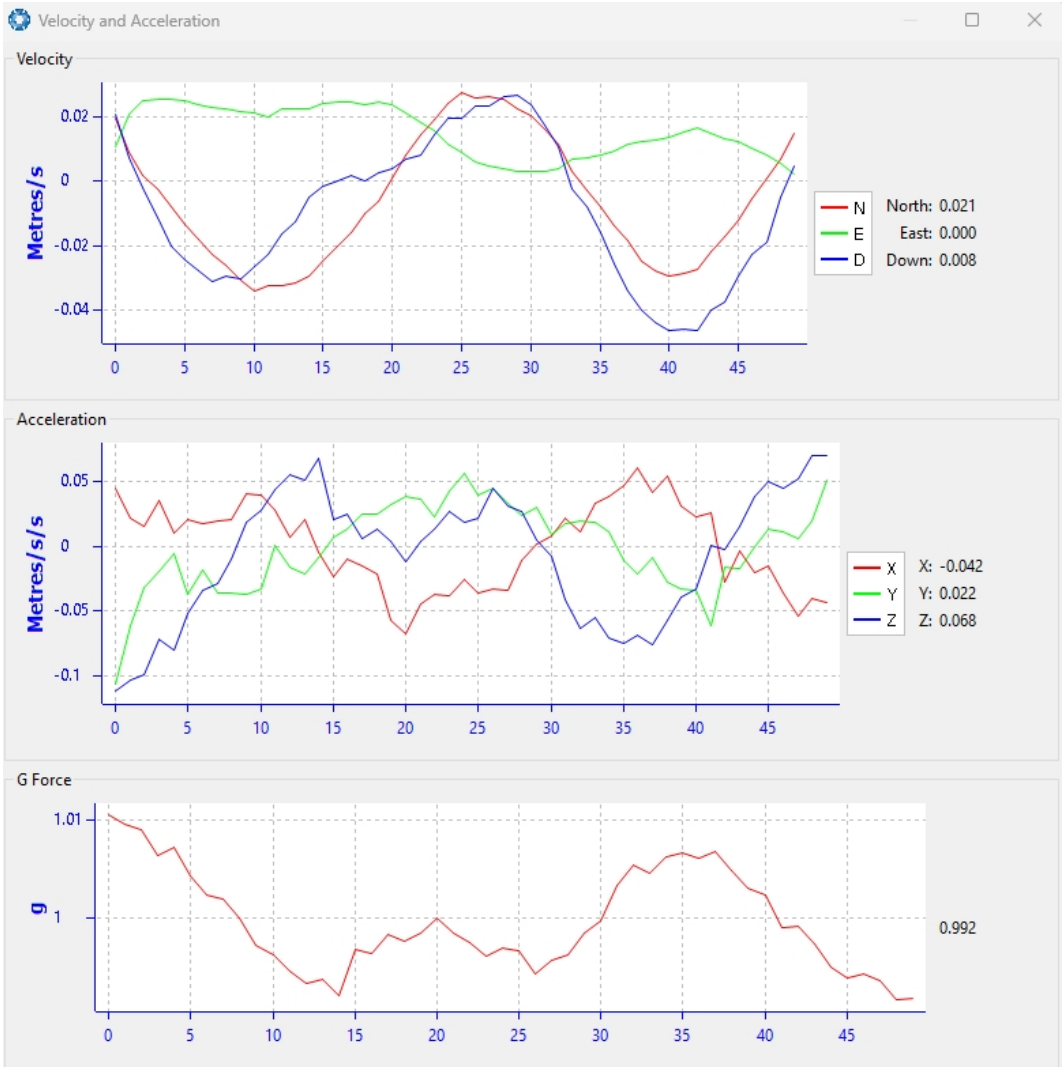


Figure 32: INS Manager Desktop Velocity and Acceleration Outputs

9.6.2.10 Time

[INS Manager - Desktop only]

`View > Time` displays the time for the connected device in Unix, UTC, or Local time formats, ensuring accurate timekeeping and synchronisation across systems.

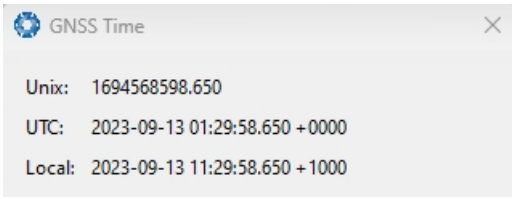


Figure 33: INS Manager Desktop Time Dialogue

9.6.2.11 Communications

[INS Manager - Desktop only]

`View > Communications` shows statistics on the data packets received from the INS (e.g., packets decoded, bytes discarded, LRC/CRC errors and is useful for diagnosing any signal quality problems on the connection, such as signal strength or connectivity.

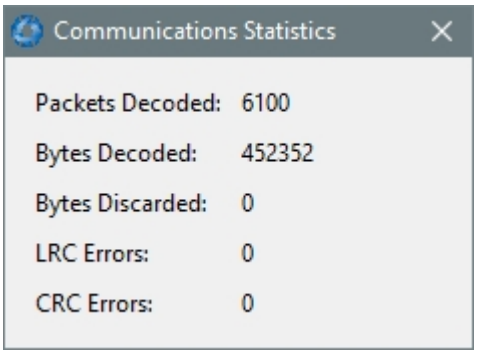


Figure 34: INS Manager Desktop Communications Dialogue

9.6.2.12 Vessel Motion

`View > Vessel Motion` This page displays the vessel motion (heave, surge, sway) at up to 4 reference points. For data to display, the Vessel Motion Packet (ID 89) must be configured in the **9.6.3 Configuration Menu** dialogue. Reference point offsets are configured in the **9.6.3 Configuration Menu** dialogue.

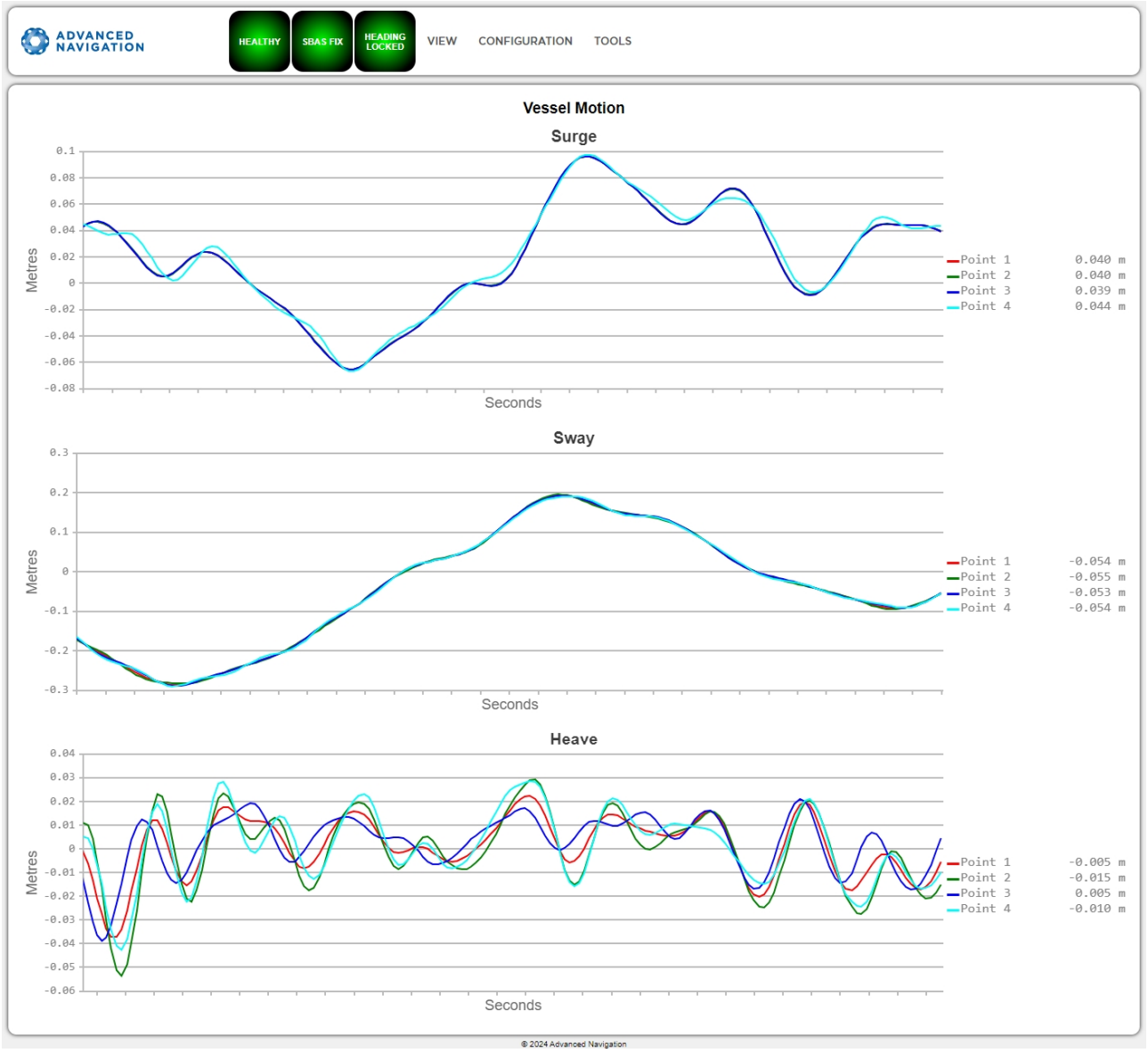


Figure 35: INS Manager Web UI Vessel Motion Page

9.6.2.13 Heave

`View > Heave` displays the heave at up to 4 reference points. For data to display, the Heave Packet (ID 58) is configured in the **9.6.3 Configuration Menu** dialogue. Reference point offsets are configured in the **9.6.3 Configuration Menu** dialogue.

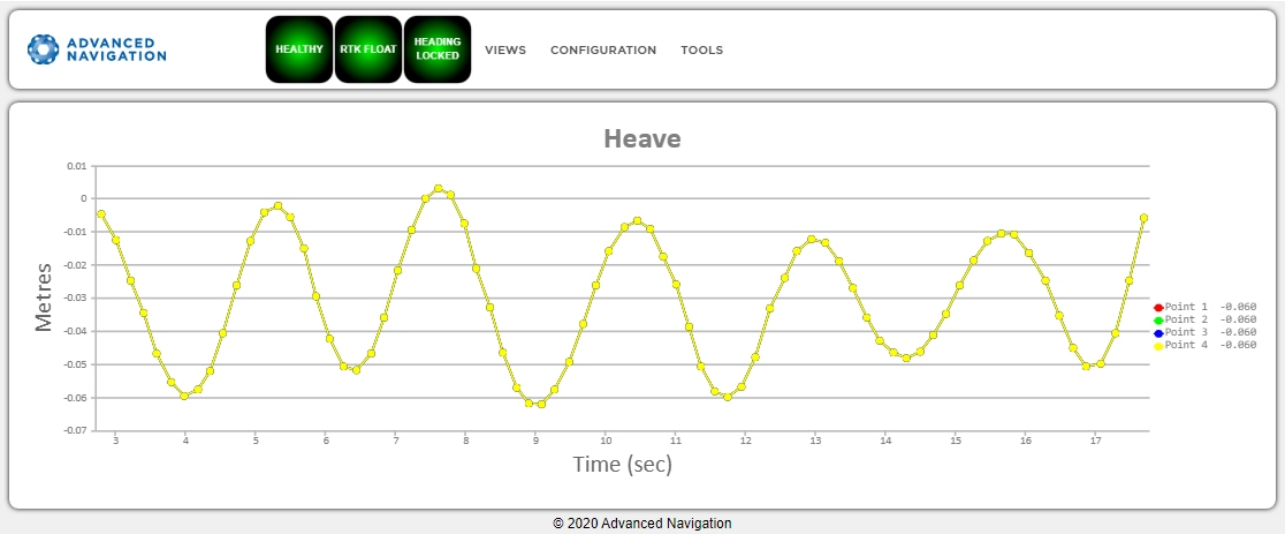


Figure 36: INS Manager Web UI Heave Page

9.6.2.14 North Seeking Status

`View > North Seeking Status` displays the current status and initialization messages.

Note: North Seeking Alignment is used to derive the heading based on gyrocompassing, which determines true north from the rotation of the earth. Gyrocompassing is utilised to establish a heading reference when alternative sources, such as dual-antenna GNSS or velocity heading, are unavailable.

Status Messages	Description
Awaiting position via GNSS or manual initialisation	An approximate position is required. Initialisation commences once position is available (via GNSS or the **Manual Initialisation** page).
Initialising, please wait	The process of determining heading has started.
Alignment initialised via alternate source	Alignment is bypassed because heading has been successfully established via an alternate source (e.g., dual antenna, velocity, or external input).
Heading alignment complete	The system has successfully determined heading.

Table 20: Heading Alignment - Status Messages

Warning Messages	Description
Initialisation restarted - excessive motion detected	The initialization has restarted due to excessive motion.
Initialisation restarted - change in latitude detected	The initialization has restarted due to a significant change in latitude.

Warning Messages	Description
Initialisation restarted - Change in COG lever arm detected	The initialization has restarted due to a change in the Centre of Gravity (COG) lever arm **configuration**.
Warning - Check latitude, value out of range	A discrepancy between the current latitude and the alignment algorithm was detected. Verify the current latitude is correct.

Table 21: Heading Alignment - Warning Messages

9.6.2.15 External Data

[INS Manager - Web only]

`View > External Data` acts as a jump page to open time-based data graphs for each supported external source. External sources must be connected and configured appropriately for the corresponding packet to report valid data.

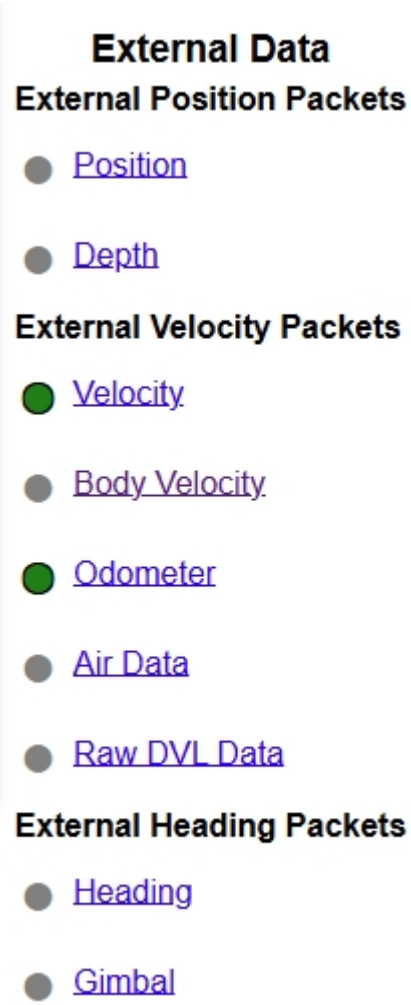


Figure 37: INS Manager Web UI External Data Selection Page

The indicator color next to the External Packet Link reflects the data validity:

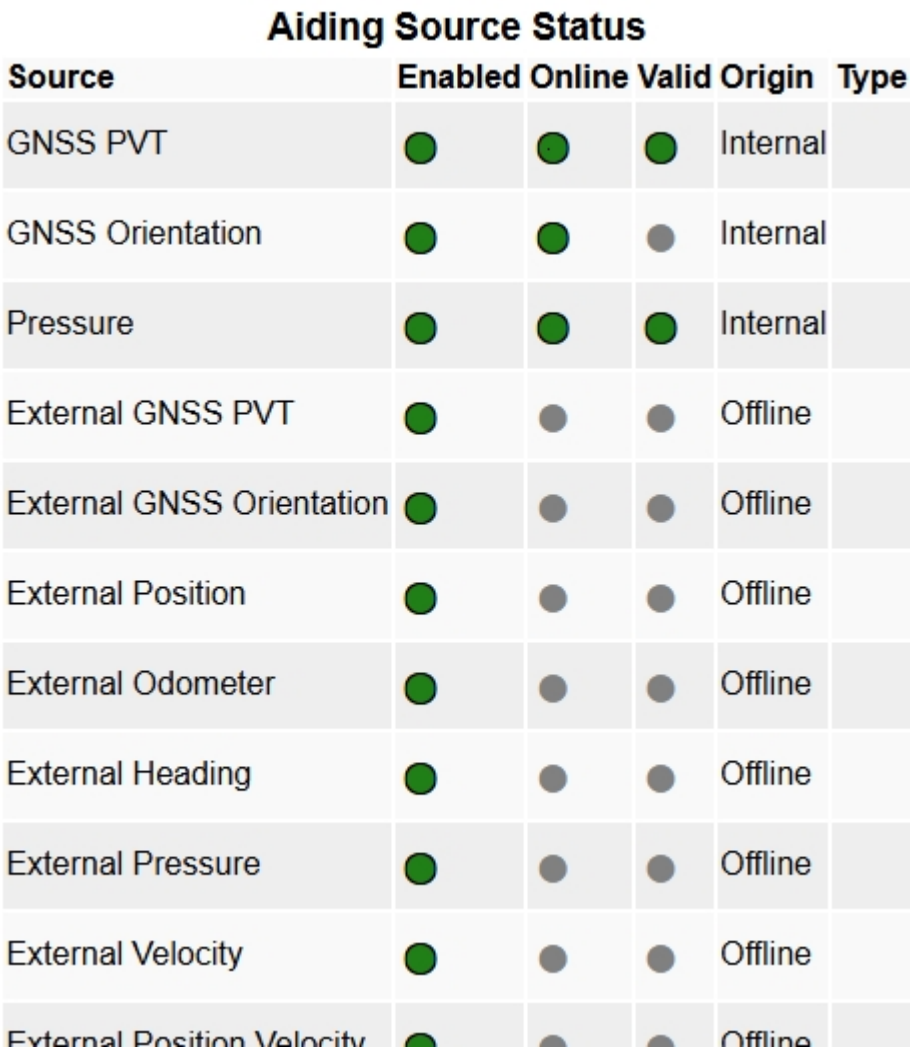
Indicator color	Validity of incoming data
Green	Valid
Grey	Invalid

9.6.2.16 Aiding Source Status

`View > Aiding Source Status` Shows the status of Aiding Sources, covering:

- Enabled
- Online: Receiving data

- Valid: Data is valid and usable by the filter
- Origin: (Internal, [Port], Offline)
- Type: [Format] such as "ANPP"



Aiding Source Status

Source	Enabled	Online	Valid	Origin	Type
GNSS PVT	●	●	●	Internal	
GNSS Orientation	●	●	○	Internal	
Pressure	●	●	●	Internal	
External GNSS PVT	●	○	○	Offline	
External GNSS Orientation	●	○	○	Offline	
External Position	●	○	○	Offline	
External Odometer	●	○	○	Offline	
External Heading	●	○	○	Offline	
External Pressure	●	○	○	Offline	
External Velocity	●	○	○	Offline	
External Position Velocity	●	○	○	Offline	

Figure 38: INS Manager Web UI Aiding Source Status

9.6.3 Configuration Menu

The Configuration menu contain options to configure the INS.

9.6.3.1 Configuration Export

`Configuration > Configuration Export` is used to export or import all the INS settings to or from a file. This is useful to restore a unit to a preset configuration at a later date or for batch configuration of multiple units.

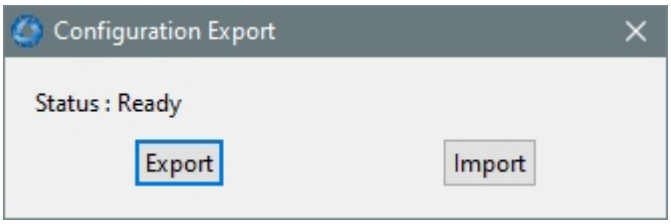


Figure 39: INS Manager Desktop Configuration Export Dialogue

9.6.3.2 Filter Options

`Configuration > Filter Options` are used to enable or disable aiding sources and optimize algorithms bases on known conditions.

For Vehicle Type, select the most appropriate option for your application.

For most applications, filter options can be left as their defaults.

See **8.2 Filter Initialisation** for more information on these specific settings.

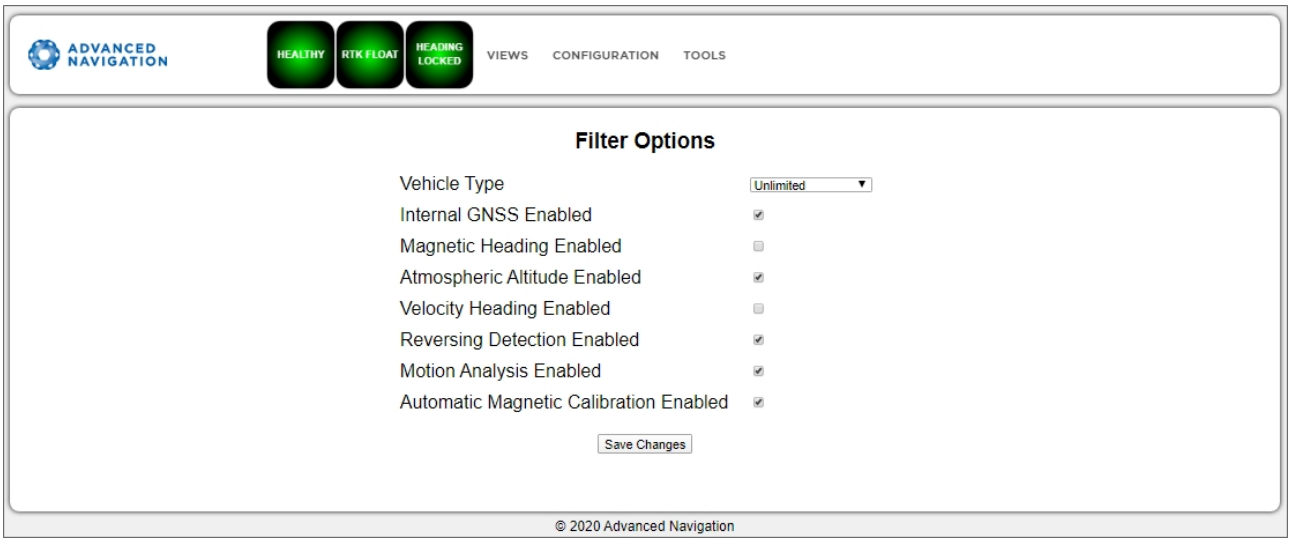


Figure 40: INS Manager Web Filter Options Page

9.6.3.3 Sensor Ranges

Note: The sensor range page is available for some of the INS products, (including Certus, Certus Mini, GNSS Compass). The sensor range can only be modified for these products; other products offer a fixed sensor range.

`Configuration > Sensor Ranges` page is used to set the dynamic range of the sensors. If you are experiencing over-range events during operation, these settings should be used to increase the range of the sensor which is reporting the over-range. The lowest ranges give the best performance, so make changes slowly and monitor the results.

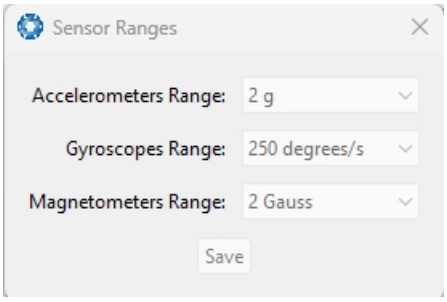


Figure 41: INS Manager Desktop Sensor Ranges Dialogue

9.6.3.4 Data Ports / Packet Rates

`Configuration > Data Ports` (Web) `Configuration > Packet Rates` (Desktop) allows you to specify which data packets or messages are output on a periodic basis and at what rate via serial or Ethernet ports.

The default packets enabled for the primary port are the **13.4.1 System State Packet** (ID #20) and the **13.4.9 Raw Sensors Packet** (ID #28) at 20 Hz and these typically provide all the data that a user will require. These two packets need to be enabled for the data graphs to update in the INS Manager. Other ANPP format state packets can be enabled as required.

Note: Packet rate configuration applies only to the data port which the Manager is connected to. To configure other ports, the Manager must be connected to that port.

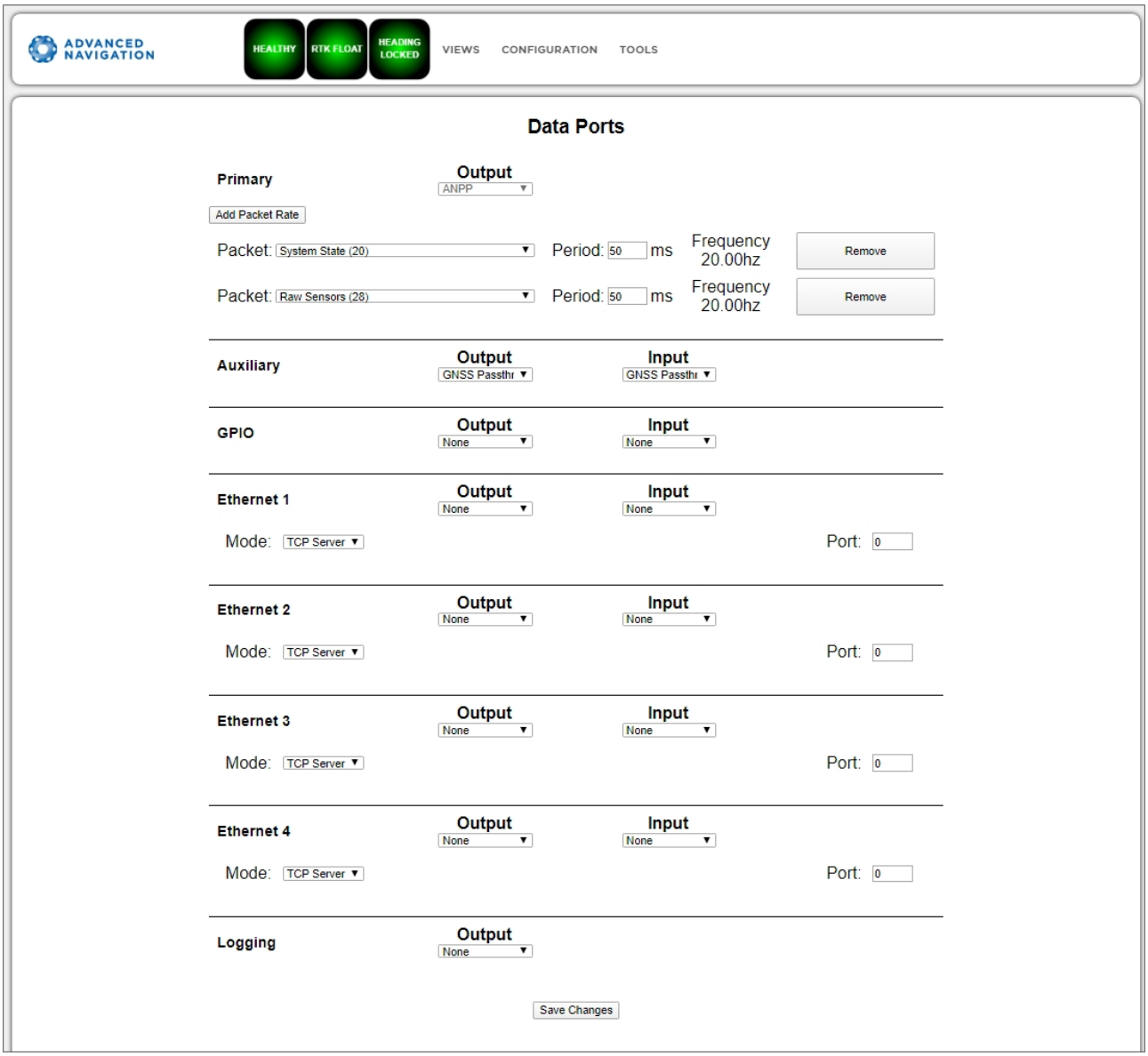


Figure 42: INS Manager Web Data Ports Page

9.6.3.5 Alignment

`Configuration > Alignment` is used to set the alignment offsets of the system installation. It is important to set the values correctly to ensure accurate results.

For most applications, only the GNSS antenna offset values need to be entered and the rest of the values can be left at their factory defaults of zero.

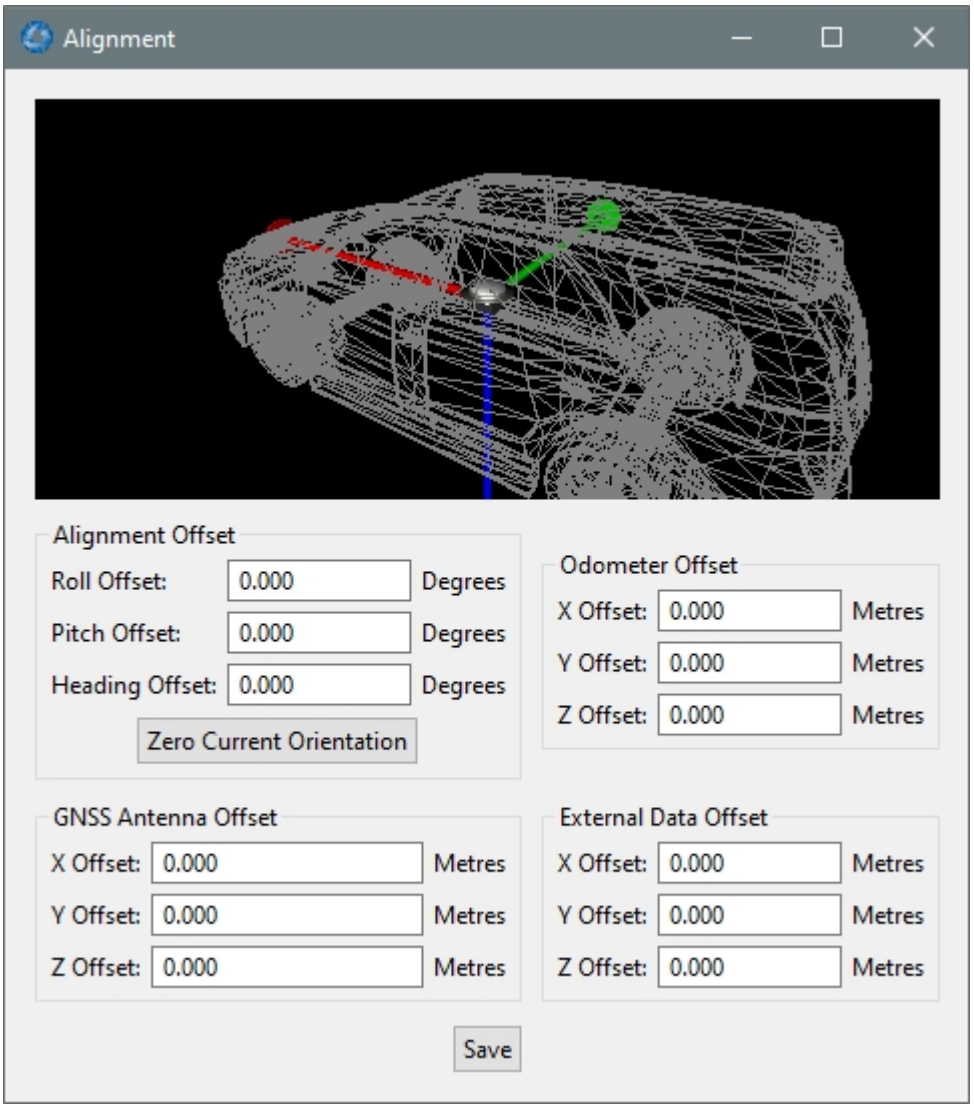


Figure 43: INS Manager Desktop Alignment Dialogue

Alignment Offset

If the INS is installed into the vehicle with the positive X axis pointing forwards and the positive Z axis pointing down, then no alignment offset is required and the roll, pitch and heading offset values can remain at the factory defaults of zero.

If the unit is installed in a different orientation then the roll, pitch and heading offset must be entered. For example, if the unit is installed on its side with the X axis pointing up and the Z axis pointing forwards and no change to the Y axis, then this would result in a pitch offset of +90 degrees with roll and heading remaining zero.

If there is a small misalignment due to mechanical mounting error this can be compensated for by setting the vehicle stationary on a level surface and clicking the `Zero Current Orientation` button.

Note: Zero Current Orientation will only correct for roll and pitch offsets, the Heading offset must be entered manually and saved after using this function.

GNSS Antenna Offset

The GNSS antenna offset is measured from the centre of the INS unit to the phase centre of the primary antenna in the orientation of the Body Coordinate Frame: (X positive forward, Z positive down).

Odometer Offset

The odometer offset is measured from the centre of the INS unit to the point at which the vehicle's tyre being measured makes contact with the road in the body co-ordinate frame (X positive forward, Z positive down). If your odometer is not measuring a specific wheel, the offset should be to the point on the ground beneath the measurement point. If, for example, your car is a front wheel drive and you are using the velocity from the car OBDII port, the measurement point would be midway between the two front wheels.

External Data Offset

These values are only required for speciality applications operating with an external source of velocity and position data. The offsets are used when providing NMEA Input, External GNSS, or with an Air Data Unit. When using a DVL, use the odometer offset.

9.6.3.6 Baud Rates/Port Mode and Baud Rates

`Configuration > Baud Rates` (Web) `Configuration > Port Mode and Baud Rates` (Desktop) is for setting the baud rates for the GPIO, primary and auxiliary ports.

The factory default baud rate value for these ports is 115200 bps.

Note: Some machines running Microsoft Windows do not support higher baud rates. When changing baud rates, it is recommended to test the baud rate configuration first, without ticking the Permanent box. This way, if it is not possible to communicate at the higher baud rate, a power cycle can be used to revert to the previous baud rate setting.

For newer INS products, the ports can be configured in software to RS232 or RS422. The factory default is dependant on your product.

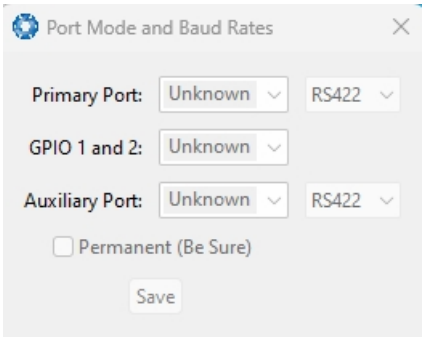


Figure 44: INS Manager Desktop Baud Rates Dialogue

9.6.3.7 GPIO

`Configuration > GPIO` allows the user to configure the input and output functions of the GPIO and Auxiliary ports. These functions are described in detail in **12.3.1 Dynamic Pin Functions**. These functions change dynamically and are effective immediately upon clicking `Save Changes`.

Note: The GPIO port functions at RS232 levels for data functions and 0 to 5 volt (or 0 to 3.3 V) levels for all other functions. The auxiliary port functions at RS232 or RS422 levels for data functions and 0 to 5 volt (or 0 to 3.3 V) levels for all other functions. The internal hardware automatically reconfigures based upon the selected function.

The default values for these settings are Inactive. The dedicated 1PPS signal is active by default.

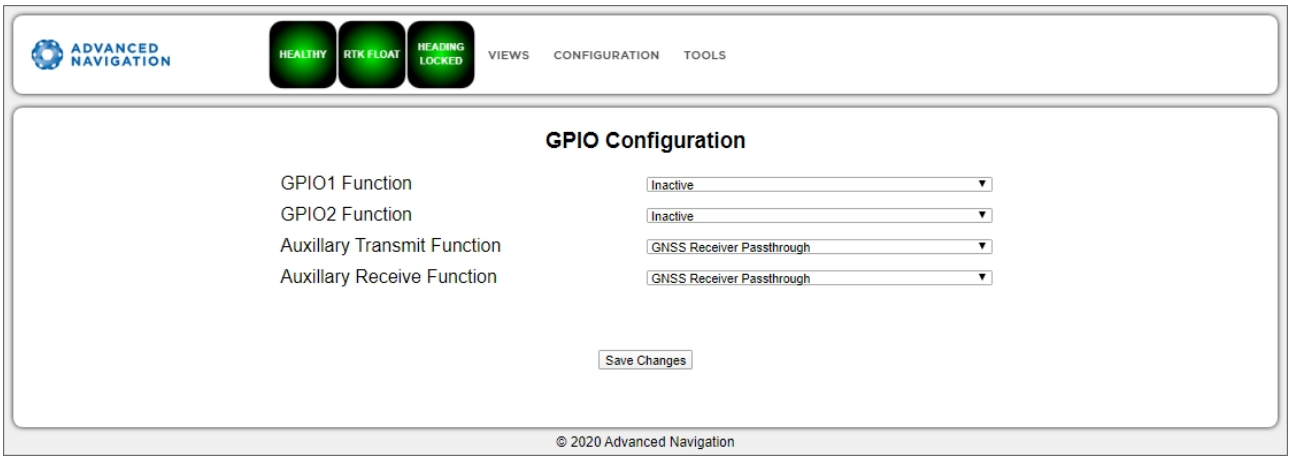


Figure 45: INS Manager Web GPIO Configuration Page

9.6.3.8 Odometer

`Configuration > Odometer` allows the user to configure the odometer pulse length and offset. See **7.4 Odometer** and **8.10 Odometer Pulse Length** for more information on the use of odometers. The odometer offset is also applied when using a DVL input.

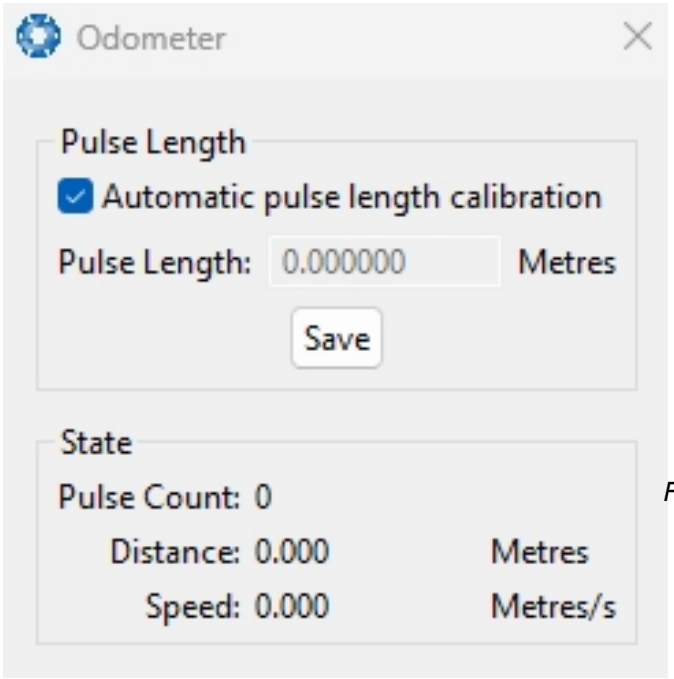


Figure 46: INS Manager Desktop Odometer Dialogue

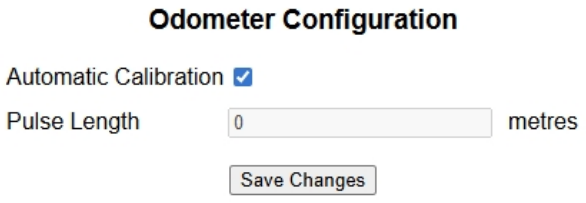


Figure 47: INS Manager Web Odometer Dialogue

9.6.3.9 Reset

`Configuration > Reset`. From this page you can select buttons:

- `Reset` Perform a powers cycle. No configuration settings or state data are lost.
- `Cold Start` Clears all filters, and connections are reset and must be re-established. No configuration settings are lost.
- `Factory Restore` Resets all INS settings back to their factory defaults, including state data and all configuration settings. Re-enable the DHCP Client and loses any static IP address settings. Erases the hot start data so that the system is forced to perform a cold start.

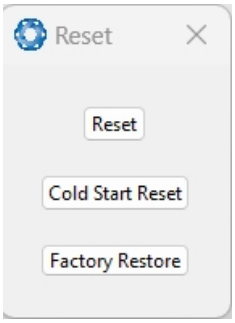


Figure 48: INS Manager Desktop Reset Dialogue

9.6.3.10 Reference Point Offsets

`Configuration > Reference Point Offsets` page allows the user to set measurement points away from its default position at the physical centre of the INS unit. The primary reference point offset applies to data from all ANPP packets as well as all peripheral output such as NMEA messages and Heave Point 1. When the values are zero the measurement point is the centre of the INS unit.

The Heave Points 2 to 4 allow the user to offset reference points for the heave values 2 to 4 in the **13.4.37 Heave Packet**.

The Centre of Gravity (COG) Lever Arm Offset should be used to define where the centre of gravity is for the vehicle. It is useful in applications, such as a surface vessels, to improve the ability of the North Seeking Gyrocompass Heading to complete alignment.

These offsets are applied to the values in the **13.4.37 Heave Packet** and **Vessel Motion Packet**.

Note: These values only apply to the Heave Packet and Vessel Motion Packet. NMEA, TSS and Simrad heave is not affected by the values in this dialogue which are always measured at the centre of the INS unit.

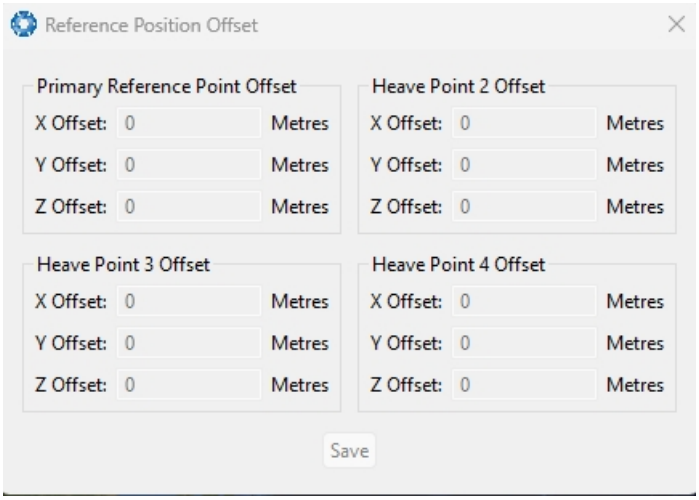


Figure 49: INS Manager Desktop Reference Position Offset Dialogue

9.6.3.11 GPIO Output

[INS Manager Desktop only]

`Configuration > GPIO Output` page allows the user to configure the NMEA0183, TSS1 and SIMRAD output rates for the GPIO and Auxiliary port. These output rates will be observed with respect to the actual output selected in the **9.6.3.7 GPIO** port configuration.

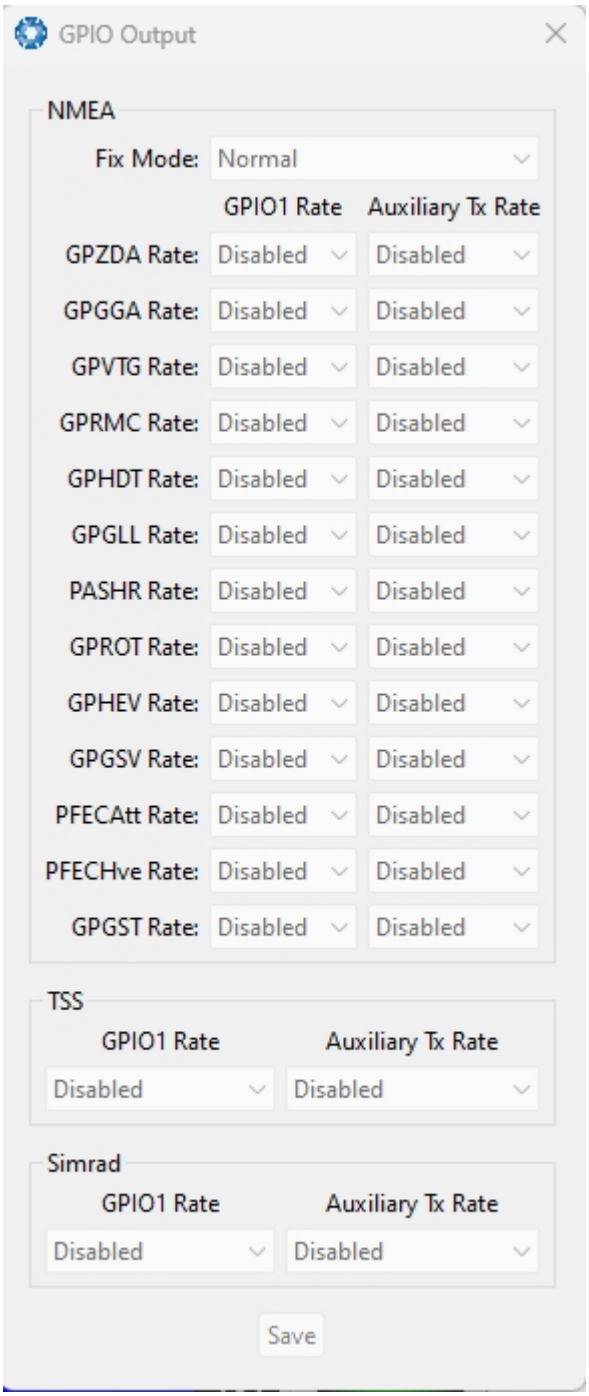


Figure 50: INS Manager Desktop GPIO Output Dialogue

9.6.3.12 Manual Initialisation

`Configuration > Manual Initialisation` is used to manually initialise the INS when a GNSS fix is not available. Setting the position will initialise the navigation filter. Setting the heading will also initialise the heading filter.

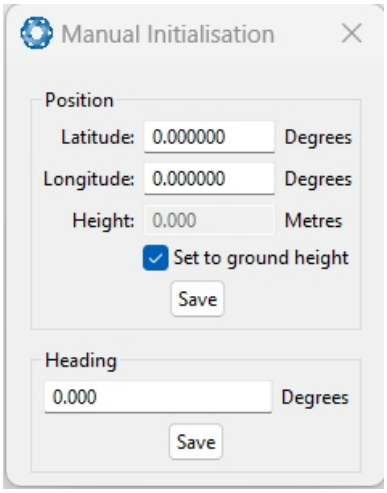


Figure 51: INS Manager Desktop Manual Initialisation Dialogue

9.6.3.13 Gimbal

[INS Manager Desktop only]

`Configuration > Gimbal` is used in gimbal specific applications.

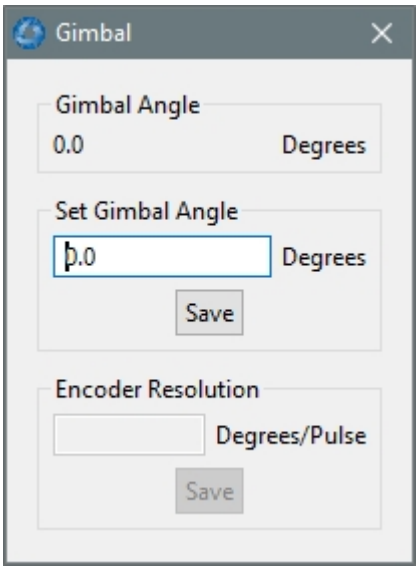


Figure 52: INS Manager Desktop Gimbal Dialogue

9.6.3.14 Ethernet Settings

`Configuration > Ethernet Settings` allows you to switch between DHCP and Static IP selection. When the DHCP Client is disabled you need to enter the TCP/IP settings if you wish to access the INS Manager Web interface.

If you are only connecting to the INS via the primary serial port these settings can be ignored.
The default is for the Ethernet mode to be set to DHCP.

Note: Ethernet settings are only available for INS devices with an ethernet interface.

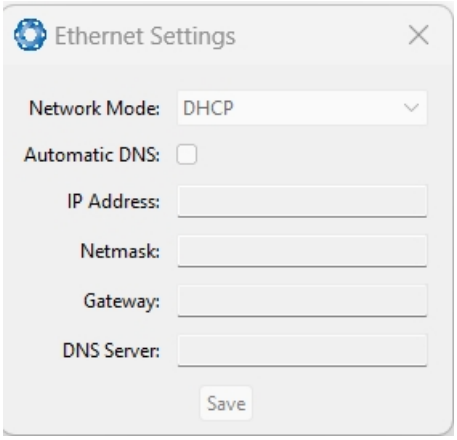


Figure 53: INS Manager Desktop Ethernet Settings Dialogue

9.6.3.15 NTRIP Client

[INS Manager Web only]

`Configuration > NTRIP` is used to connect to a network DGPS (Differential GPS, an older standard for GNSS corrections of lower performance than RTK) or RTK service to stream correction data to the INS for DGPS or RTK. The NTRIP client requires the INS has an available internet connection to the NTRIP Caster to function.

The source of the corrections data can be either your own base station, configured as an NTRIP Caster, or a third-party service. Either way, you will need the login details for connecting to and authenticating with the NTRIP Caster, and identifying the nearest mountpoint. For more details on the connection steps, refer to the Manual Initialisation page.

Troubleshooting NTRIP Caster Connections

Some mount points require the INS send its position. To do this, click the `Report Position` check box, and then click `Save Changes`. The current INS position will then be sent to the NTRIP Caster every 10 seconds.

If you cannot get the list of mount points, ensure the Host Address and Port number are correct for your NTRIP Caster.

If the Connection Status is not green, and says UNAUTHORISED this indicates an incorrect username or password.



Figure 54: INS Manager Web NTRIP Client Page

9.6.3.16 CAN Settings

`Configuration > CAN Settings` allows you to enable or disable the CAN functionality. If you are only connecting to the INS via the primary serial port then these settings can be ignored.

The default for CAN is Enabled at 1,000,000 baud rate. A CANOpen EDS file is bundled with the firmware download or can be downloaded from the Advanced Navigation webpage.

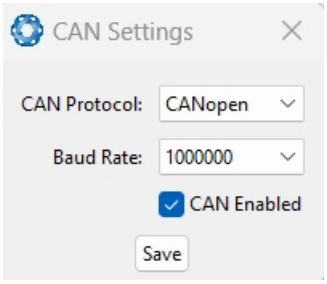


Figure 55: INS Manager Desktop CAN Settings Dialogue

9.6.3.17 Advanced

[INS Manager Web only]

`Configuration > Advanced` page allows you to set the voltage level of the GPIO and 1PPS signals lines along with the packet rates.

The options are 5 V, 3.3 V, or Disabled. The default is 5 V.

Note: The dedicated 1PPS signal is active whenever the voltage is set to 5 V or 3.3 V, and is therefore active by default.

The CANopen output packet rates can be configured as the period between packets in milliseconds within the range of 10 ms to 10,000 ms. The default is 50 ms.

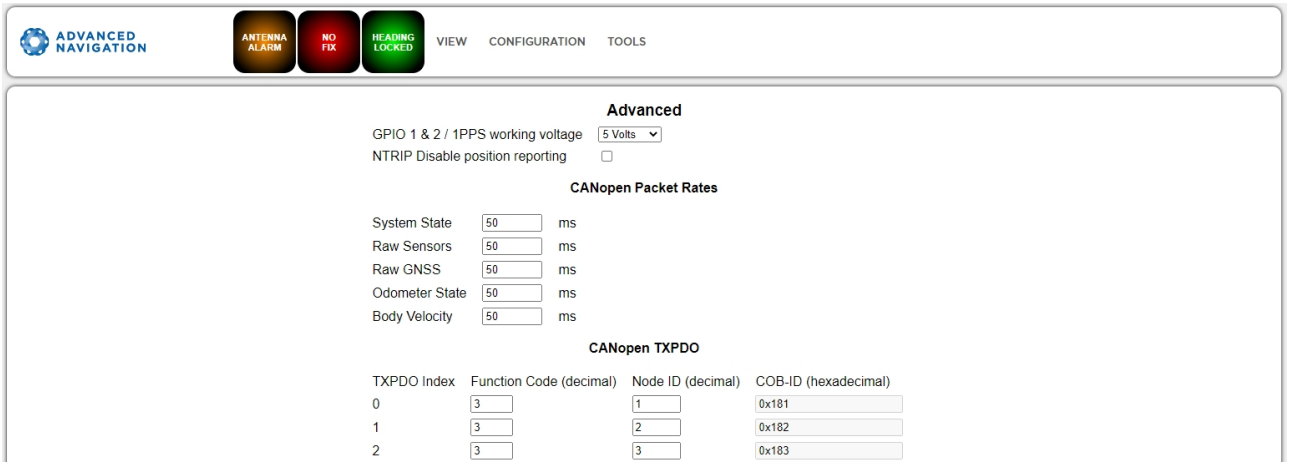


Figure 56: INS Manager Web Advanced Page

9.6.3.18 GNSS Configuration

`Configuration > GNSS Configuration` allows you to select constellation configurations based on the type of GNSS receiver installed.

If an invalid configuration is entered, the system will revert to the previous configuration when you try to save the changes. To confirm that changes have been successfully saved, close and re-open the GNSS Configuration page.

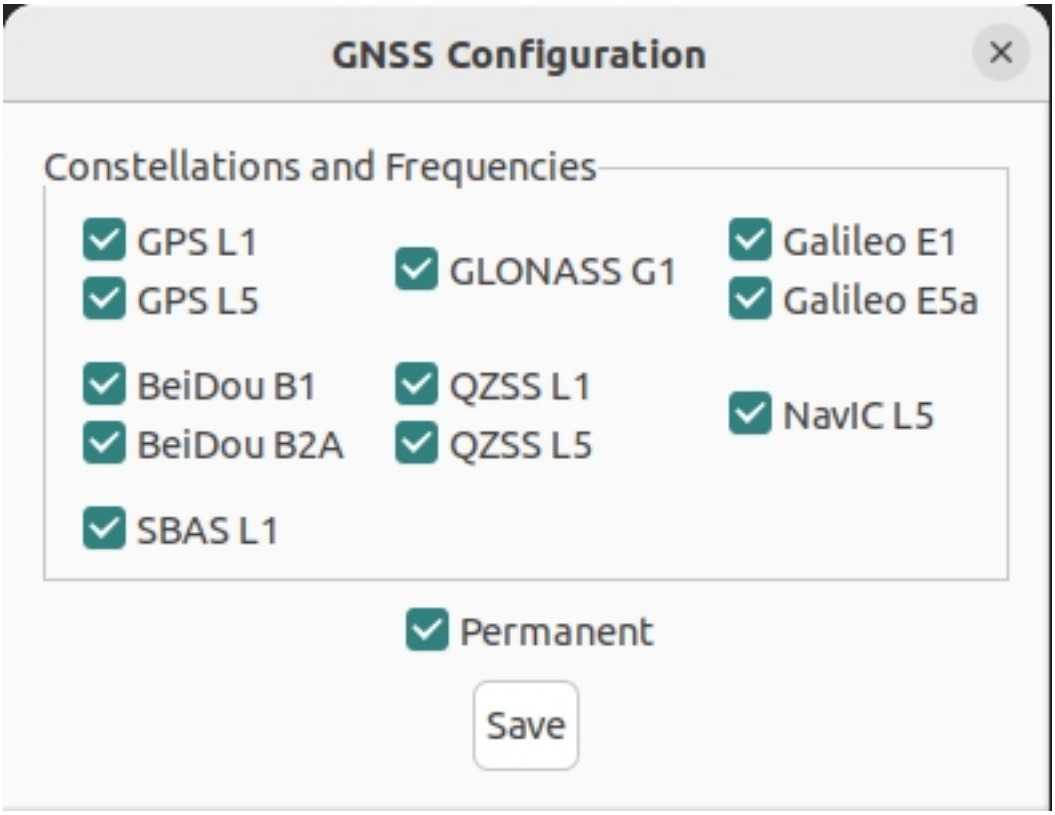


Figure 57: INS Manager Desktop GNSS Configuration Dialogue

U-blox GNSS Receiver Configuration

When using a u-blox GNSS receiver, the following configuration rules are applicable:

- By default, GPS, Galileo and GLONASS are enabled, along with SBAS and QZSS correction services.
- When selecting a GNSS constellation, all frequencies for that constellation must be selected, with the exception of GPS and Beidou.
- GPS and Beidou can be enabled with L1/B1 frequency only, and a second frequency is optional.
- SBAS and QZSS correction services can also be enabled for all constellation configurations, except when using GPS, Galileo and BeiDou together, or when GPS L1 is not enabled.

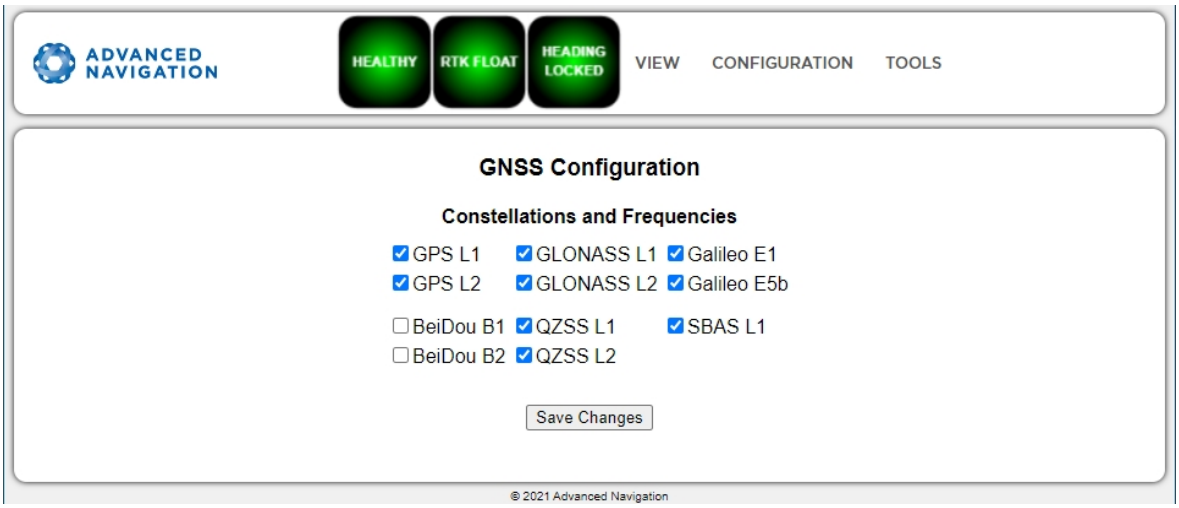


Figure 58: INS Manager Web U-blox GNSS Configuration Page

The table below lists the supported constellations and the corresponding GNSS sample period depending upon the number of constellations selected. When setting the **9.6.3 Configuration Menu**, the packet rate of the **13.4.10 Raw GNSS Packet** and **Raw Satellite Data Packet** are automatically adjusted to the next lower integer multiple of the sample period.

# of Constellations Configured	Sample Period	Example Effective Packet 60 Rate	Supported Constellation Configurations
1	40 ms	1 ms - 39 ms = 40 ms (25 Hz)	GPS
	50 ms	1 ms - 99 ms = 50 ms (20 Hz) 100 ms - 149 ms = 100 ms (10 Hz) 150 ms - 199 ms = 150 ms (6.67 Hz)	Galileo GLONASS BeiDou NVC
2	67 ms	1 ms - 133 ms = 67 ms = 15 Hz	GPS & GLONASS
	72 ms	1 ms - 143 ms = 72 ms = 14 Hz	GPS & Galileo

# of Constellations Configured	Sample Period	Example Effective Packet 60 Rate	Supported Constellation Configurations
	100 ms	1 ms - 199 ms = 100 ms (10 Hz) 200 ms - 299 ms = 200 ms (5 Hz)	GPS & BeiDou All other combinations
3	125 ms	1 ms - 249 ms = 125 ms (8 Hz) 250 ms - 374 ms = 250 ms (4 Hz)	GPS, GLONASS & Galileo GPS, GLONASS & BeiDou (no correction services) All combinations
4	143 ms	1 ms - 285 ms = 143 ms (7 Hz)	GPS, Galileo, GLONASS & BeiDou All combinations
5	200 ms	1 ms - 399 ms = 200 ms (5 Hz)	All combinations

Table 22: U-blox Supported GNSS Constellation Configurations

Trimble BD992 GNSS Receiver Configuration

When using a Trimble BD992 GNSS receiver, the following configuration rules are applicable:

- For any given constellation, the first (1) frequency must be enabled in order to receive additional frequencies
- Galileo and BeiDou are licensed separately.
- L Band corrections are available on this receiver, however require an additional subscription to Omnistar or RTX. Advanced Navigation does not supply these, contact Trimble for further details.

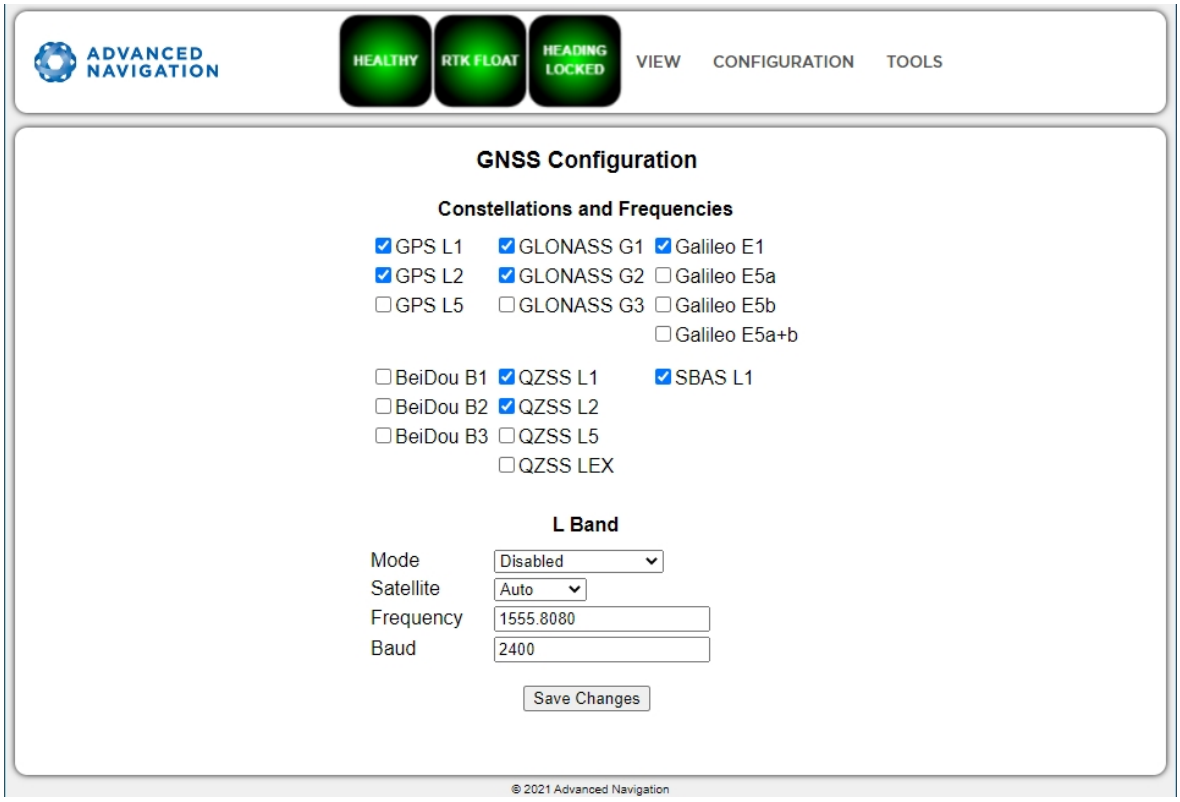


Figure 59: INS Manager Web BD992 GNSS Configuration Page

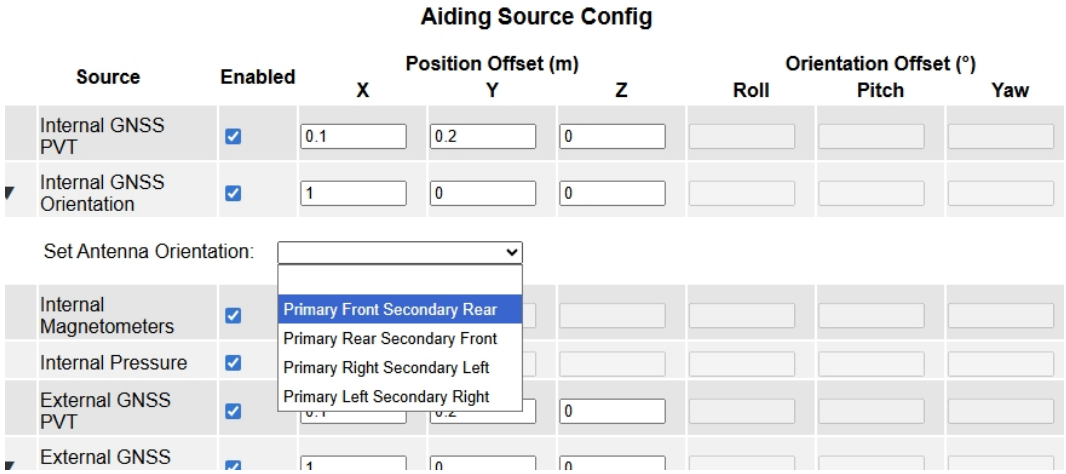
9.6.3.19 Aiding Source Configuration

`Configuration > Aiding Source` allows you to configure aiding sources:

- Enable: Off/On
- Position Offset
- Orientation Offset

Note: The Position Offset refers to the position of the sensor from the *INS defined body point* (typically the center of the INS), in body coordinates.

Note: For dual antenna GNSS, the GNSS antenna orientation values represent the position of the primary antenna from the secondary antenna, in body coordinates.

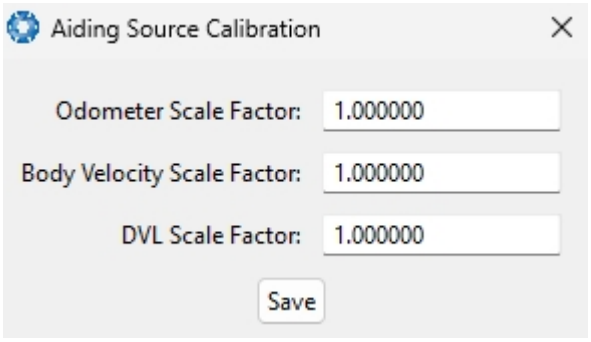


Note: Remember to select *Save Changes* after entering new parameters.

9.6.3.20 Aiding Source Calibration

`Configuration > Aiding Source Calibration` allows you to configure a scale factor correction to a sensor input for:

- Odometer Scale Factor
- Body Velocity Scale Factor
- DVL Scale Factor



9.6.4 Tools Menu

The Tools menu contains tools for performing functions with the INS and INS data.

9.6.4.1 Device Discovery

[INS Manager Desktop only]

`Tools > Device Discovery` allows network discovery and reconfiguration of devices on the local network regardless of network settings.

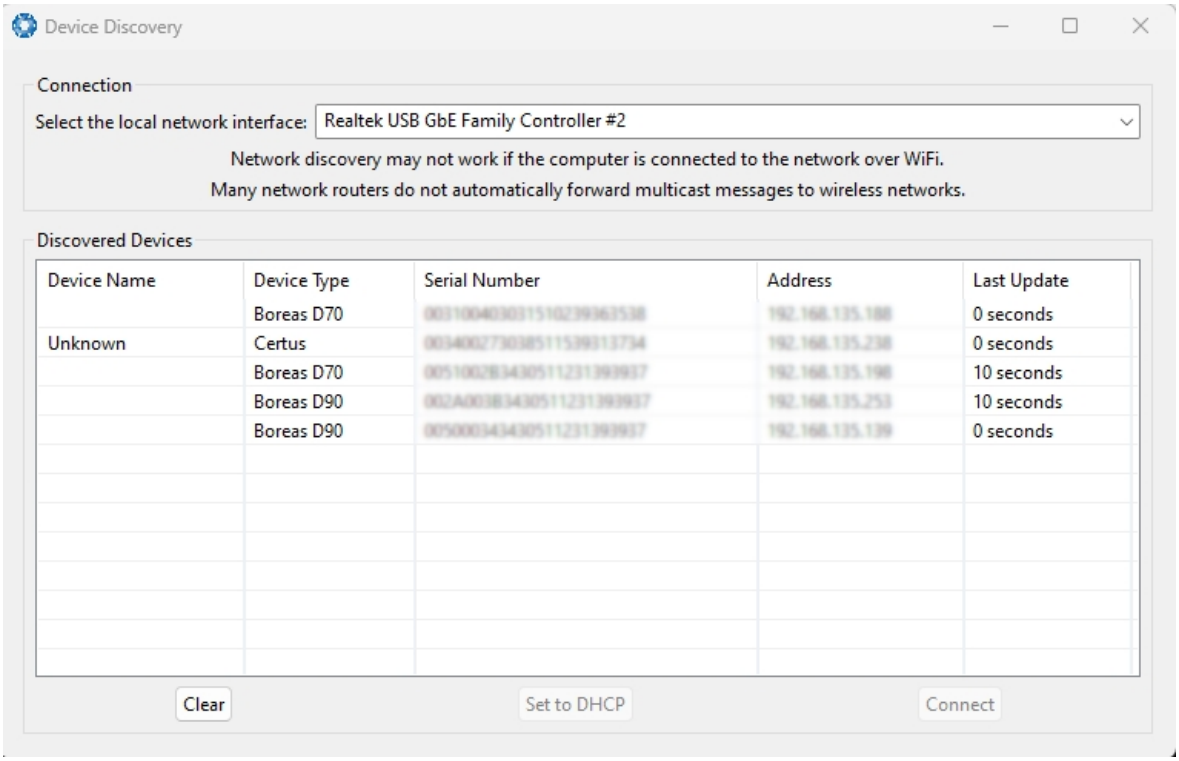


Figure 60: INS Manager Desktop Device Discovery Dialogue

Set to DHCP

Once devices are discovered, any devices on uncontactable network addresses can be reset to DHCP by pressing the “Set to DHCP” button for that device.

9.6.4.2 Firmware Update

`Tools > Firmware Update` is used to update the INS device firmware.

Advanced Navigation regularly provides firmware updates to improve performance and add features. These are described on the INS product page of the Advanced Navigation website. Advanced Navigation recommends that you upgrade to the latest version whenever available. The firmware update function is robust, such that power loss, file corruption or other issues cannot brick the device.

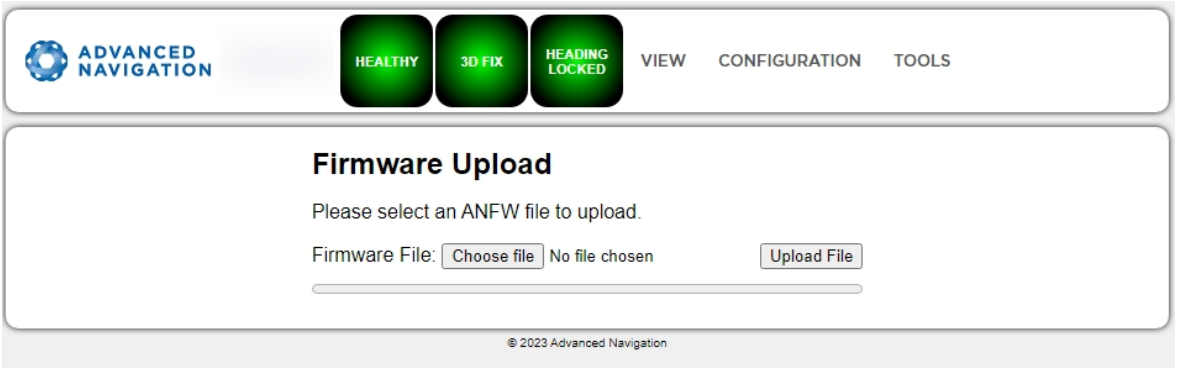


Figure 61: INS Manager Web Firmware Update Page

Click "Choose File" and select the appropriate ANFW file for upload.

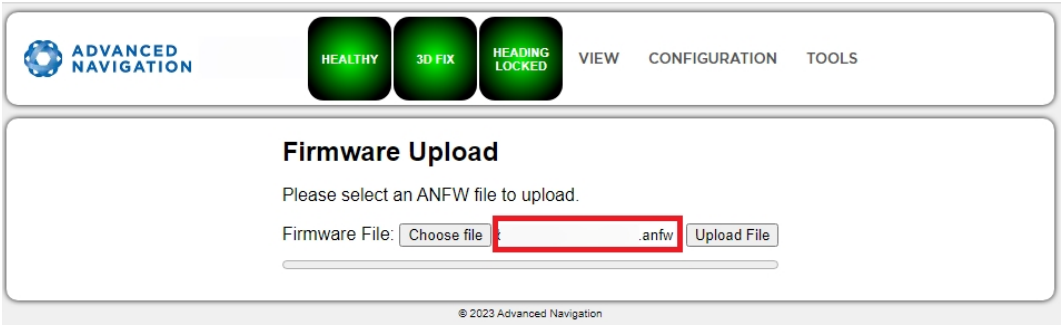


Figure 62: INS Manager Web Firmware Update Choose File

Wait while the system uploads the firmware.

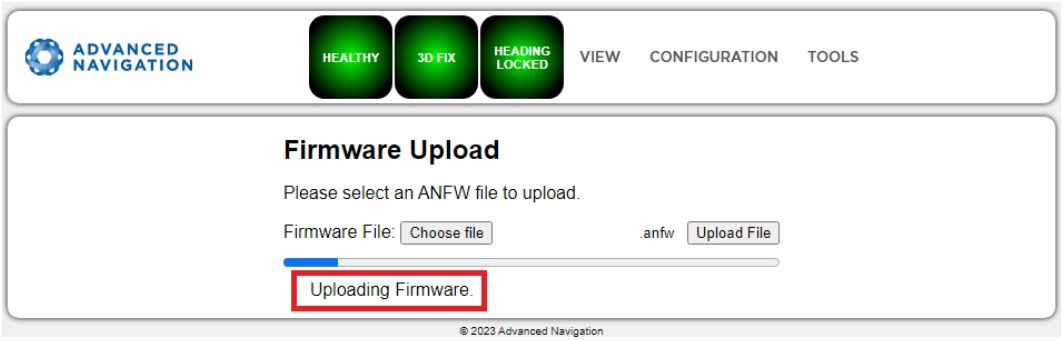


Figure 63: INS Manager Web Firmware Update Uploading Firmware

Once uploaded, the system will automatically verify the firmware prior to reboot.

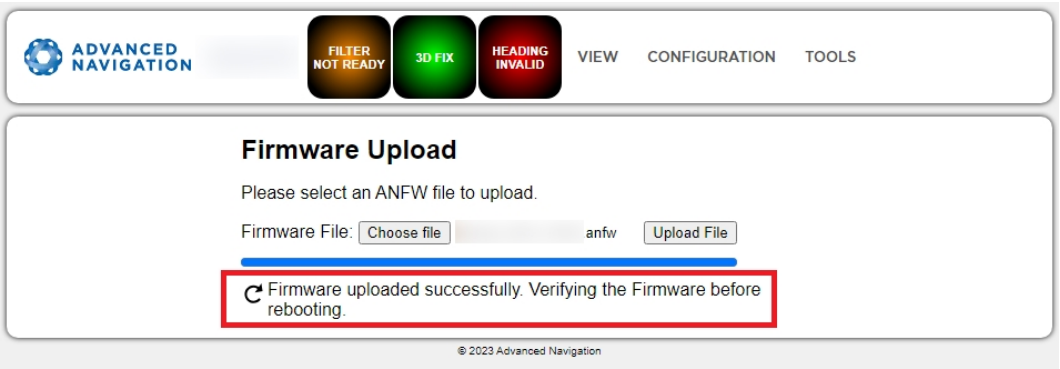


Figure 64: INS Manager Web Firmware Update Verifying

Upon successful verification, a dialog box will appear indicating the device is rebooting to apply the update.



Figure 65: INS Manager Web Firmware Update Complete

9.6.4.3 Magnetic Calibration

`Tools > Magnetic Calibration` allows you to perform magnetic calibration, view and modify the magnetic calibration values. The actual magnetic calibration is performed inside the INS unit.

This page is available for INS units which include a magnetic sensor. This includes the Certus series, and excludes the Boreas FOG series.

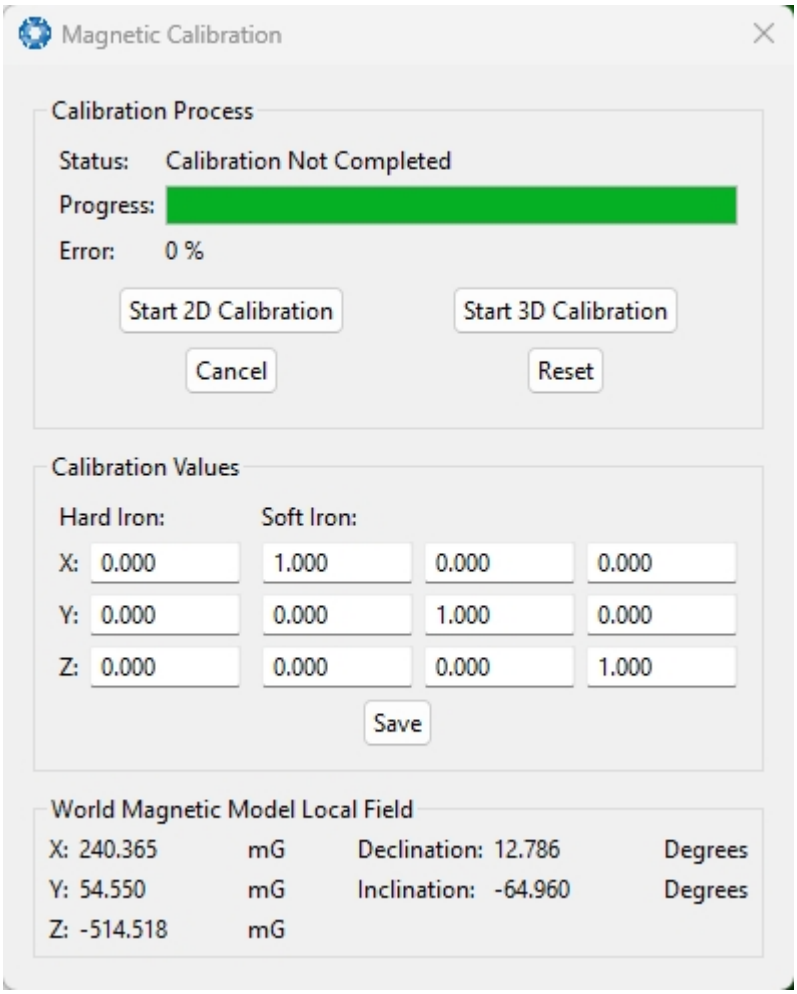


Figure 66: INS Manager Desktop Magnetic Calibration Dialogue

9.6.4.4 Logs

[INS Manager Web only]

Applicable to: Certus and Boreas products. Doesn't include Certus Mini.

`Tools > Logs`

Our higher-end INS products include an on-board high capacity data logging system. Logging is enabled by default, but by default no specific data is configured to be logged. To enable logging of specific data packets or messages, see the page section: `Configuration > Data Ports > Logging`.

The log file name includes the date and time the log file recording commenced, using UTC time, in the format `[Product Name]Log_yy-mm-dd_hh-mm-ss.xxx`. The log file extension identifies the file type.

If the time and date in the log file name is incorrect, this usually indicates the does not have a GNSS fix and has therefore not yet determined the correct time.

The logging configuration and recorded log files can only be accessed from INS Manager Web (via the ethernet interface). You cannot use the serial port interfaces.

Note: If there is a requirement not to store potentially sensitive information, it is possible to order the INS without internal storage. If this is the case, no internal logging will be possible.

Enabling and Disabling Logging

To enable or disable the logging functionality, select or deselect the `Enable Logging` check box and click `Save Changes`. The log file currently being recorded is highlighted in green. If recording an ANPP-format log file, the log file will include the current INS device configuration.

Limiting the Maximum Log File Size

If you wish to limit the maximum file size for individual log files, select the `Enable Splitting` check box, enter a `Max Log Size` (in MiB), and then click the `Save Changes` button. This will not affect pre-existing log files but will affect the currently recording log file. Note this will round down to an integer if you enter a number with decimal places.

Downloading Log Files

You can download individual log files by simply clicking on them. Note, only one log file can be downloaded at a time using this method. To download multiple log files, first select the files required, and then click the `Download Selected` button. This will download all the files to your computer and compress them into one ZIP file. Note, it may appear that nothing is happening during this process, and it can take some time to complete. You can select all log files using the `Select All` button, or individually select log files using the check box on the left of the log file name.

Deleting Log Files

You can delete log files by selecting the specific log file(s) using the check box to the left of the log file name, or using the `Select All` button, and then click the `Delete Selected` button. For bulk erasing ALL log files you can use the `Format` button. This will erase all log files irrecoverably. Note, it can take some time.

Starting New Log Files

The `Start New Log` button will stop writing the current log file and commence writing a new file. This new file will also include the current INS device configuration. If logging is currently not enabled, this button will not do anything.

The ANPP log files are binary data and are therefore not in a human-readable format. To convert the ANPP log files you can use the Log Converter function of `INS Manager Desktop > Tools > Log Converter`. This will create CSV and KML files for your analysis. All the ANPP packets are described in detail in **13 Advanced Navigation Packet Protocol**. NMEA0183 log files are ASCII text and can be opened in any text editor.

9.6.4.5 Log Converter

[INS Manager Desktop only]

`Tools > Log Converter` allows the user to convert INS log files into various standard formats, including:

- CSV: Comma Separated Value
- GPX: GPS Exchange Format
- KML: Keyhole Markup Language
- IMR: Inertial Measurement and Reference
- RINEX files

The Position Offset settings are used to project the exported position of origin to a point other than the actual centre of the INS unit. For most users, these values should be left at zero.

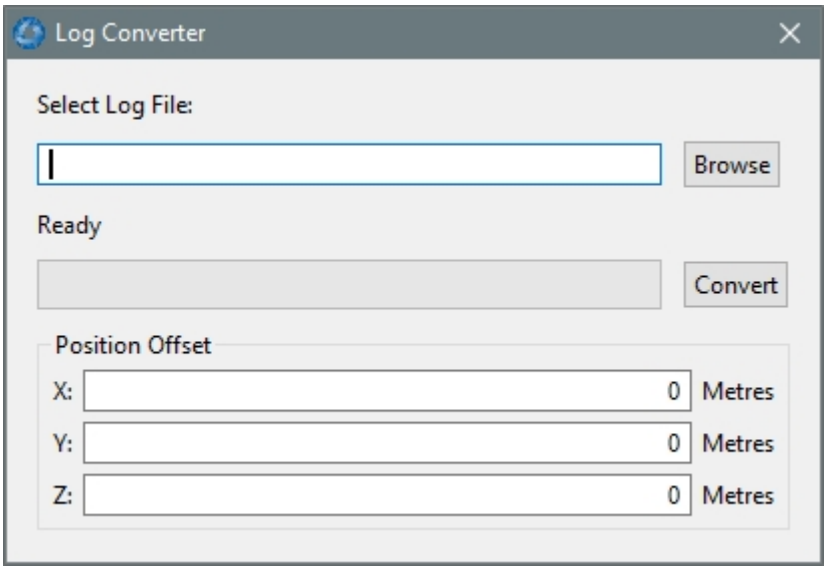


Figure 67: INS Manager Desktop Log Converter Dialogue

9.6.4.6 NTRIP Client

[INS Manager Desktop only]

`Tools > NTRIP Client`

This page can be used to connect to a network DGPS or RTK service to stream correction data to the INS for DGPS or RTK. The NTRIP client requires an internet connection to function.

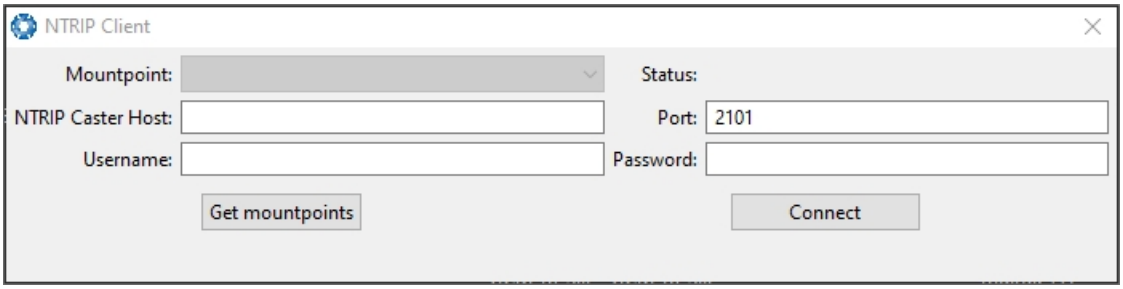


Figure 68: INS Manager Desktop NTRIP Client Dialogue

9.6.4.7 Network Connect

[INS Manager Desktop only]

`Tools > Network Connect` allows INS Manager to make a connection to the INS over a TCP/IP network rather than the primary serial port connection.

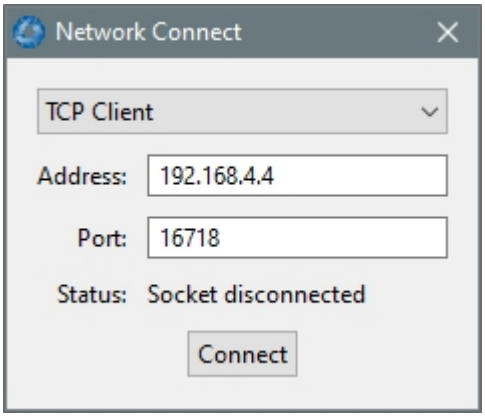


Figure 69: INS Manager Desktop Network Connect Dialogue

9.6.4.8 Serial Passthrough

[INS Manager Desktop only]

`Tools > Serial Passthrough` allows pass-through communication through either GPIO 1 and 2 or the Auxiliary port.

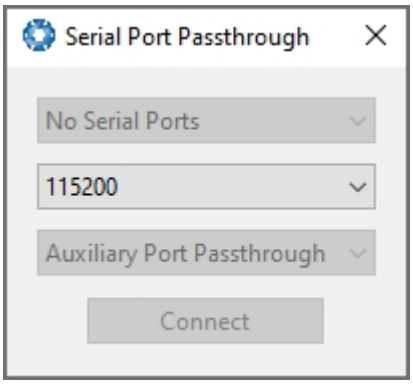


Figure 70: INS Manager Desktop Serial Port Passthrough Dialogue

9.7 Using ANPP

Advanced Navigation units can be configured and monitored using the Advanced Navigation Packet Protocol (ANPP), which is detailed fully in **13 Advanced Navigation Packet Protocol**.

A software development kit (SDK) is available to assist you when developing your own applications. It contains example code in a number of programming languages, ready to compile and test. It can be downloaded from the Advanced Navigation website.

10 OEM Integration

10.1 OEM PCB Mechanical Mounting

Motus OEM mounts to its target PCB through the use of four mounting holes on the aluminium frame. The aluminium frame is designed to achieve effective mating height for the board to board connector without any additional components. Once the Motus OEM is connected to the target PCB, four M1.6 screws and washers and nuts on the back side of the target PCB are used to hold the Motus OEM in place, see **Figure 71: Motus OEM exploded mounting view**. The length of the M1.6 screws will depend on the thickness of the target PCB. M1.6×6 mm screws are recommended for 1.6 mm thickness target PCBs.

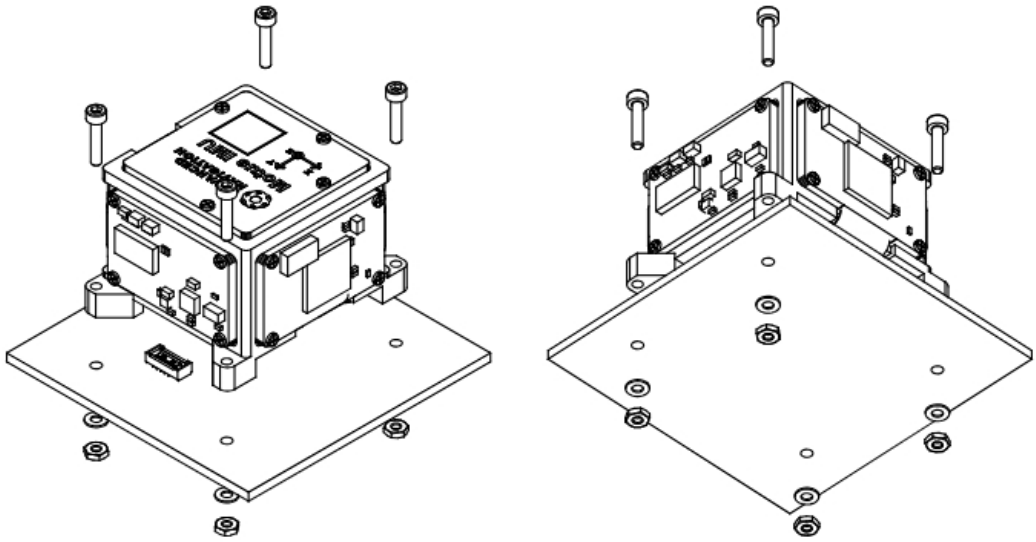


Figure 71: Motus OEM exploded mounting view

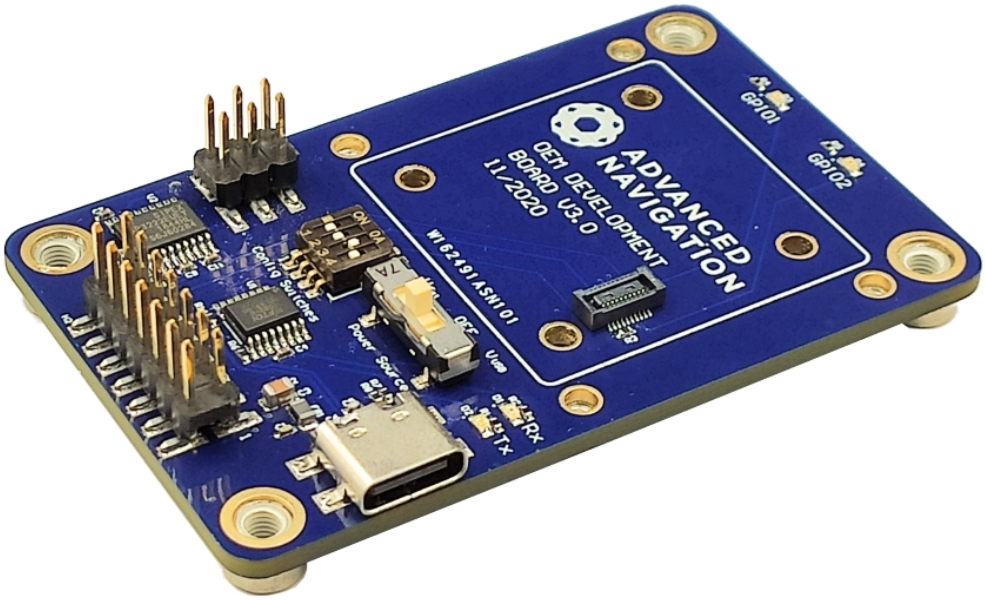


Figure 72: Motus OEM development board

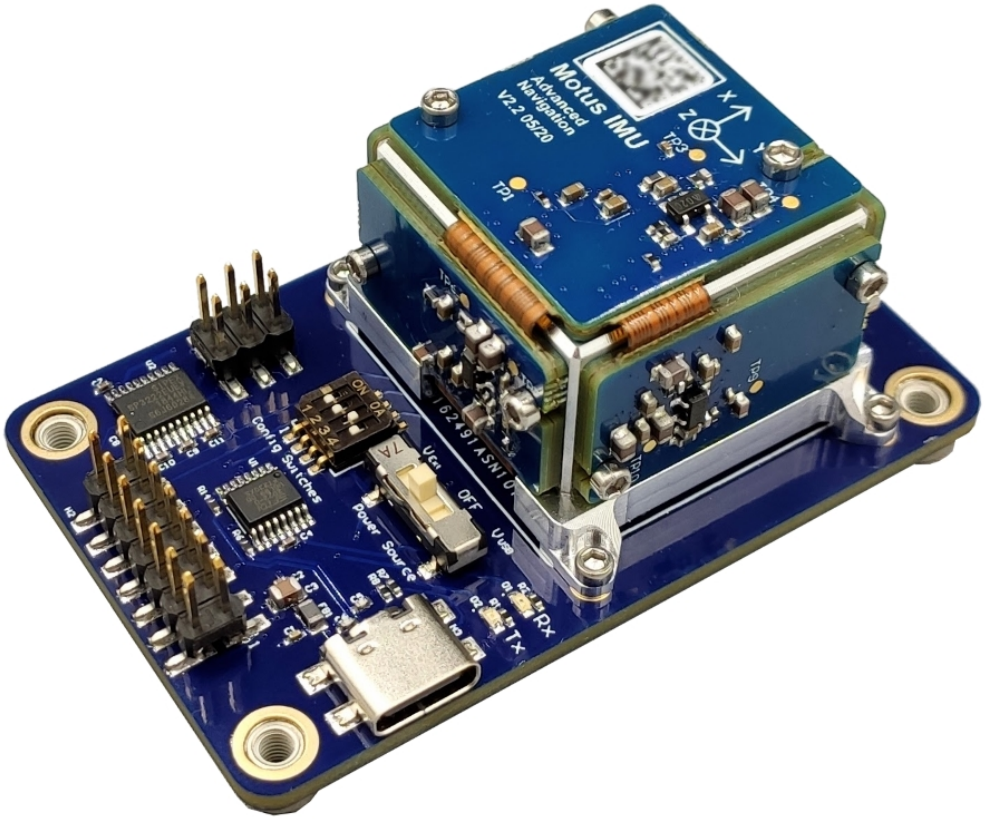


Figure 73: Motus OEM mounted on development board

10.2 OEM Electrical Connector

The electrical connection to the Motus OEM PCB is through a Hirose DF40 series 20 pin micro pitch board to board connector. These connectors have shock absorbing features, a large self-alignment distance and high contact reliability. A positive click should be felt when successful mating occurs between Motus OEM and the target board. It should be noted that this connector is not designed for the Motus OEM to be plugged in whilst power is applied to the user's PCB and connector mating cycles should not exceed 50 cycles to ensure connector reliability.

The manufacturer's part number for the required PCB connector is:

- Hirose Part #: DF40C(2.0)-20DS-0.4V(51)

The connector can be ordered from the following distributors using the following numbers:

- Digikey Part #: H11905CT-ND
- Mouser Part #: 798-DF40C20DS0.4V51

See **Figure 74: Motus OEM connector pin numbering** for the connector pin numbering.

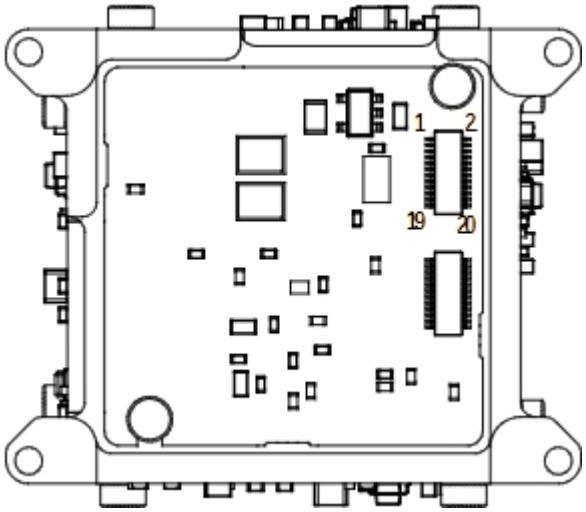


Figure 74: Motus OEM connector pin numbering

Pin	Function
1	Supply Voltage
2	Ground
3	Supply Voltage
4	Ground
5	Supply Voltage
6	Ground
7	3.3 V translator supply, see **10.2.1 Interoperability with Different Voltage Systems**
8	Ground
9	Primary UART receive
10	Reset line, see **10.2.2 Reset Line**
11	Primary UART transmit
12	GPIO1
13	No connection
14	GPIO2
15	No connection
16	Auxiliary UART transmit
17	No connection
18	Auxiliary UART receive
19	No connection
20	No connection

Table 23: Pin allocation table

10.2.1 Interoperability with Different Voltage Systems

All signals are 3.3 V level, however inputs are tolerant to 5 V signals from the target interface. If you require a different voltage level to be compatible with your target system it is recommended that you install a voltage level translator between the signals of each device. To power the translator 3.3 V is provided on pin 7 of the Motus OEM connector. This supply is only capable of driving up to 25 mA and should be appropriately decoupled when powering a circuit.

10.2.2 Reset Line

The reset line is internally pulled high on Motus OEM. This pin must be left floating for normal operation. If a user wishes to reset Motus OEM the pin can be pulled low externally to force a reset. It is important that this signal is not driven high under any circumstances as it will prevent the Motus OEM from functioning correctly. Please keep the reset line track length to a minimum to avoid noise.

10.2.3 Basic Connection

See **Figure 75: Motus OEM basic electrical connections** for the basic electrical connections required by Motus OEM.

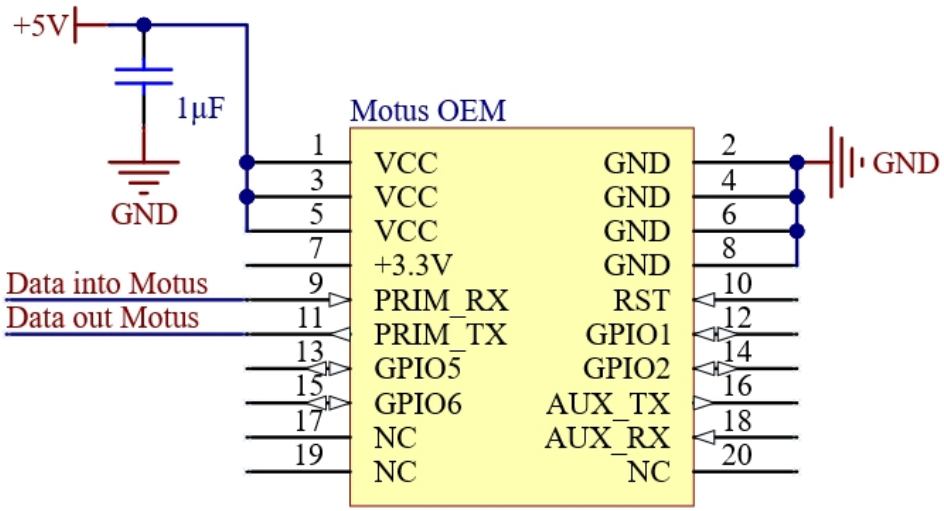


Figure 75: Motus OEM basic electrical connections

10.3 Recommended Footprint

The footprint for Motus OEM incorporates four through-holes for M1.6 screws to pass through and 20 SMD pads for the board to board connector. The board to board connector should be the only component within the keep-out area as shown in **Figure 76: Photo of Motus OEM mounting PCB footprint** and **Figure 77: Recommended footprint for Motus OEM**.

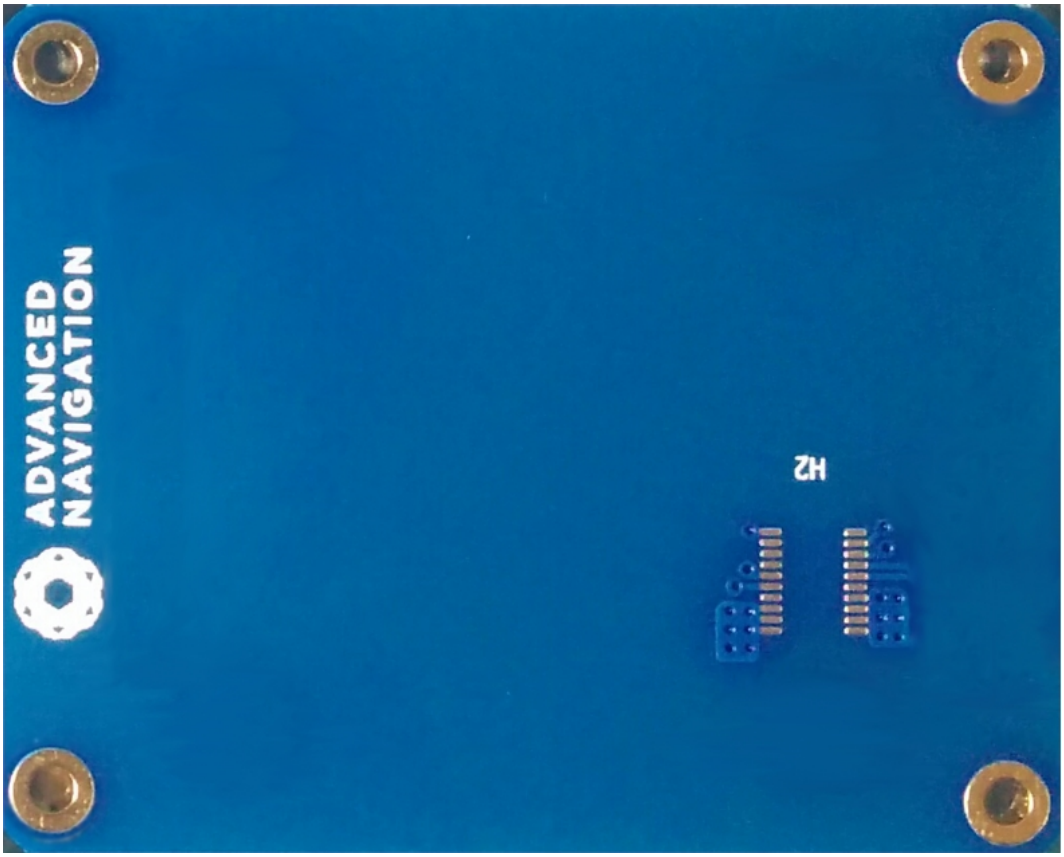


Figure 76: Photo of Motus OEM mounting PCB footprint

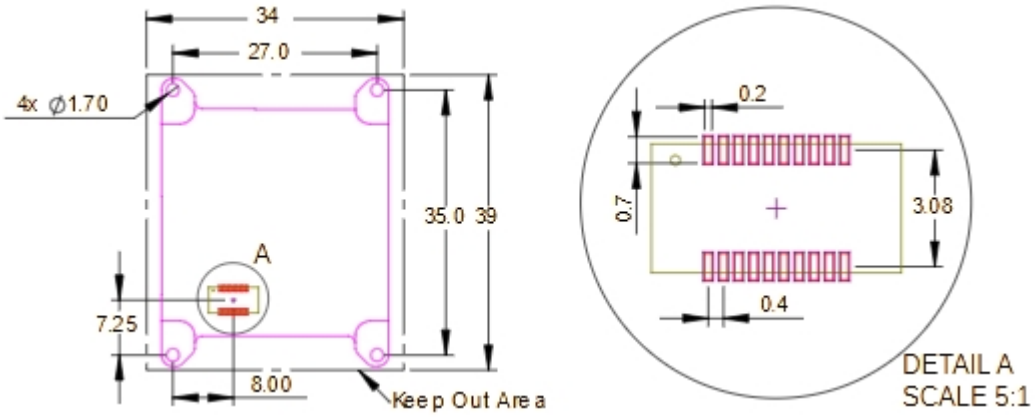


Figure 77: Recommended footprint for Motus OEM

See **6.3.2 OEM Mechanical Drawings** for dimensional drawings. PCB footprint CAD files may be available from the **Motus product page** on the Advanced Navigation website. Contact Advanced Navigation support if you need footprint files or to verify your footprint is correct.

10.4 OEM Power Supply

The power supply is a nominal 5 V DC. A high level of power supply filtering has been built into Motus OEM, however it is still recommended that the power supply be low noise (< 20 mV peak-to-peak). The unit is only rated for inputs from 4.5 V to 5.5 V; exceeding 6 V can cause permanent damage to the unit.

On start-up of the Motus OEM the device will initially draw an increased impulse current (~400 mA for 2 microseconds) whilst it charges on-board capacitors. It is important that the user's circuit is tolerant to these small impulses on start-up to avoid unsuccessful power up. It is recommended that a supply is selected that can provide a maximum current of 0.5 A. Switch-mode supplies are suitable as Motus OEM has on-board filters to remove any supply noise and ripple.

10.5 OEM Pin Protection

Users should ensure that any signals that are connected externally from the PCB, including GPIO and UART pins, are adequately protected. The use of a 5 V zener diode and 1 kΩ resistor should provide enough protection for the majority of use cases, see **Figure 78: Motus OEM external signal protection example**.

External interface pins on Motus have on-board pull-up/pull-down resistors enabled to avoid signals oscillating when not connected. The user shouldn't need to place any pull-up or pull-down resistors on their PCB.

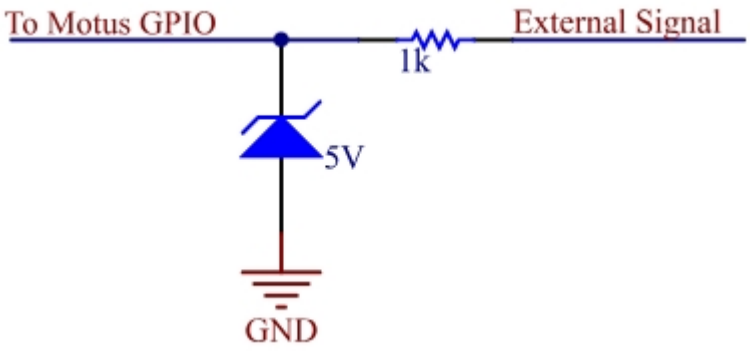


Figure 78: Motus OEM external signal protection example

11 OEM Development Board

The OEM development board is a configurable PCB that provides developers with easy access to all of the features of the Motus OEM unit through a number of different interfaces. See **Figure 79: Motus OEM development board** for an image of the OEM development board.



Figure 79: Motus OEM development board

11.1 Serial UART to USB Converter

An FTDI FT232R USB to serial UART converter chip is included on board to allow the Motus OEM to interface directly to a computer through USB. Two status LEDs indicate the flow of data to and from the USB port.

USB drivers for all operating systems can be downloaded directly from FTDI's website at **http://www.ftdichip.com/Products/ICs/FT232R.htm**.

All development boards are shipped with this interface enabled as standard.

11.2 Dual RS232 Transceiver

An Exar Corporation SP3222EU dual channel serial UART to RS232 transceiver is included on the development board to allow developers to communicate with the Motus OEM unit via RS232. The SP3222EU allows baud rates of up to 1000000 baud.

Channel 1 of the SP3222EU is for use with the primary serial port on the Motus OEM unit and Channel 2 of the SP3222EU is for use with GPIO 3 (RS232 TX 2) and GPIO 4 (RS232 RX 2).

Connection to the RS232 level signals is through a 3×2 2.54 mm header, see **Figure 80: RS232 header with functions labelled**. This header is on the left of the board and allows developers to attach wires or headers as needed.

Please note that by default the SP3222EU is disabled. Switch 1 turns it on, please see **11.4 Configuration Switches**.

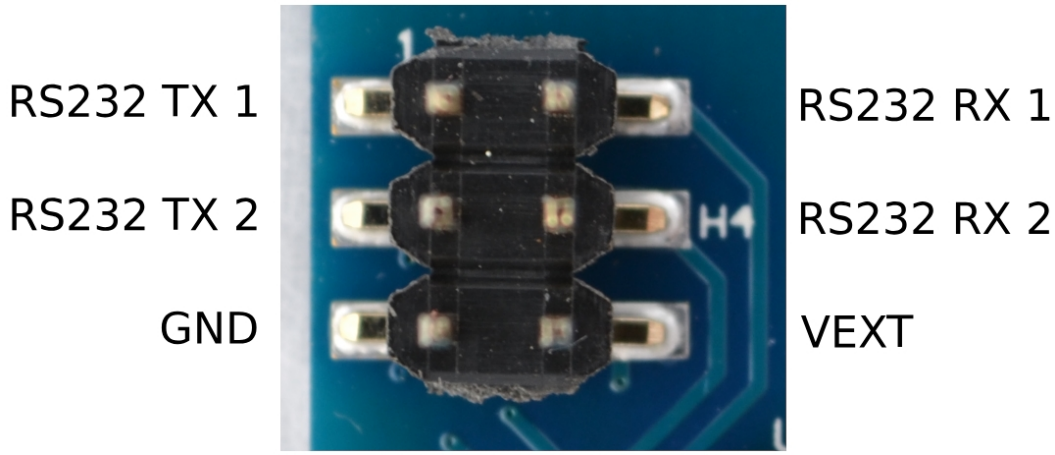


Figure 80: RS232 header with functions labelled

11.3 GPIO Header and Status LEDs

Users have access to 4 multi-purpose GPIO pins from Motus OEM. These are available on the 7×2 2.54 mm header, see **Figure 81: GPIO header with functions labelled**. Two of these pins (GPIO 1 and 2) are also connected to LEDs on the development board. When using a GPIO in certain input functions, the LED can cause interference with the signal. In this situation, the LED's resistor should be removed. When using the GPIO lines for off-board functions it is important to note that these signals are not protected against static charges, over-voltage or over-current events. Failure to protect the signals may result in damage to Motus OEM. Other pins on the GPIO header include a regulated 3.3 V supply from Motus that can supply up to 50 mA, Motus's reset line and the external voltage input (VEXT).

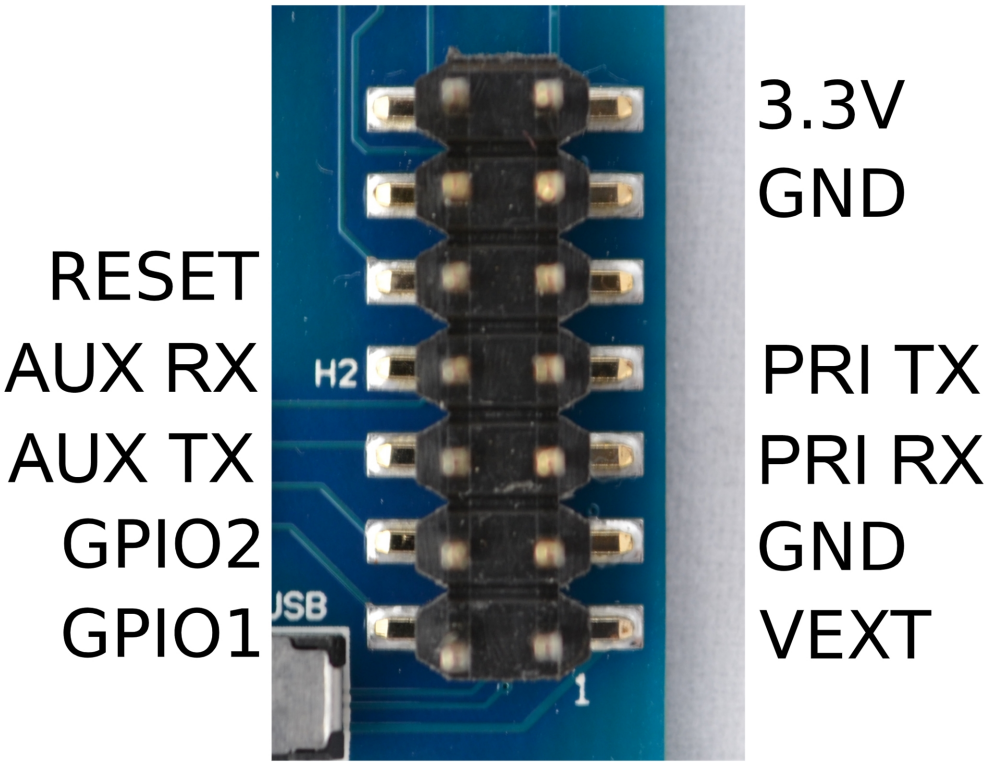


Figure 81: GPIO header with functions labelled

11.4 Configuration Switches

The OEM development board contains a DIP switch array that allows the user to route signals between the USB to serial converter, RS232 converter and GPIO header. The configuration switches are shown below in **Figure 82: Configuration switches in their default position** in their default positions. The default position routes the Motus primary serial port to the USB to serial converter and has GPIO 3 and GPIO 4 connected only to the GPIO header. See **Table 24: Configuration switch functions** for the configuration switch functions.

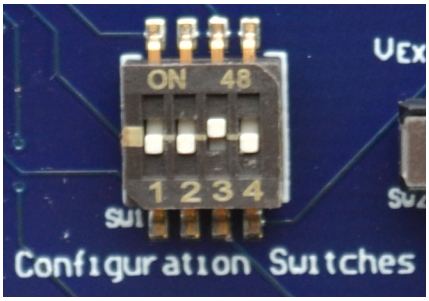


Figure 82: Configuration switches in their default position

Switch	Default	Function
1	Off	Power on RS232 transceiver. When the RS232 transceiver is powered on GPIO 3 and 4 will be connected to RS232 TX 2 and RS232 RX 2 respectively. It will no longer be possible to use GPIO 4 in a TTL level mode. The primary serial port TX will also be connected to RS232 TX 1 but the primary serial port RX will only be connected to RS232 RX 1 if switch 2 is turned on.

Switch	Default	Function
2	Off	Connect primary serial port RX to RS232 transceiver (RS232 RX 1). Please note that only one of the switches 2, 3 or 4 should be turned on at once to prevent short circuit.
3	On	Connect primary serial port RX to USB serial converter. Please note that only one of the switches 2, 3 or 4 should be turned on at once to prevent short circuit.
4	Off	Connect primary serial port RX to GPIO header (UART RX). Please note that only one of the switches 2, 3 or 4 should be turned on at once to prevent short circuit.

Table 24: Configuration switch functions

11.4.1 Primary Serial Port to USB

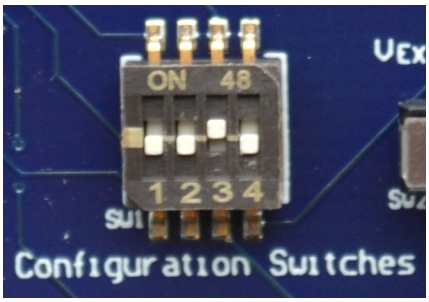


Figure 83: Switch positions for primary serial port to USB (default)

11.4.2 Primary Serial Port to RS232 1 and GPIO 3 & 4 to RS232 2

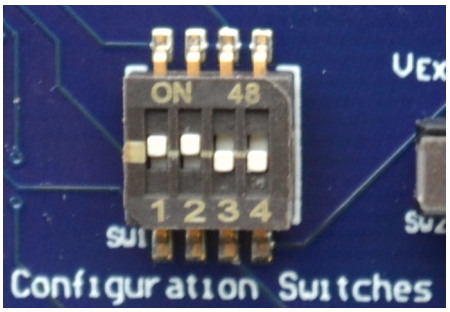


Figure 84: Switch positions for primary serial port to RS232 1 and GPIO 3 & 4 to RS232 2

11.4.3 Primary Serial Port and GPIO 3 & 4 to GPIO Header at TTL levels

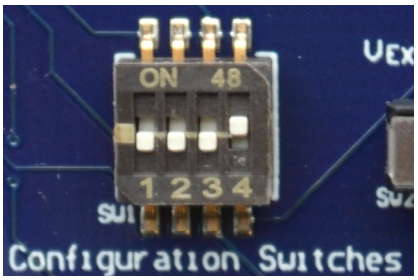


Figure 85: Switch positions for primary serial port and GPIO 3 & 4 to GPIO header at TTL levels

11.5 Power Source

The development board has two power source options: USB power (5 V nominal) or an external power source. By default, the development board is configured to use USB power.

The external power source can be fed into the board through VExt on either the GPIO header or RS232 header. The external power is a 5 V input. See **OEM Power Supply** for further details.

The power source is selected by a three way switch, see **Figure 86: Power switch position for USB power (default)** and **Figure 87: Power switch position for external power**.

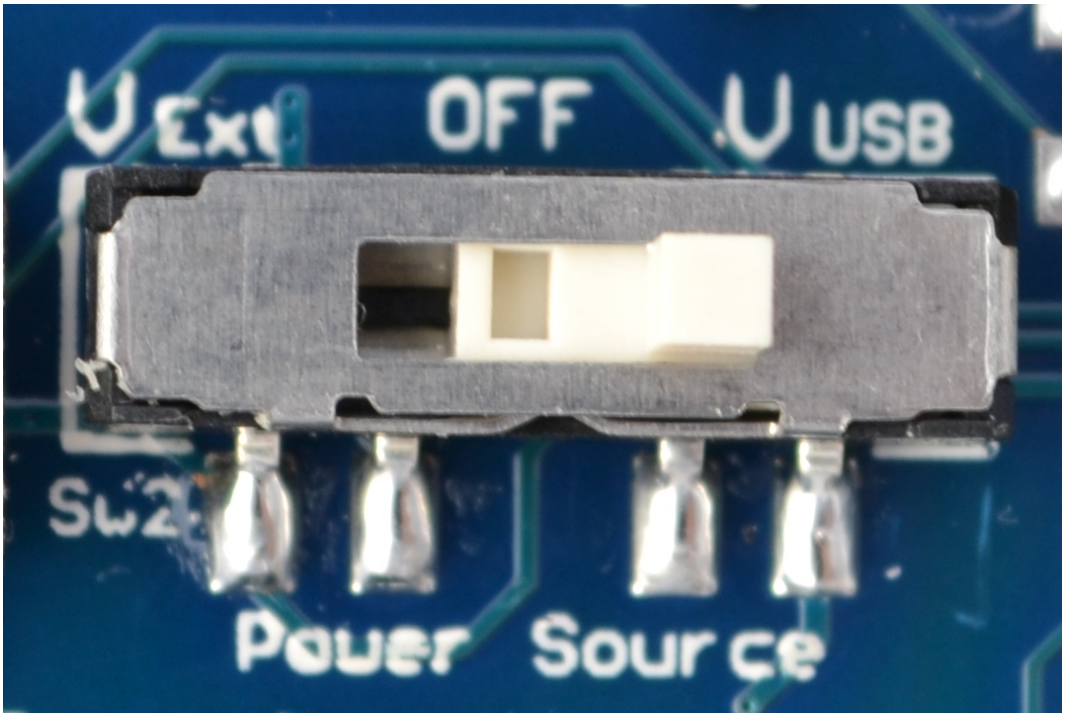


Figure 86: Power switch position for USB power (default)

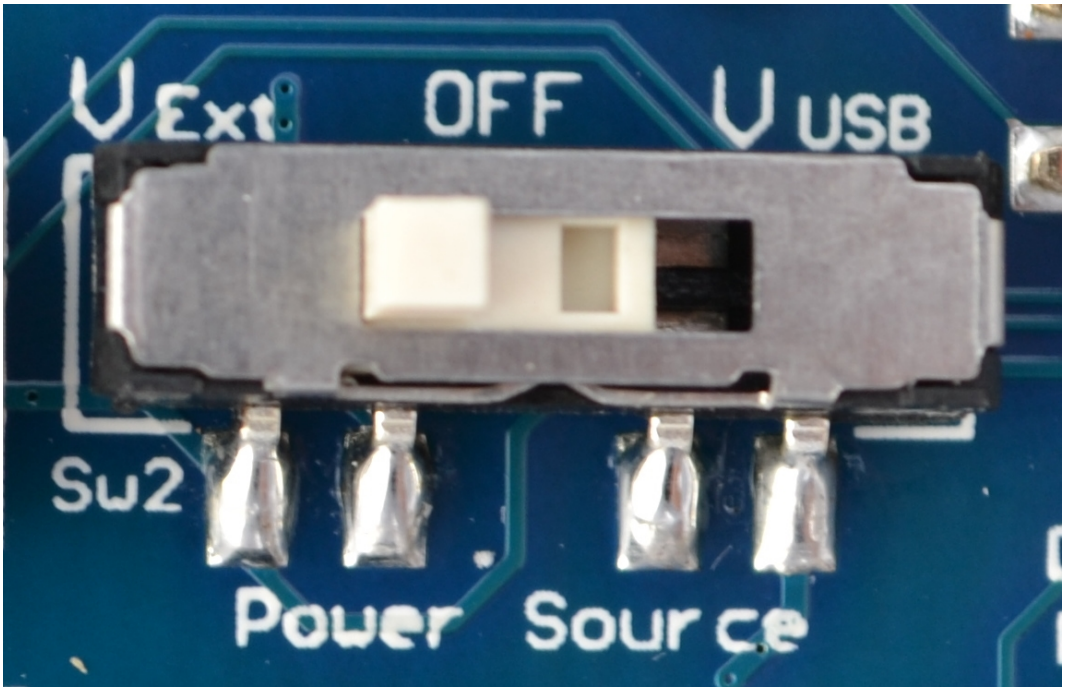
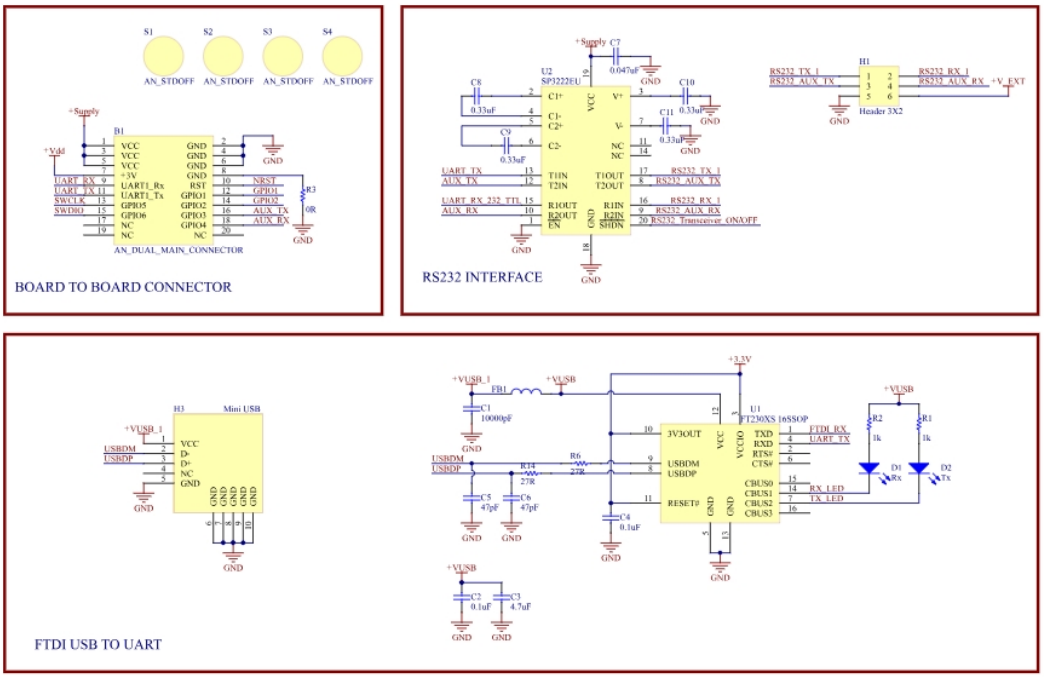
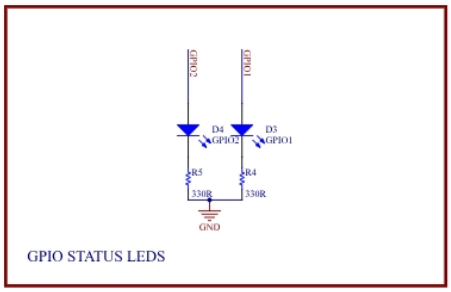


Figure 87: Power switch position for external power

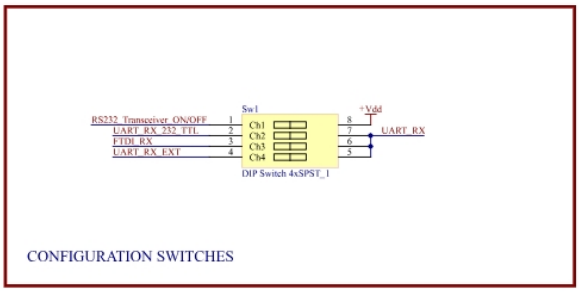
11.6 Schematics

See the following illustrations for the development board schematics.

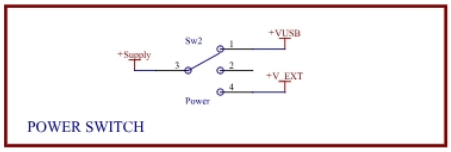




GPIO STATUS LEDS



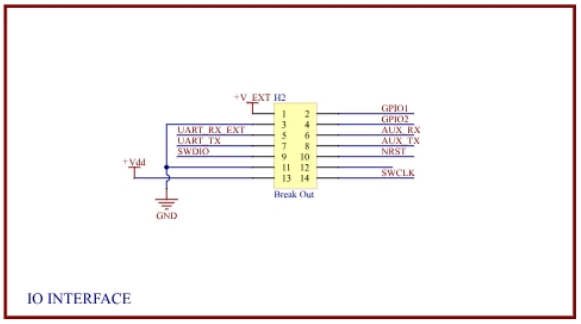
CONFIGURATION SWITCHES



POWER SWITCH



M3 STANDOFFS



IO INTERFACE

12 Interfacing

Communication with Motus is possible on the following ports:

- 12.1 Serial Interface
- 12.3 GPIO Pins and the Auxiliary Port

Each port has different input and output capabilities.

12.1 Serial Interface

There are two serial ports available for use. The GPIO port can also be used as a serial port, with less functionality.

Model	Primary Port	Auxiliary Port	GPIO Port
Motus (Rugged)	ANPP only over RS232 or RS422. **No** dynamic pin functions	ANPP over RS232, 12.3.1 Dynamic Pin Functions	ANPP over RS232, 12.3.1 Dynamic Pin Functions

Table 25: I/O Ports Available

12.1.1 Baud Rate

The primary and auxiliary port baud rate can be set in the INS Manager. It is important to select a baud rate that is capable of carrying the amount of data that Motus has been configured to output. See 13.1.4 Packet Rates for more details on the data output calculation. The data rate in bytes per second can be calculated by dividing the baud rate by 10. For example if the baud rate is 115,200, the data rate is 11,520 bytes per second. If the selected data rate is insufficient, data may be lost without warning.

12.1.2 Format

The format is fixed at 1 start bit, 8 data bits, 1 stop bit and no parity. See 13 Advanced Navigation Packet Protocol for details on the ANPP protocol.

12.2 External Data

External sources of position, velocity and/or heading can be integrated into the Motus filter solution. The data can be sent to Motus in the ANPP format over the primary (RS232 or RS422), auxiliary (RS232 or RS422) or GPIO ports (RS232 only) or in a number of different formats. If using the ANPP format, use the table below to find the relevant section.

Packet
13.4.25 External Position and Velocity Packet
13.4.26 External Position Packet
13.4.27 External Velocity Packet
13.4.28 External Body Velocity Packet
13.4.29 External Heading Packet
13.4.33 External Time Packet

Packet
13.4.34 External Depth Packet
13.4.38 External Odometer Packet
13.4.39 External Air Data Packet
13.4.43 External Magnetometers Packet

Table 26: ANPP External Data Packets

12.3 GPIO Pins and the Auxiliary Port

In addition to the primary port, Motus contains two general purpose input/output (GPIO) pins and an auxiliary port on the IO Breakout Connector. These pins are multi function and can be used to extend Motus with additional peripherals, sensors and data formats. Both GPIO pins have digital input, digital output, frequency input and frequency output functionality.

GPIO1 can also function as an RS232 (only) serial transmit line and GPIO2 can function as an RS232 (only) serial receive line. The GPIO serial baud rate can be configured anywhere from 4800 to 250 K baud. The auxiliary serial baud rate can be configured anywhere from 4800 to 4 M baud. See 13.5.3.1 Supported Baud Rate. Changes to these baud rates can be made by using the Baud Rates page in INS Manager.

The available GPIO and auxiliary functions are listed in the section 12.3.1 Dynamic Pin Functions. The function of a GPIO pin or the auxiliary port can be changed at any time using the GPIO Configuration page in Manager. GPIO 1 and GPIO 2 operate at standard voltage levels for digital input, digital output, frequency input and frequency output, but when a serial data function is selected, this GPIO pin automatically switches to RS232 voltage level.

12.3.1 Dynamic Pin Functions

Function Type	Name
Tristate	12.3.1.1 Inactive
Digital Output	12.3.1.2 1PPS Output
	12.3.1.3 GNSS Fix Output
Digital Input	12.3.1.4 Zero Velocity Input
	12.3.1.5 Disable Magnetometers
	12.3.1.6 Disable GNSS
	12.3.1.7 Disable Pressure
	12.3.1.8 Set Zero Orientation Alignment
	12.3.1.9 System State Packet Trigger
	12.3.1.10 Raw Sensors Packet Trigger
	12.3.1.11 1PPS Input
	12.3.1.12 Event 1 Input
	12.3.1.13 Event 2 Input
	12.3.1.14 Odometer Direction, Forward Low
	12.3.1.15 Odometer Direction, Forward High
	12.3.1.16 Reverse Alignment, Forward Low
	12.3.1.17 Reverse Alignment, Forward High
	12.3.1.18 Zero Angular Velocity Input
Frequency Input	12.3.1.19 Odometer Input
	12.3.1.20 Pitot Tube Input
	12.3.1.21 Pressure Depth Transducer (deprecated)
	12.3.1.22 Left Wheel Speed Sensor
	12.3.1.23 Right Wheel Speed Sensor
	12.3.1.24 Wheel Speed Sensor
	12.3.1.25 Wheel Encoder Phase A
	12.3.1.26 Wheel Encoder Phase B
	12.3.1.27 Gimbal Encoder Phase A
	12.3.1.28 Gimbal Encoder Phase B

Function Type	Name
Serial Receive	12.3.1.29 NMEA Input
	12.3.1.30 Novatel GNSS Input
	12.3.1.31 Topcon GNSS Input
	12.3.1.33 ANPP Input
	12.3.1.34 RTCM Differential GNSS Corrections Input
	12.3.1.35 Trimble GNSS Input
	12.3.1.36 u-blox GNSS Input
	12.3.1.37 Hemisphere GNSS Input
	12.3.1.38 Teledyne DVL Input
	12.3.1.39 Tritech USBL Input
	12.3.1.40 Linkquest DVL Input
	12.3.1.41 Linkquest USBL Input
	12.3.1.43 Water Linked DVL Input
	12.3.1.42 Nortek DVL Input
	12.3.1.44 Nortek Nucleus DVL Input
	12.3.1.32 Septentrio GNSS Input
Serial Transmit	12.3.1.46 NMEA Output
	12.3.1.47 ANPP Output
	12.3.1.48 TSS1 Output
	12.3.1.49 Simrad 1000 Output
	12.3.1.50 Simrad 3000 Output
	12.3.1.51 Nortek Nucleus DVL Output
Serial	12.3.1.52 Serial Port Passthrough
	12.3.1.53 GNSS Receiver Passthrough

Table 27: Dynamic Pin Functions

12.3.1.1 Inactive

Type: Tristate
GPIO Port: 1, 2
Auxiliary: All

12.3.1.2 1PPS Output

Type: Digital Output
GPIO Port: 1, 2

Auxiliary: Transmit

Description: There is a difference between the output of the GPIO and auxiliary ports when set to 1PPS output mode.

1PPS Source	Voltage Level	Trigger
Auxiliary	-5 V to 5 V	Falling Edge
GPIO	0 to 5 V	Rising Edge

Figure 88: 1PPS Specifications

The pin is normally low and pulses high for 50 milliseconds to signal the precise second. The 1PPS line starts pulsing approximately 100 milliseconds after power up and always fires irrespective of whether Motus has accurate time or not. It is important to note that when Motus acquires time corrections from it's GNSS receiver, the 1PPS signal may fire at an interval of less than 1 second. This typically only occurs the first time the GNSS receiver obtains a fix after startup. The time initialised status flag can be used to determine whether the time and 1PPS line is accurate or not.

12.3.1.3 GNSS Fix Output

Type: Digital Output

GPIO Port: 1, 2

Auxiliary: Transmit

Description: In this function, the pin is low when there is no GNSS fix or a 2D fix and high when there is a 3D, SBAS, Differential or RTK GNSS fix.

12.3.1.4 Zero Velocity Input

Type: Digital Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: When using this function, a high state indicates to Motus that it is stationary. The low state indicates that the vehicle is not stationary. Use of this function can prevent drift during periods when a GNSS signal is not available.

12.3.1.5 Disable Magnetometers

Type: Digital Input

GPIO Port: All

Auxiliary: Receive

Description: This function accepts a digital input with a low state enabling the magnetometers and a high state disabling the magnetometers.

12.3.1.6 Disable GNSS

Type: Digital Input

GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function accepts a digital input with a low state enabling the GNSS and a high state disabling the GNSS.

12.3.1.7 Disable Pressure

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function accepts a digital input with a low state enabling the atmospheric pressure sensor and a high state disabling the atmospheric pressure sensor.

12.3.1.8 Set Zero Orientation Alignment

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function accepts a digital input. The input is normally low and a transition from low to high causes Motus to set its alignment so that the current orientation is zero. Due to the risk of exhausting the flash cycles, the change is not permanent and will disappear on reset. To make it permanent the **13.5.4 Installation Alignment Packet** must be read and then sent back to Motus with the permanent flag set. This function requires de-bouncing if attached to a switch.

12.3.1.9 System State Packet Trigger

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function accepts a digital input. The input is normally low and a transition from low to high causes Motus to send the System State Packet. This function requires de-bouncing if attached to a switch.

12.3.1.10 Raw Sensors Packet Trigger

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function accepts a digital input. The input is normally low and a transition from low to high causes Motus to send the **13.4.9 Raw Sensors Packet**. This function requires de-bouncing if attached to a switch.

12.3.1.11 1PPS Input

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function is designed to allow external GNSS receivers to synchronise time with Motus. It triggers on a transition from low to high.

12.3.1.12 Event 1 Input

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function is designed to allow external events to be recorded by Motus. The event is recorded in the Filter Status (see **13 Advanced Navigation Packet Protocol**), and resets after the next packet is output. The event triggers on a transition from low to high.

12.3.1.13 Event 2 Input

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function is designed to allow external events to be recorded by Motus. The event is recorded in the Filter Status (see **13 Advanced Navigation Packet Protocol**), and resets after the next packet is output. The event triggers on a transition from low to high.

12.3.1.14 Odometer Direction, Forward Low

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function is designed to take a reversing indication input for direction with an odometer or wheel encoder. It should be used in combination with Odometer Input or Wheel Speed Sensor.

12.3.1.15 Odometer Direction, Forward High

Type:	Digital Input
GPIO Port:	1, 2
Auxiliary:	Receive
Description:	This function is designed to take a reversing indication input for direction with an odometer or wheel encoder. It should be used in combination with Odometer Input or Wheel Speed Sensor.

12.3.1.16 Reverse Alignment, Forward Low

Type: Digital Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: This function is designed to reverse the alignment settings for vehicles that can have two vehicle forward directions, such as locomotives.

12.3.1.17 Reverse Alignment, Forward High

Type: Digital Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: This function is designed to reverse the alignment settings for vehicles that can have two vehicle forward directions, such as locomotives.

12.3.1.18 Zero Angular Velocity Input

Type: Digital Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: In this function, a high state indicates to Motus that the vehicle is not rotating. The low state indicates that the vehicle could be rotating. Use of this function can significantly improve heading drift performance when a GNSS signal is not available.

12.3.1.19 Odometer Input

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: This function is designed for low-resolution vehicle speed sensors and odometers. It expects a normally low input with a high state for the trigger. If the pulse length is more than 0.1 metres then this odometer input function should be used. If it is less than 0.1 metres, the wheel speed sensor function should be used.

Parameter	Value
Trigger	Low → High
Maximum Frequency	600 Khz
Maximum Pulse Rate	4294967 pulses/metre

Table 28: Odometer Specifications

12.3.1.20 Pitot Tube Input

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: This function allows fixed wing aircraft to measure airspeed to improve navigation performance. It requires a differential pressure sensor that has a frequency output such as the Kavlico P992 (frequency output option) or the Paroscientific series 5300.

12.3.1.21 Pressure Depth Transducer (deprecated)

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: This function is designed for interfacing with a legacy pressure depth transducer. These are no longer available and the function is now deprecated.

12.3.1.22 Left Wheel Speed Sensor

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: This function is designed for the left wheel of a vehicle with dual wheel speed sensors.

12.3.1.23 Right Wheel Speed Sensor

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: This function is designed for the right wheel of a vehicle with dual wheel speed sensors.

12.3.1.24 Wheel Speed Sensor

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: Receive

Description: This function is designed for high resolution vehicle speed sensors and wheel speed sensors. It expects a normally low input with a high state for the trigger. If the pulse length is more than 0.1 metres the odometer input function should be used, if it is less than 0.1 metres the wheel speed sensor function should be used.

Parameter	Value
Trigger	Low → High
Maximum Frequency	40 MHz
Maximum Pulse Rate	4,294,967,295 pulses/metre

Table 29: Wheel Speed Sensor Specifications

12.3.1.25 Wheel Encoder Phase A

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: N/A

Description: This function is designed for rotary incremental quadrature encoders. It should be used in combination with Wheel Encoder Phase B.

12.3.1.26 Wheel Encoder Phase B

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: N/A

Description: This function is designed for rotary incremental quadrature encoders. It should be used in combination with Wheel Encoder Phase A.

12.3.1.27 Gimbal Encoder Phase A

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: N/A

Description: This function is designed for interfacing with a rotary incremental quadrature encoder to measure the azimuth angle of a gimbal in which Motus is installed. It should be used in combination with Gimbal Encoder Phase B.

12.3.1.28 Gimbal Encoder Phase B

Type: Frequency Input

GPIO Port: 1, 2

Auxiliary: N/A

Description: This function is designed for interfacing with a rotary incremental quadrature encoder to measure the azimuth angle of a gimbal in which Motus is installed. It should be used in combination with **Gimbal Encoder Phase A**.

12.3.1.29 NMEA Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts external data in the NMEA 0183 format. Advanced Navigation recommends against using NMEA 0183 input wherever possible due to the inefficiency, low accuracy and precision and weak error checking of the format. All NMEA messages received must have a valid checksum. Supported messages are listed below. The recommended combination of messages are RMC with optional messages GPGSV and GPGSA. If RMC is not available, GGA and VTG can be used.

Message ID	Description	Notes
GPGGA GNGGA	3D position	All constellations
GPGLL GNGLL	2D position	All constellations
GPRMC GNRMC	2D position, 2D velocity and coarse time	All constellations
GPVTG GNVTG	2D velocity	All constellations
GPHDT GNHDT HEHDT	Heading	All constellations
GPGSV GNGSV	Satellites	Only GPS (GPGSV)
GPGSA GNGSA	Dilution of Precision	
xxPIPS	Depth below surface	
xxVBW	Water speed, ground speed	
xxHDM	Magnetic heading	
PSIMSSB	SSBL position	
xxVHW	Water speed and heading	
xxDBS	Depth below surface	

Table 30: Supported NMEA messages

12.3.1.30 Novatel GNSS Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function is designed for interfacing Motus with a Novatel GNSS receiver. It accepts data in the Novatel binary format and requires messages BESTPOSB and BESTVELB at rates higher than 1 Hz (20 Hz recommended). The message BESTSATSB is optional to display detailed satellite information. The message HEADING is also supported for ALIGN capable receivers.

12.3.1.31 Topcon GNSS Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function is designed for interfacing Motus with a Topcon GNSS receiver. It accepts data in the GRIL TPS binary format and expects messages PG and VG at rates higher than 1 Hz.

12.3.1.32 Septentrio GNSS Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function is designed for interfacing Motus with a Septentrio GNSS receiver. It accepts data in the Septentrio Binary Format (SBF) and requires blocks PVTGeodetic, PosCovGeodetic and VelCovGeodetic at rates higher than 1 Hz (10 Hz recommended). The blocks DOP, ReceiverTime, ChannelStatus, ReceiverStatus, RFStatus and MeasEpoch are optional to display detailed status and signal information. The blocks BaseVectorGeod and AttCovEuler are also supported for dual-antenna Septentrio receivers.

12.3.1.33 ANPP Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts data in the ANPP format as specified in **13 Advanced Navigation Packet Protocol**.

When this function is enabled, the GPIO and auxiliary ports function the same as the primary port, with two exceptions:

- The GPIO and auxiliary ports cannot perform a firmware update.
- When setting the ANPP packet and message output rates, the changes only apply to the port you are connected to. For example, you cannot change the primary port or GPIO port packet and message rates when connected to the auxiliary port.

12.3.1.34 RTCM Differential GNSS Corrections Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts RTCM v3 GNSS corrections messages. This allows for Differential GNSS or RTK with the Motus internal GNSS receiver.

12.3.1.35 Trimble GNSS Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function is designed for interfacing Motus with an external Trimble GNSS receiver. It accepts data in the Trimble binary format GSOF and expects packet 0x40 with records 1, 2, 8, and 12 at rates higher than 1Hz (20 Hz recommended) and optional records 9 and 34 at 1 to 2 Hz.

12.3.1.36 u-blox GNSS Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function is designed for interfacing Motus with an external u-blox GNSS receiver. It accepts data in the u-blox binary format and expects message NAV-PVT at rates higher than 1 Hz.

12.3.1.37 Hemisphere GNSS Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function is designed for interfacing Motus with an external Hemisphere GNSS receiver. It accepts data in the Hemisphere binary format and expects message Bin1 at rates higher than 1 Hz. For Hemisphere receivers that provide heading using two antennas, NMEA should be used instead as the binary format does not allow for transmission of heading information.

12.3.1.38 Teledyne DVL Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts DVL data from a Teledyne DVL.

12.3.1.39 Tritech USBL Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts USBL data from a Tritech USBL.

12.3.1.40 Linkquest DVL Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts DVL data from a Linkquest DVL.

12.3.1.41 Linkquest USBL Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts USBL data from a Linkquest USBL.

12.3.1.42 Nortek DVL Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts DVL data in the Nortek DVL binary format. It accepts the packets 0x1B (DVL Bottom Track) and 0x1D (DVL Water Track). The recommended update rate is 8 Hz.

12.3.1.43 Water Linked DVL Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts DVL data in Water Linked DVL. The recommended update rate is 8 Hz.

12.3.1.44 Nortek Nucleus DVL Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts DVL data in the Nortek Nucleus DVL binary format. It accepts the packets 0xB4 (DVL Bottom Track) and 0xBE (DVL Water Track). The recommended update rate is 8 Hz.

12.3.1.45 Valeport SVS Input

Type: Serial Receive

GPIO Port: 2

Auxiliary: Receive

Description: This function accepts data from a Valeport SVS. It accepts input from the Valeport standard format expecting pressure, temperature and speed of sound.

12.3.1.46 NMEA Output

Type: Serial Transmit

GPIO Port: 1

Auxiliary: Transmit

Description: This function outputs a configurable combination of the NMEA messages GPZDA, GPGGA, GPVTG, GPRMC, GPHDT, PASHR, GPROT, GPHEV, GPGSV, PFECGPAtt, and PFECGPHve at up to 50 Hz. The messages output and the output rate can be configured using the web UI or INS Manager. An example output is shown below.

```
$GPZDA,031644.460,07,05,2013,00,00*52

$GPGGA,031644.460,3352.3501851,S,15112.2355488,E,6,00,1.4,150
.0,M,0.0,M,,*7E

$GPVTG,089.19,T,089.19,M,000.00,N,000.00,K,E*27

$GPRMC,031644.460,A,3352.3501851,S,15112.2355488,E,0.0,89.2,0
70513,12.5,W,E*02

$GPHDT,89.2,T*06

$PASHR,031644.460,089.19,T,-00.01,-00.47,-00.00,,,,0,0*2E
```

12.3.1.47 ANPP Output

Type: Serial Transmit

GPIO Port: 1

Auxiliary: Transmit

Description: This function outputs data in the ANPP format as specified in **13 Advanced Navigation Packet Protocol**. This function is for ANPP peripheral devices only.

When this function is enabled, the GPIO and auxiliary ports function except for:

- The GPIO and auxiliary ports cannot perform a firmware update.
- When setting the ANPP packet and message output rates, the changes only apply to the port you are connected to. For example, you cannot change the primary port or GPIO port packet and message rates when connected to the auxiliary port.

12.3.1.48 TSS1 Output

Type: Serial Transmit

GPIO Port: 1

Auxiliary: Transmit

Description: This function outputs the TSS1 format at a user configurable rate of up to 50 Hz. The output rate can be configured using the page in the web UI.

12.3.1.49 Simrad 1000 Output

Type:	Serial Transmit
GPIO Port:	1
Auxiliary:	Transmit
Description:	This function outputs the Simrad 1000 format at a user configurable rate of up to 50 Hz. The output rate can be configured using the **9.6.3 Configuration Menu** dialogue in the INS Manager.

12.3.1.50 Simrad 3000 Output

Type:	Serial Transmit
GPIO Port:	1
Auxiliary:	Transmit
Description:	This function outputs the Simrad 3000 format at a user configurable rate up to 50 Hz.The output rate can be configured using the **9.6.3 Configuration Menu** dialogue in the INS Manager.

12.3.1.51 Nortek Nucleus DVL Output

Type:	Serial Transmit
GPIO Port:	1
Auxiliary:	Transmit
Description:	Upon receiving the login prompt via the Nortek Nucleus DVL Input function, this function transmits the default 'nortek' password string followed by the 'start' string required to place the DVL into measurement mode.

12.3.1.52 Serial Port Passthrough

Type:	Serial
GPIO Port:	1, 2
Auxiliary:	All
Description:	This function allows passthrough communication through either GPIO 1 and 2 or the Auxiliary port. To transmit data use the **13.3 System Packets**. When data is received it will be sent back using the same **13.3 System Packets**.

12.3.1.53 GNSS Receiver Passthrough

Type:	Serial
GPIO Port:	1, 2
Auxiliary:	All
Description:	This function allows for pass through connection to the internal GNSS receiver. This is useful for firmware updates of the internal GNSS receiver.

13 Advanced Navigation Packet Protocol

The Advanced Navigation Packet Protocol (ANPP) is a binary protocol designed with high error checking, high efficiency and safe design practices. It has a publicly defined and flexible specification. Each Advanced Navigation product uses a subset of all the ANPP packets. This manual describes ANPP packets relevant to this product.

The following packet types are available:

- System Packets
- State Packets
- Configuration Packets

Note: You can access you're device via ANPP messages via one our web based or desktop applications, such as *INS Manager*.

System Packets

Packet ID	Name	Length	R/W
System Packets			
0	13.3.1 Acknowledge Packet	4	R
1	13.3.2 Request Packet	Varies	W
2	13.3.3 Boot Mode Packet	1	R/W
3	13.3.4 Device Information Packet	24	R
4	13.3.5 Restore Factory Settings Packet	4	W
5	13.3.6 Reset Packet	4	W
10	13.3.7 Serial Port Passthrough Packet	Varies	R/W

System Packets

State Packets

Packet ID	Name	Length	R/W
State Packets			
20	13.4.1 System State Packet	100	R
21	13.4.2 Unix Time Packet	8	R
22	13.4.3 Formatted Time Packet	14	R
23	13.4.4 Status Packet	4	R
24	13.4.5 Position Standard Deviation Packet	12	R
25	13.4.6 Velocity Standard Deviation Packet	12	R
26	13.4.7 Euler Orientation Standard Deviation Packet	12	R
27	13.4.8 Quaternion Orientation Standard Deviation Packet	16	R

Packet ID	Name	Length	R/W
28	13.4.9 Raw Sensors Packet	48	R
29	13.4.10 Raw GNSS Packet	74	R/W
30	13.4.11 Satellites Packet	13	R
31	13.4.12 Detailed Satellites Packet	Varies	R
32	13.4.13 Geodetic Position Packet	24	R
33	13.4.14 ECEF Position Packet	24	R
34	13.4.15 UTM Position Packet	26	R
35	13.4.16 NED Velocity Packet	12	R
36	13.4.17 Body Velocity Packet	12	R
37	13.4.18 Acceleration Packet	12	R
38	13.4.19 Body Acceleration Packet	16	R
39	13.4.20 Euler Orientation Packet	12	R
40	13.4.21 Quaternion Orientation Packet	16	R
41	13.4.22 DCM Orientation Packet	36	R
42	13.4.23 Angular Velocity Packet	12	R
43	13.4.24 Angular Acceleration Packet	12	R
44	13.4.25 External Position and Velocity Packet	60	R/W
45	13.4.26 External Position Packet	36	R/W
46	13.4.27 External Velocity Packet	24	R/W
47	13.4.28 External Body Velocity Packet	16 or 24	R/W
48	13.4.29 External Heading Packet	8	R/W
49	13.4.30 Running Time Packet	8	R
50	13.4.31 Local Magnetic Field Packet	12	R
51	13.4.32 Odometer State Packet	20	R
52	13.4.33 External Time Packet	8	W
53	13.4.34 External Depth Packet	8	R/W
54	13.4.35 Geoid Height Packet	4	R
57	13.4.36 Wind Packet	12	R/W
58	13.4.37 Heave Packet	16	R
67	13.4.38 External Odometer Packet	13	W
68	13.4.39 External Air Data Packet	25	R/W
70	13.4.40 Raw DVL Data Packet	60	R
72	13.4.41 Gimbal State Packet	8	R/W
73	13.4.42 Automotive Packet	24	R

Packet ID	Name	Length	R/W
75	13.4.43 External Magnetometers Packet	17	R/W
80	Basestation Packet	45	R
83	13.4.44 Zero Angular Velocity Packet	8	W
85	13.4.45 Sensor Temperature Packet	32	R
90	13.4.46 Automatic Magnetic Calibration Status Packet	78	R
92	13.4.47 GNSS Position Velocity Time Packet	76	R/W
93	13.4.48 GNSS Orientation Packet	36	R/W
95	13.4.49 Aiding Source Status Packet	64	R/W

State Packets

Configuration Packets

Packet ID	Name	Length	R/W
Configuration Packets			
180	13.5.1 Packet Timer Period Packet	4	R/W
181	13.5.2 Packets Period Packet	Varies	R/W
182	13.5.3 Baud Rates Packet	17	R/W
185	13.5.4 Installation Alignment Packet	73	R/W
186	13.5.5 Filter Options Packet	17	R/W
188	13.5.6 GPIO Configuration Packet	13	R/W
189	13.5.7 Magnetic Calibration Values Packet	49	R/W
190	13.5.8 Magnetic Calibration Configuration Packet	1	W
191	13.5.9 Magnetic Calibration Status Packet	3	R
192	13.5.10 Odometer Configuration Packet	8	R/W
193	13.5.11 Set Zero Orientation Alignment Packet	5	W
194	13.5.12 Reference Point Offsets Packet	49	R/W
195	13.5.13 GPIO Output Configuration Packet	183	R/W
198	13.5.14 User Data Packet	64	R/W
199	13.5.15 GPIO Input Configuration Packet	65	R/W
207	13.5.16 Aiding Source Configuration Packet 1	243	R/W
208	13.5.17 Aiding Source Configuration Packet 2	243	R/W

Table 31: Configuration Packets

13.1 About Packets

13.1.1 Packet Structure

The ANPP packet structure and the header format are shown in the tables below. Example source code is provided in a SDK, available from the Advanced Navigation website.

Header				
Header LRC	Packet ID	Packet Length	CRC16	Packet Data

Table 32: ANPP Packet Structure

Field #	Bytes Offset	Data Type	Size	Description
1	0	u8	1	Header LRC
2	1	u8	1	Packet ID
3	2	u8	1	Packet Length
4	3	u16	2	CRC

Table 33: ANPP Header Format

13.1.1.1 Header LRC

The header LRC (Longitudinal Redundancy Check) provides error checking on the packet header. It also allows the decoder to find the start of a packet by scanning for a valid LRC. Only the lower 8 bits of the calculation below are provided in the LRC field of the packet header. The LRC can be found using the following:

$$LRC = (PacketID + PacketLength + crc_0 + crc_1) \oplus \mathrm{0xFF} + 1$$

13.1.1.2 Packet ID

The packet ID is used to distinguish the contents of the packet. Packet IDs range from 0 to 255. See 13 Advanced Navigation Packet Protocol.

System packets have packet IDs in the range 0 to 19. These packets are implemented the same by every device using ANPP.

State packets are packets that contain data that changes with time, i.e. temperature. State packets can be set to output at a certain rate. State packets are packet IDs in the range 20 to 179.

Configuration packets are used for reading and writing device configuration. Configuration packets are packet IDs in the range 180 to 255.

13.1.1.3 Packet Length

The packet length denotes the length of the packet data, i.e. from byte index 5 onwards inclusive. Packet length has a range of 0 – 255.

13.1.1.4 CRC

The CRC is a CRC16-CCITT. The starting value is 0xFFFF. The CRC covers only the packet data.

13.1.2 Packet Requests

Any of the state and configuration packets can be requested at any time using the 13.3 System Packets.

13.1.3 Packet Acknowledgement

When configuration packets are sent to Motus, it will reply with an acknowledgement packet that indicates whether the configuration change was successful or not. See **13.3 System Packets**.

External data packets will also generate negative acknowledgement packets if there is a problem with the packet. Positive acknowledgements will not be sent. See **12.2 External Data** for a list of External Data Packets.

13.1.4 Packet Rates

The packet rates can be configured either using the web UI, or INS Manager or through the **13.5.2 Packets Period Packet**. By default Motus is configured to output the **13.4.1 System State Packet** at 20 Hz and the **13.4.9 Raw Sensors Packet** at 20 Hz. When configuring packet rates it is essential to ensure the baud rate is capable of handling the data throughput. This can be calculated using the rate and packet size. The packet size is calculated by adding the packet overhead to the packet length; the packet overhead is a fixed value of 5. For example to output the system state packet at 50Hz the calculation would be:

Data throughput = (100 (packet length) + 5 (fixed packet overhead)) * 50 (rate)

Data throughput = 5250 bytes per second

Minimum baud rate = data throughput x 10 bits per byte (including framing bits) x 1.1 (10% margin) = 57750 Baud

Closest standard baud rate = 115200 Baud

When multiple packets are set to output at the same rate, the order the packets output is from lowest ID to highest ID.

13.1.5 Packet Timing

Packets are output in order of packet ID from lowest ID to highest ID and all packets that are output in one sequence have their data matched to the same time of validity. The time of validity can be found in either the **13.4.1 System State Packet**, the **13.4.2 Unix Time Packet** or the **13.4.3 Formatted Time Packet**. For example if the **13.4.2 Unix Time Packet**, **13.4.4 Status Packet** and **13.4.16 NED Velocity Packet** packet were all set to output at 10 Hz, at each 0.1 second period the three packets would output consecutively by order of packet ID with all data synchronised between them and the **13.4.2 Unix Time Packet** providing the time of validity for the other two packets.

13.2 Data Types

The following data types are used in the packet protocol. All data types in the protocol are little endian byte ordering.

Abbreviation	Bytes	Also known as
u8	1	unsigned char, unsigned byte, uint8_t
s8	1	char, byte, int8_t
u16	2	unsigned short, uint16_t
s16	2	short, int16_t
u32	4	unsigned int, unsigned long, uint32_t
s32	4	int, long, int32_t
u64	8	unsigned long long, uint64_t

Abbreviation	Bytes	Also known as
s64	8	long long, int64_t
fp32	4	float
fp64	8	double

Table 34: Data Type Abbreviations used in the ANPP

13.3 System Packets

13.3.1 Acknowledge Packet

Acknowledgement Packet				
Packet ID				0
Length				4
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Packet ID being acknowledged
2	1	u16	2	CRC of packet being acknowledged
3	3	u8	1	13.3.1.1 Acknowledge Result

Table 35: Acknowledge Packet

13.3.1.1 Acknowledge Result

Value	Description
0	Acknowledge success
1	Acknowledge failure, CRC error
2	Acknowledge failure, packet size incorrect
3	Acknowledge failure, values outside of valid ranges
4	Acknowledge failure, system flash memory failure
5	Acknowledge failure, system not ready
6	Acknowledge failure, unknown packet

Table 36: Acknowledge Result

13.3.2 Request Packet

Request Packet				
Packet ID				1
Length				1 x number of packets requested
Read / Write				Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Packet ID requested
+				Field 1 repeats for additional packet requests

Table 37: Request Packet

13.3.3 Boot Mode Packet

Boot Mode Packet				
Packet ID				2
Length				1
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	13.3.3.1 Boot Mode Types

Table 38: Boot Mode Packet

13.3.3.1 Boot Mode Types

Value	Description
0	Bootloader
1	Main Program

Table 39: Boot Mode Types

13.3.4 Device Information Packet

Device Information Packet				
Packet ID				3
Length				24
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	Software version
2	4	u32	4	13.3.4.1 Device ID
3	8	u32	4	Hardware revision
4	12	u32	4	Motus Serial number part 1

Device Information Packet				
5	16	u32	4	Motus Serial number part 2
6	20	u32	4	Motus Serial number part 3

Table 40: Device Information Packet

13.3.4.1 Device ID

Value	Description
1	Spatial
4	Spatial FOG
5	Spatial Dual
11	Orientus v3
13	Air Data Unit 1
14	Subsonus
16	Spatial FOG Dual
17	Motus
19	GNSS Compass
21	Subsonus Tag
22	Poseidon
26	Certus
28	Boreas D90
41	Boreas D70
43	Boreas A90
44	Boreas A70
49	Certus Mini A
50	Certus Mini N
51	Certus Mini D
54	Boreas D50
56	Boreas A50
69	Air Data Unit 2

Table 41: Device IDs

13.3.5 Restore Factory Settings Packet

Restore Factory Settings Packet	
Packet ID	4
Length	4

Restore Factory Settings Packet				
Read / Write				Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	Verification sequence (set to 0x85429E1C)

Table 42: Restore Factory Settings Packet

13.3.6 Reset Packet

Reset Packet				
Packet ID				5
Length				4
Read / Write				Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	13.3.6.1 Verification Sequence Values

Table 43: Reset Packet

13.3.6.1 Verification Sequence Values

Value	Description
0x21057A7E	Standard hot start reset
0x9A5D38B7	Cold start reset

Table 44: Verification Sequence Values

13.3.7 Serial Port Passthrough Packet

Serial Port Passthrough Packet				
Packet ID				10
Length				Variable, up to 255 bytes
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Passthrough Route
2	1			Passthrough Data

Table 45: Serial Port Passthrough Packet

13.3.7.1 Passthrough Route

Value	Description
1	GPIO 1 and 2
2	Auxiliary

Table 46: Passthrough Route

13.4 State Packets

13.4.1 System State Packet

System State Packet				
Packet ID				20
Length				100
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u16	2	**13.4.1.1 System Status**
2	2	u16	2	**13.4.1.2 Filter Status**
3	4	u32	4	**13.4.1.4 Unix Time Seconds**
4	8	u32	4	**13.4.1.5 Microseconds**
5	12	fp64	8	Latitude (rad)
6	20	fp64	8	Longitude (rad)
7	28	fp64	8	Height (m)
8	36	fp32	4	Velocity north (m/s)
9	40	fp32	4	Velocity east (m/s)
10	44	fp32	4	Velocity down (m/s)
11	48	fp32	4	Body acceleration X (m/s/s)
12	52	fp32	4	Body acceleration Y (m/s/s)
13	56	fp32	4	Body acceleration Z (m/s/s)
14	60	fp32	4	G force (g)
15	64	fp32	4	Roll (radians)
16	68	fp32	4	Pitch (radians)
17	72	fp32	4	Heading (radians)
18	76	fp32	4	Angular velocity X (rad/s)
19	80	fp32	4	Angular velocity Y (rad/s)
20	84	fp32	4	Angular velocity Z (rad/s)
21	88	fp32	4	Latitude standard deviation (m)
22	92	fp32	4	Longitude standard deviation (m)
23	96	fp32	4	Height standard deviation (m)

Table 47: System State Packet

13.4.1.1 System Status

This field contains 16 bits that indicate problems with the system. These are boolean fields with a zero indicating false and one indicating true.

Bit	Description
0	System Failure
1	Accelerometer Sensor Failure
2	Gyroscope Sensor Failure
3	Magnetometer Sensor Failure
4	Reserved (set to zero)Pressure Sensor Failure
5	Reserved (set to zero)
6	Accelerometer Over Range
7	Gyroscope Over Range
8	Magnetometer Over Range
9	Reserved (set to zero)Pressure Over Range
10	Minimum Temperature Alarm
11	Maximum Temperature Alarm
12	Low Voltage Alarm
13	High Voltage Alarm
14	Reserved (set to zero)
15	Data Output Overflow Alarm

Table 48: System Status

13.4.1.2 Filter Status

This field contains 16 bits that indicate the status of the filters. These are boolean fields with a zero indicating false and one indicating true.

Bit	Description
0	Orientation Filter Initialised
1	Navigation Filter Initialised
2	Heading Initialised
3	UTC Time Initialised
4	13.4.1.3 GNSS Fix Status
5	
6	
7	Event 1 Occurred
8	Event 2 Occurred
9	Internal GNSS Enabled
10	Heading Active
11	Velocity Heading Enabled
12	Atmospheric Altitude Enabled
13	External Position Active
14	External Velocity Active
15	External Heading Active

Table 49: Filter Status

13.4.1.3 GNSS Fix Status

Value	Description
0	No GNSS fix
1	2D GNSS fix
2	3D GNSS fix
3	SBAS GNSS fix
4	Differential GNSS fix
5	PPP GNSS fix
6	RTK Float GNSS fix
7	RTK Fixed GNSS fix

Table 50: GNSS Fix Status

13.4.1.4 Unix Time Seconds

This field provides the Unix time in seconds, using the standard epoch of January 1, 1970.

13.4.1.5 Microseconds

This field provides the sub-second component of time. It is represented as microseconds since the last second. Minimum value is 0 and maximum value is 999999.

13.4.2 Unix Time Packet

Unix Time Packet				
Packet ID				21
Length				8
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	Unix time stamp (seconds). See **13.4.1.4 Unix Time Seconds**
2	4	u32	4	**Microseconds**

Table 51: Unix Time Packet

13.4.3 Formatted Time Packet

Formatted Time Packet				
Packet ID				22
Length				14
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	Microseconds
2	4	u16	2	Year
3	6	u16	2	Year day, 0 - 365
4	8	u8	1	Month, 0 - 11
5	9	u8	1	Month Day, 1 - 31
6	10	u8	1	Week Day, 0 – 6 (0 = Sunday)
7	11	u8	1	Hour, 0 - 23
8	12	u8	1	Minute, 0 - 59
9	13	u8	1	Second, 0 - 59

Table 52: Formatted Time Packet

13.4.4 Status Packet

Status Packet				
Packet ID				23
Length				4
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u16	2	System Status
2	2	u16	2	Filter Status

Table 53: Status Packet

13.4.5 Position Standard Deviation Packet

Position Standard Deviation Packet				
Packet ID				24
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Latitude standard deviation (m)
2	4	fp32	4	Longitude standard deviation (m)
3	8	fp32	4	Height standard deviation (m)

Table 54: Position Standard Deviation Packet

13.4.6 Velocity Standard Deviation Packet

Velocity Standard Deviation Packet				
Packet ID				25
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Velocity north standard deviation (m/s)
2	4	fp32	4	Velocity east standard deviation (m/s)
3	8	fp32	4	Velocity down standard deviation (m/s)

Table 55: Velocity Standard Deviation Packet

13.4.7 Euler Orientation Standard Deviation Packet

Euler Orientation Standard Deviation Packet				
Packet ID				26
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Roll standard deviation (rad)
2	4	fp32	4	Pitch standard deviation(rad)
3	8	fp32	4	Heading standard deviation(rad)

Table 56: Euler Orientation Standard Deviation Packet

13.4.8 Quaternion Orientation Standard Deviation Packet

Quaternion Orientation Standard Deviation Packet				
Packet ID				27
Length				16
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Q0 standard deviation
2	4	fp32	4	Q1 standard deviation
3	8	fp32	4	Q2 standard deviation
4	12	fp32	4	Q3 standard deviation

Table 57: Quaternion Orientation Standard Deviation Packet

13.4.9 Raw Sensors Packet

13.4.9.1 Raw Sensors Packet - INS

Raw Sensors Packet				
Packet ID				28
Length				48
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Accelerometer X (m/s/s)
2	4	fp32	4	Accelerometer Y (m/s/s)
3	8	fp32	4	Accelerometer Z (m/s/s)
4	12	fp32	4	Gyroscope X (rad/s)
5	16	fp32	4	Gyroscope Y (rad/s)
6	20	fp32	4	Gyroscope Z (rad/s)
7	24	fp32	4	Magnetometer X (mG)
8	28	fp32	4	Magnetometer Y (mG)
9	32	fp32	4	Magnetometer Z (mG)
10	36	fp32	4	IMU Temperature (deg C)
11	40	fp32	4	Pressure (Pascals)
12	44	fp32	4	Pressure Temperature (deg C)

Table 58: Raw Sensors Packet - INS

13.4.10 Raw GNSS Packet

This packet represents the raw data as it is received from the GNSS receiver. The position is not corrected for antenna position offset and the velocity is not compensated for the antenna lever arm offset. The INS position and velocity that are in the other packets are corrected for antenna position offset and lever arm.

Note: This packet is supported, but is superseded by **13.4.47 GNSS Position Velocity Time Packet** and **13.4.48 GNSS Orientation Packet** which provide additional information.

Note: Raw GNSS is only available if your device is interfaced to an external GNSS receiver.

Raw GNSS Packet				
Packet ID				29
Length				74
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	Unix time stamp (seconds)
2	4	u32	4	Microseconds
3	8	fp64	8	Latitude (rad)
4	16	fp64	8	Longitude (rad)
5	24	fp64	8	Height (m)
6	32	fp32	4	Velocity north (m/s)
7	36	fp32	4	Velocity east (m/s)
8	40	fp32	4	Velocity down (m/s)
9	44	fp32	4	Latitude standard deviation (m)
10	48	fp32	4	Longitude standard deviation (m)
11	52	fp32	4	Height standard deviation (m)
12	56	fp32	4	Tilt (rad)
13	60	fp32	4	Heading (rad)
14	64	fp32	4	Tilt standard deviation (rad)
15	68	fp32	4	Heading standard deviation (rad)
16	72	u16	2	Status. See **13.4.10.1 Raw GNSS Status**

Table 59: Raw GNSS Packet

13.4.10.1 Raw GNSS Status

Bit	Description
0	GNSS Fix Status
1	
2	
3	Doppler velocity valid
4	Time valid
5	External GNSS
6	Tilt valid
7	Heading valid
8	Floating ambiguity heading
9-15	Reserved (set to zero)

Table 60: Raw GNSS Status

13.4.11 Satellites Packet

Satellites Packet				
Packet ID				30
Length				13
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	HDOP
2	4	fp32	4	VDOP
3	8	u8	1	GPS satellites
4	9	u8	1	GLONASS satellites
5	10	u8	1	BeiDou satellites
6	11	u8	1	GALILEO satellites
7	12	u8	1	SBAS satellites

Table 61: Satellites Packet

13.4.12 Detailed Satellites Packet

Detailed Satellites Packet				
Packet ID				31
Length				7 x number of satellites
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Satellite System
2	1	u8	1	Satellite number (PRN)
3	2	u8	1	13.4.12.2 Satellite Frequencies
4	3	u8	1	Elevation (deg)
5	4	u16	2	Azimuth (deg)
6	6	u8	1	SNR
+				Fields 1-6 repeat for additional satellites

Table 62: Detailed Satellites Packet

Each packet can contain information for up to 36 satellites. If there are more than 36 satellites to report, only one packet will be sent per epoch.

13.4.12.1 Satellite Systems

Value	System
0	Unknown
1	GPS
2	GLONASS
3	BeiDou
4	Galileo
5	SBAS
6	QZSS
7-9	Reserved (set to zero)

Table 63: Satellite Systems

13.4.12.2 Satellite Frequencies

Bit	Description
0	L1 C/A
1	L1 C
2	L1 P
3	L1 M
4	L2 C

Bit	Description
5	L2 P
6	L2 M
7	L5

Table 64: Satellite Frequencies

13.4.13 Geodetic Position Packet

Geodetic Position Packet				
Packet ID				32
Length				24
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp64	8	Latitude (rad)
2	8	fp64	8	Longitude (rad)
3	16	fp64	8	Height (m)

Table 65: Geodetic Position Packet

13.4.14 ECEF Position Packet

ECEF Position Packet				
Packet ID				33
Length				24
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp64	8	ECEF X (m)
2	8	fp64	8	ECEF Y (m)
3	16	fp64	8	ECEF Z (m)

Table 66: ECEF Position Packet

13.4.15 UTM Position Packet

UTM Position Packet				
Packet ID				34
Length				26
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp64	8	Northing (m)

UTM Position Packet				
2	8	fp64	8	Easting (m)
3	16	fp64	8	Height (m)
4	24	u8	1	Zone number
5	25	s8	1	Zone character

Table 67: UTM Position Packet

13.4.16 NED Velocity Packet

NED Velocity Packet				
Packet ID				35
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Velocity north (m/s)
2	4	fp32	4	Velocity east (m/s)
3	8	fp32	4	Velocity down (m/s)

Table 68: NED Velocity Packet

13.4.17 Body Velocity Packet

Body Velocity Packet				
Packet ID				36
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Velocity X (m/s)
2	4	fp32	4	Velocity Y (m/s)
3	8	fp32	4	Velocity Z (m/s)

Table 69: Body Velocity Packet

13.4.18 Acceleration Packet

This packet includes the acceleration due to gravity.

Acceleration Packet				
Packet ID				37
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**

Acceleration Packet				
1	0	fp32	4	Acceleration X (m/s/s)
2	4	fp32	4	Acceleration Y (m/s/s)
3	8	fp32	4	Acceleration Z (m/s/s)

Table 70: Acceleration Packet

13.4.19 Body Acceleration Packet

This packet does not include the acceleration due to gravity.

Body Acceleration Packet				
Packet ID				38
Length				16
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Body acceleration X (m/s/s)
2	4	fp32	4	Body acceleration Y (m/s/s)
3	8	fp32	4	Body acceleration Z (m/s/s)
4	12	fp32	4	G force (g)

Table 71: Body Acceleration Packet

13.4.20 Euler Orientation Packet

Euler Orientation Packet				
Packet ID				39
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Roll (rad)
2	4	fp32	4	Pitch (rad)
3	8	fp32	4	Heading (rad)

Table 72: Euler Orientation Packet

13.4.21 Quaternion Orientation Packet

Quaternion Orientation Packet				
Packet ID				40
Length				16
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	QS
2	4	fp32	4	QX
3	8	fp32	4	QY
4	12	fp32	4	QZ

Table 73: Quaternion Orientation Packet

13.4.22 DCM Orientation Packet

DCM Orientation Packet				
Packet ID				41
Length				36
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	DCM[0][0]
2	4	fp32	4	DCM[0][1]
3	8	fp32	4	DCM[0][2]
4	12	fp32	4	DCM[1][0]
5	16	fp32	4	DCM[1][1]
6	20	fp32	4	DCM[1][2]
7	24	fp32	4	DCM[2][0]
8	28	fp32	4	DCM[2][1]
9	32	fp32	4	DCM[2][2]

Table 74: DCM Orientation Packet

13.4.23 Angular Velocity Packet

Angular Velocity Packet				
Packet ID				42
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Angular velocity X (rad/s)
2	4	fp32	4	Angular velocity Y (rad/s)
3	8	fp32	4	Angular velocity Z (rad/s)

Table 75: Angular Velocity Packet

13.4.24 Angular Acceleration Packet

Angular Acceleration Packet				
Packet ID				43
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Angular acceleration X (rad/s/s)
2	4	fp32	4	Angular acceleration Y (rad/s/s)
3	8	fp32	4	Angular acceleration Z (rad/s/s)

Table 76: Angular Acceleration Packet

13.4.25 External Position and Velocity Packet

External Position and Velocity Packet				
Packet ID				44
Length				60
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp64	8	Latitude (rad)
2	8	fp64	8	Longitude (rad)
3	16	fp64	8	Height (m)
4	24	fp32	4	Velocity north (m/s)
5	28	fp32	4	Velocity east (m/s)
6	32	fp32	4	Velocity down (m/s)
7	36	fp32	4	Latitude standard deviation (m)
8	40	fp32	4	Longitude standard deviation (m)
9	44	fp32	4	Height standard deviation (m)
10	48	fp32	4	Velocity north standard deviation (m/s)
11	52	fp32	4	Velocity east standard deviation (m/s)
12	56	fp32	4	Velocity down standard deviation (m/s)

Table 77: External Position and Velocity Packet

13.4.26 External Position Packet

External Position Packet				
Packet ID				45
Length				36
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp64	8	Latitude (rad)
2	8	fp64	8	Longitude (rad)
3	16	fp64	8	Height (m)
4	24	fp32	4	Latitude standard deviation (m) Must be non-zero positive value
5	28	fp32	4	Longitude standard deviation (m) Must be non-zero positive value
6	32	fp32	4	Height standard deviation (m) Must be non-zero positive value

Table 78: External Position Packet

13.4.27 External Velocity Packet

External Velocity Packet				
Packet ID				46
Length				24
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Velocity north (m/s)
2	4	fp32	4	Velocity east (m/s)
3	8	fp32	4	Velocity down (m/s)
4	12	fp32	4	Velocity north standard deviation (m/s)
5	16	fp32	4	Velocity east standard deviation (m/s)
6	20	fp32	4	Velocity down standard deviation (m/s)

Table 79: External Velocity Packet

13.4.28 External Body Velocity Packet

This packet has been defined to support a single velocity standard deviation for all three axes, or, different values for all three axes. Modify the packet length in the packet header accordingly, depending on which format you are using.

External Body Velocity Packet				
Packet ID				47
Length				16 or 24
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Velocity X (m/s)
2	4	fp32	4	Velocity Y (m/s)
3	8	fp32	4	Velocity Z (m/s)
4	12	fp32	4	Velocity X standard deviation (m/s)
5	16	fp32	4	OPTIONAL: Velocity standard deviation Y (m/s)
6	20	fp32	4	OPTIONAL: Velocity standard deviation Z (m/s)

Table 80: External Body Velocity Packet

13.4.29 External Heading Packet

External Heading Packet				
Packet ID				48
Length				8
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Heading (rad)
2	4	fp32	4	Heading standard deviation (rad)

Table 81: External Heading Packet

13.4.30 Running Time Packet

This packet provides the running time in seconds since the most recent power on.

Running Time Packet				
Packet ID				49
Length				8
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	Running time (seconds)
2	4	u32	4	**13.4.1.5 Microseconds**

Table 82: Running Time Packet

13.4.31 Local Magnetic Field Packet

Local Magnetic Field Packet				
Packet ID				50
Length				12
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Local magnetic field X (mG)
2	4	fp32	4	Local magnetic field Y (mG)
3	8	fp32	4	Local magnetic field Z (mG)

Table 83: Local Magnetic Field Packet

13.4.32 Odometer State Packet

Odometer State Packet				
Packet ID				51
Length				20
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	s32	4	Odometer pulse count
2	4	fp32	4	Odometer distance (m)
3	8	fp32	4	Odometer speed (m/s)
4	12	fp32	4	Odometer slip (m)
5	16	u8	1	Odometer active
6	17		3	Reserved (set to zero)

Table 84: Odometer State Packet

13.4.33 External Time Packet

External Time Packet				
Packet ID				52
Length				8
Read / Write				Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	**13.4.1.4 Unix Time Seconds**
2	4	u32	4	**13.4.1.5 Microseconds**

Table 85: External Time Packet

13.4.34 External Depth Packet

External Depth is the measured distance below Mean Sea Level

External Depth Packet				
Packet ID				53
Length				8
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Depth (m)
2	4	fp32	4	Depth standard deviation (m)

Table 86: External Depth Packet

13.4.35 Geoid Height Packet

This packet provides the offset at the current location between the WGS84 ellipsoid and the EGM96 geoid model. This can be used to determine the current height above mean sea level and also depth through the following equations:

Height Above Mean Sea Level = WGS84 Height – Geoid Height

Depth = Geoid Height – WGS84 Height

WGS84 Height is the height at the current location in the System State packet, see **13.4.1 System State Packet**

Geoid Height Packet				
Packet ID				54
Length				4
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Geoid height (m)

Table 87: Geoid Height Packet

13.4.36 Wind Packet

This packet provides the Motus current 2D wind velocity. These values are only valid when external air data is provided to Motus. This can be either through the **13.4.39 External Air Data Packet** or when a pitot tube is interfaced to one of the GPIO pins. When this packet is written to the device, it disables the wind estimation filter and the wind values provided are used statically.

Wind Packet				
Packet ID				57
Length				12
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**

Wind Packet				
1	0	fp32	4	Wind velocity north (m/s)
2	4	fp32	4	Wind velocity east (m/s)
3	8	fp32	4	Wind velocity standard deviation (m/s)

Table 88: Wind Packet

13.4.37 Heave Packet

Heave Packet				
Packet ID				58
Length				16
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Heave point 1 (m)
2	4	fp32	4	Heave point 2 (m)
3	8	fp32	4	Heave point 3 (m)
4	12	fp32	4	Heave point 4 (m)

Table 89: Heave Packet

13.4.38 External Odometer Packet

External Odometer Packet				
Packet ID				67
Length				13
Read / Write				Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Estimated delay (s)
2	4	fp32	4	Speed (m/s)
3	8		4	Reserved (set to zero)
4	12	u8	1	Odometer flags

Table 90: External Odometer Packet

13.4.38.1 Odometer Flags

Bit	Description
0	Reversing detection supported
1-7	Reserved (set to zero)

Table 91: Odometer Flags

13.4.39 External Air Data Packet

External Air Data Packet				
Packet ID				68
Length				25
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Barometric altitude delay (s)
2	4	fp32	4	Airspeed delay (s)
3	8	fp32	4	Barometric altitude (m)
4	12	fp32	4	True Airspeed (TAS) (m/s) when temperature sensor **is** connected. Indicated Airspeed (IAS) (m/s) when temperature sensor **is not** connected.
5	16	fp32	4	Barometric altitude standard deviation (m)
6	20	fp32	4	Airspeed standard deviation (m/s)
7	24	u8	1	Flags

Table 92: External Air Data Packet

13.4.39.1 External Air Data Flags

Bit	Description
0	Barometric altitude set and valid
1	Airspeed set and valid
2	Barometric altitude sensor over-range
3	Airspeed sensor over-range
4	Barometric altitude sensor failure
5	Airspeed sensor failure
6-7	Reserved (set to zero)

Table 93: External Air Data Flags

Barometric altitude does not need to be referenced to any co-ordinate frame or QNH. If the barometric altitude reference is changed during operation, the barometric altitude reference reset flag should be set for the next packet.

13.4.40 Raw DVL Data Packet

Raw DVL Data Packet				
Packet ID				70
Length				60
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u32	4	Unix timestamp (seconds)
2	4	u32	4	Microseconds
3	8	u32	4	Flags, see 13.4.40.1 Raw DVL Data Flags
4	12	fp32	4	Bottom velocity X (m/s)
5	16	fp32	4	Bottom velocity Y (m/s)
6	20	fp32	4	Bottom velocity Z (m/s)
7	24	fp32	4	Bottom velocity standard deviation (m/s)
8	28	fp32	4	Water velocity X (m/s)
9	32	fp32	4	Water velocity Y (m/s)
10	36	fp32	4	Water velocity Z (m/s)
11	40	fp32	4	Water velocity standard deviation (m/s)
12	44	fp32	4	Water velocity layer depth (m)
13	48	fp32	4	Depth (m)
14	52	fp32	4	Altitude (m)
15	56	fp32	4	Temperature (deg C)

Table 94: Raw DVL Data Packet

13.4.40.1 Raw DVL Data Flags

Bit	Description
0	Bottom velocity valid
1	Water velocity valid
2	Temperature valid
3	Depth valid
4	Altitude valid

Table 95: Raw DVL Data Flags

13.4.41 Gimbal State Packet

Motus will accept (write) Current angle values between -2pi and 2pi exclusive.

Motus will output (read) values between -pi and pi for current angle.

Gimbal State Packet				
Packet ID				72
Length				8
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Current angle (rad)
2	4		4	Reserved (set to zero)

Table 96: Gimbal State Packet

13.4.42 Automotive Packet

The value of the slip angle will be zero unless velocity is greater than 3.45 m/s and heading is valid.

Automotive Packet				
Packet ID				73
Length				24
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Virtual odometer distance (m)
2	4	fp32	4	Slip Angle (rad) Value reported is zero, unless heading is valid and the velocity is greater than 3.45 m/s
3	8	fp32	4	Velocity X (m/s)
4	12	fp32	4	Velocity Y (m/s)
5	16	fp32	4	Distance standard deviation (m)
6	20		4	Reserved (set to zero)

Table 97: Automotive Packet

13.4.43 External Magnetometers Packet

Heading can be derived using the magnetometers of an additional external Advanced Navigation product to input external magnetometer values.

The 2D, 3D or automatic magnetic calibration of Motus cannot be used to calibrate the external magnetometer values. External magnetometers need to be calibrated before feeding into Motus.

To use External Magnetic Heading, disable Magnetic Heading Enabled and Automatic Magnetic Calibration Enabled options in the **9.6.3 Configuration Menu**of the Configuration Menu.

When sending external magnetometers the external heading flag will become active.

External Magnetometers Packet	
Packet ID	75
Length	17

External Magnetometers Packet				
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Delay (s)
2	4	fp32	4	Magnetometer X (milligauss)
3	8	fp32	4	Magnetometer Y (milligauss)
4	12	fp32	4	Magnetometer Z (milligauss)
6	16	u8	1	External Magnetometer Flags

Table 98: External Magnetometers Packet

13.4.43.1 External Magnetometers Flags

Bit	Description
0	Magnetometers failure
1	Magnetometers over-range

Table 99: External magnetometers Flags

13.4.44 Zero Angular Velocity Packet

Zero Angular Velocity Packet				
Packet ID				83
Length				8
Read / Write				Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	fp32	4	Duration. See **13.4.44.1 Zero Angular Velocity Duration**
2	4		4	Reserved (set to zero)

Table 100: Zero Angular Velocity Packet

13.4.44.1 Zero Angular Velocity Duration

Duration is the duration that the unit has been stationary for about the heading axis. The minimum value is 0.1 seconds and the maximum value is 2 seconds.

13.4.45 Sensor Temperature Packet

Sensor Temperatures Packet				
Packet ID				85
Length				32
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**

Sensor Temperatures Packet				
1	0	fp32[3]	12	Accelerometer temperature
2	12	fp32[3]	12	Gyroscope temperature
3	24	fp32	4	Magnetometer temperature
4	28	fp32	4	Pressure sensor temperature

Table 101: Sensor temperature packet

13.4.46 Automatic Magnetic Calibration Status Packet

Automatic Magnetic Calibration Status Packet				
Packet ID				90
Length				78
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	**Method**
2	1	u8	1	**Flags**
3	2	fp32	4	Convergence
4	6	fp32	4	Scale Factor X
5	10	fp32	4	Scale Factor Y
6	14	fp32	4	Scale Factor Z
7	18	fp32	4	Soft Iron X
8	22	fp32	4	Soft Iron Y
9	26	fp32	4	Soft Iron Z
10	30	fp32	4	Hard Iron X
11	34	fp32	4	Hard Iron Y
12	38	fp32	4	Hard Iron Z
13	42	fp32	4	Scale Factor SD X
14	46	fp32	4	Scale Factor SD Y
15	50	fp32	4	Scale Factor SD Z
16	54	fp32	4	Soft Iron SD X
17	58	fp32	4	Soft Iron SD Y
18	62	fp32	4	Soft Iron SD Z
19	66	fp32	4	Hard Iron SD X
20	70	fp32	4	Hard Iron SD Y
21	74	fp32	4	Hard Iron SD Z

Table 102: Automatic Magnetic Calibration Status Packet

13.4.46.1 Automatic Magnetic Calibration Status Method

Value	Description
0	Disabled
1	GNSS Aided
2	Online

Table 103: Automatic Magnetic Calibration Status Method

13.4.46.2 Automatic Magnetic Calibration Status Flags

Bit	Description
0	Ready
1	Coarse Complete
2	Existing Calibration
3	Awaiting Filters
4	Manual Calibration in Progress
5	Invalid Vehicle Type
6	Magnetic Heading Disabled
7	Reserved (set to zero)

Table 104: Automatic Magnetic Calibration Status Flags

13.4.47 GNSS Position Velocity Time Packet

This packet provides the raw Position, Velocity, and Time (PVT) data as it is received from the GNSS receiver.

The position and velocity reported describe the location of the GNSS primary antenna (not the vehicle reference point).

Note: Raw GNSS is only available if your device is interfaced to an external GNSS receiver.

Note: This packet partly supersedes the Raw GNSS Packet (Packet ID 29).

GNSS Position Velocity Time Packet				
Packet ID				92
Length				76
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	13.4.47.1 GNSS ID
2	1		1	Reserved (set to zero)
3	2	u16	2	13.4.47.2 GNSS Position Velocity Time Status Bit-field
4	4	u32	4	Posix time (s)
5	8	u32	4	Posix time (µs)
6	12	fp64	8	Latitude (rad)
7	20	fp64	8	Longitude (rad)
8	28	fp64	8	Altitude above ellipsoid (m)
9	36	fp32	4	Position standard deviation, North (m)
10	40	fp32	4	Position standard deviation, East (m)
11	44	fp32	4	Position standard deviation, Down (m)
12	48	fp32	4	Velocity, North (m/s)
13	52	fp32	4	Velocity, East (m/s)
14	56	fp32	4	Velocity, Down (m/s)
15	60	fp32	4	Velocity standard deviation, North (m/s)
16	64	fp32	4	Velocity standard deviation, East (m/s)
17	68	fp32	4	Velocity standard deviation, Down (m/s)
18	72	u32	4	Latency (µs)

Table 105: GNSS PVT Packet

13.4.47.1 GNSS ID

Value	Description
0	Internal GNSS
Other values	External GNSS sources

Table 106: GNSS ID Status Bitfield

13.4.47.2 GNSS Position Velocity Time Status Bitfield

This packet provides the raw Position, Velocity, and Time (PVT) solution data from the GNSS receiver. This packet supersedes the Raw GNSS Packet (ID 29).

Boolean fields have zero indicating false, and one indicating true.

Bit Range	Num Bits	Description
0-2	3	**13.4.1.3 GNSS Fix Status**
3-5	3	Spoofing Status, per: **13.4.47.3 Spoofing and Interference Status**
6-8	3	Interference Status, per: **13.4.47.3 Spoofing and Interference Status**
9	1	Velocity valid
10	1	Time valid
11	1	Antenna disconnected
12	1	Antenna short
13	1	GNSS Failure
14-15	2	Reserved (set to zero)

Table 107: GNSS PVT Status Bitfield

13.4.47.3 Spoofing and Interference Status

Status values for Spoofing and Interference, as set in the *GNSS Position Velocity Time* and *GNSS Orientation Status* Bitfield.

Value	Description
0	Unknown
1	None
2	Detected and mitigated
3	Detected and unmitigated
4-7	Reserved (set to zero)

Table 108: Spoofing and Interference Status

13.4.48 GNSS Orientation Packet

13.4.48.1 GNSS Orientation Packet

This packet represents the raw orientation data as it is received from the GNSS receiver.

GNSS Orientation Packet is only applicable for Dual Antenna installations. It does not apply to Single Antenna installations.

The orientation is that of the line between the two dual GNSS antennas (not of the vehicle body) and is described in the NED frame.

Note: Raw GNSS is only available if your device is interfaced to an external GNSS receiver.

Note: This packet partly supersedes the Raw GNSS Packet (Packet ID 29).

GNSS Orientation Packet				
Packet ID				93
Length				36
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	13.4.47.1 GNSS ID
2	1		1	Reserved (set to zero)
3	2	u16	2	GNSS Orientation Status Bitfield
4	4	u32	4	Posix time (s)
5	8	u32	4	Posix time (µs)
6	12	fp32	4	Azimuth of GNSS antennas baseline (rad)
7	16	fp32	4	Azimuth Standard Deviation (rad)
8	20	fp32	4	Tilt of GNSS antennas baseline (rad)
9	24	fp32	4	Tilt Standard Deviation (rad)
10	28	fp32	4	Baseline length (m)
11	32	u32	4	Latency (µs)

Table 109: GNSS Orientation Packet

GNSS Orientation Status Bitfield

Bit Range	Num Bits	Description
0-2	3	13.4.1.3 GNSS Fix Status
3	1	Antenna disconnected
4	1	Antenna short
5	1	GNSS Failure
6-8	3	Spoofing Status, per: 13.4.47.3 Spoofing and Interference Status
9-11	3	Interference Status, per: 13.4.47.3 Spoofing and Interference Status
12-15	4	Reserved (set to zero)

Table 110: GNSS Orientation Status Bitfield

13.4.49 Aiding Source Status Packet

Aiding Source Status Packet				
Packet ID				95
Length				64
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u16	2	13.4.49.1 Aiding Source Status Field
2	2	u16	2	Internal GNSS Orientation Status
3	4	u16	2	Internal Magnetometers Status
4	6	u16	2	Internal Pressure Status
5	8	u16	2	External GNSS PVT Status
6	10	u16	2	External GNSS Orientation Status
7	12	u16	2	External Position Status
8	14	u16	2	External Odometer Status
9	16	u16	2	External Heading Status
10	18	u16	2	External Pressure Status
11	20	u16	2	External Velocity Status
12	22	u16	2	External Position Velocity Status
13	24	u16	2	External Body Velocity Status
14	26	u16	2	External Air Data Status
15	28	u16	2	External Magnetometers Status
16	30	u16	2	External LVS Status
17	32	u16	2	Internal Depth Sensor Status

Aiding Source Status Packet				
18	34	u16	2	External Subsonus Status
19	36	u16	2	External DVL Data Status
20	38	u16	2	External Depth Status
21	40	u16	2	External USBL Status
22	42		22	Reserved (set to zero)

13.4.49.1 Aiding Source Status Field

All aiding source status fields share the same definition:

Bit	Description
0	Online - Communicating with device
1	Valid - Providing valid filter information
2	Fault - Reporting a fault
3-9	Reserved (set to zero)
10-15	**Aiding Source Origin** • 0: Internal • 1: Primary Port • 2: Aux Port • 3: GPIO • 4: Data Stream 1 • 5: Data Stream 2 • 6: Data Stream 3 • 7: Data Stream 4

13.5 Configuration Packets

Configuration packets can be both read from and written to the Motus. On many of the configuration packets the first byte is a permanent flag. A zero in this field indicates that the settings will be lost on reset, a one indicates that they will be permanent (i.e. stored in flash).

Note: The number of configuration packets with the permanent flag set to high should be minimised to reduce the risk of exhausting the flash cycles.

13.5.1 Packet Timer Period Packet

Packet Timer Period Packet	
Packet ID	180
Length	4

Packet Timer Period Packet				
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u8	1	13.5.1.1 UTC Synchronisation
3	2	u16	2	13.5.1.2 Packet Timer Period

Table 111: Packet Timer Period Packet

13.5.1.1 UTC Synchronisation

This is a boolean value that determines whether or not the packet timer is synchronised with UTC time, with zero for disabled and one for enabled. For UTC Synchronisation to be enabled the packet timer period must multiply into 1000000 evenly. For example if the packet timer period is 10000 (10 ms), 1000000/10000 = 100 which is valid for UTC synchronisation. If the packet timer period is 15000 (15 ms), 1000000/15000 = 66.6666 which is not valid for UTC synchronisation. To get the rate, use the following formula:

Packet Timer Rate = 1000000/(Packet Timer Period) Hz

13.5.1.2 Packet Timer Period

This is a value in microseconds that sets the master packet timer period. The minimum value is 1000 (1 ms) or 1000 Hz and the maximum value is 65000 (65 ms) or 15.38 Hz. The packet timer period must be in increments of 1000.

13.5.2 Packets Period Packet

This packet allows the user to configure, add, remove, modify and clear state packets (packets 20 through 180).

Note: This packet will only affect the port from which the packet is sent, leaving the other ports unaffected. e.g. If you are connected to the Primary serial port, sending this packet will only change the ANPP output for the Primary serial port.

Packets Period Packet				
Packet ID				181
Length				2 + (5 x number of packet periods)
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u8	1	Clear Existing Packet Periods
3	2	u8	1	Packet ID
4	3	u32	4	Packet Period
+				Fields 3-4 repeat for additional packet periods

Table 112: Packets Period Packet

13.5.2.1 Clear Existing Packets

This is a boolean field, when set to one it deletes any existing packet rates. When set to zero existing packet rates remain. Only one packet rate can exist per packet ID, so new packet rates will overwrite existing packet rates for the same packet ID.

13.5.2.2 Packet Period

This indicates the period in units of the packet timer period. The packet rate can be calculated as follows.

Packet Rate = 1000000/(Packet Period x Packet Timer Period) Hz

For example if the packet timer period is set to 1000 (1 ms). Setting packet ID 20 with a packet period of 50 will give the following.

Packet 20 Rate = 1000000/(50 x 1000)

Packet 20 Rate = 20 Hz

13.5.3 Baud Rates Packet

Baud Rates Packet				
Packet ID				182
Length				17
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u32	4	Primary RS232 and RS422 see 13.5.3.1 Supported Baud Rate
3	5	u32	4	GPIO 1 & 2 **Maximum Baud Rate 250000** see 13.5.3.1 Supported Baud Rate
4	9	u32	4	Auxiliary RS232 see 13.5.3.1 Supported Baud Rate
5	13		4	Reserved (set to zero)

Table 113: Baud Rates Packet

13.5.3.1 Supported Baud Rate

Note: Not all Advanced Navigation products support minimum and maximum baud rates.

Supported Baud Rates - Bits 0-30
2400
4800
9600
19200
38400
57600

Supported Baud Rates - Bits 0-30
115200
230400
250000
460800
500000
800000
921600
1000000
1250000
2000000
4000000

Table 114: Supported Baud Rates

13.5.4 Installation Alignment Packet

Installation Alignment Packet				
Packet ID				185
Length				73
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	fp32	4	Alignment DCM[0][0]
3	5	fp32	4	Alignment DCM[0][1]
4	9	fp32	4	Alignment DCM[0][2]
5	13	fp32	4	Alignment DCM[1][0]
6	17	fp32	4	Alignment DCM[1][1]
7	21	fp32	4	Alignment DCM[1][2]
8	25	fp32	4	Alignment DCM[2][0]
9	29	fp32	4	Alignment DCM[2][1]
10	33	fp32	4	Alignment DCM[2][2]
11 <add [3] jhere]	37	fp32	4	GNSS antenna offset X (m)[1]
12 <add [3] jhere]	41	fp32	4	GNSS antenna offset Y (m)[1]
13 <add	45	fp32	4	GNSS antenna offset Z (m)

Installation Alignment Packet				
[3] jhere]				
14	49	fp32	4	Odometer offset X (m)[2]
15	53	fp32	4	Odometer offset Y (m)[2]
16	57	fp32	4	Odometer offset Z (m)[2]
17	61	fp32	4	External data offset X (m)[3]
18	65	fp32	4	External data offset Y (m)[3]
19	69	fp32	4	External data offset Z (m)[3]

Table 115: Installation Alignment Packet

[1]Update to GNSS Antenna offset results in an update to fields:

- **13.5.16 Aiding Source Configuration Packet 1**: #3 Internal GNSS PVT - Position Offset
- **13.5.16 Aiding Source Configuration Packet 1**: #5 External GNSS PVT - Position Offset

[2]Update to Odometer offset results in an update to fields:

- **13.5.16 Aiding Source Configuration Packet 1**: #8 External Odometer - Position Offset
- **13.5.17 Aiding Source Configuration Packet 2**: #6 External DVL Data - Position Offset

[3]Update to External data offset results in an update to fields:

- **13.5.16 Aiding Source Configuration Packet 1**: #7 External Position - Position Offset
- **13.5.16 Aiding Source Configuration Packet 1**: #11 External Velocity - Position Offset
- **13.5.16 Aiding Source Configuration Packet 1**: #12 External Position Velocity - Position Offset
- **13.5.16 Aiding Source Configuration Packet 1**: #13 External Body Velocity - Position Offset
- **13.5.17 Aiding Source Configuration Packet 2**: #3 Internal Depth Sensor - Position Offset
- **13.5.17 Aiding Source Configuration Packet 2**: #4 External Subsonus - Position Offset
- **13.5.17 Aiding Source Configuration Packet 2**: #9 External USBL Position Offset

13.5.4.1 Alignment DCM

The alignment DCM (Direction Cosine Matrix) is used to represent an alignment offset of Motus from its standard alignment. A DCM is used rather than Euler angles for accuracy reasons. To convert Euler angles to DCM, use the formula below with angles in radians.

DCM[0][0] = cos(heading) * cos(pitch)

DCM[0][1] = sin(heading) * cos(pitch)

DCM[0][2] = -sin(pitch)

DCM[1][0] = -sin(heading) * cos(roll) + cos(heading) * sin(pitch) * sin(roll)

DCM[1][1] = cos(heading) * cos(roll) + sin(heading) * sin(pitch) * sin(roll)

DCM[1][2] = cos(pitch) * sin(roll)

DCM[2][0] = sin(heading) * sin(roll) + cos(heading) * sin(pitch) * cos(roll)

DCM[2][1] = -cos(heading) * sin(roll) + sin(heading) * sin(pitch) * cos(roll)

DCM[2][2] = cos(pitch) * cos(roll)

13.5.5 Filter Options Packet

Filter Options Packet				
Packet ID				186
Length				17
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u8	1	13.5.5.1 Vehicle Types
3	2	u8	1	Internal GNSS enabled (boolean)
4	3		1	Magnetic Heading enabled (boolean)
5	4		1	Atmospheric altitude enabled (boolean)
6	5	u8	1	Velocity heading enabled (boolean)
7	6	u8	1	Reversing detection enabled (boolean)
8	7	u8	1	Motion analysis enabled (boolean)
9	8		1	Automatic magnetic calibration enabled (boolean)
10	9		8	Reserved (set to zero)

Table 116: Filter Options Packet

13.5.5.1 Vehicle Types

Value	Description
0	Unlimited
1	Bicycle or Motorcycle
2	Car
3	Hovercraft

Value	Description
4	Submarine
5	3D Underwater Vehicle
6	Fixed Wing Plane
7	3D Aircraft
8	Human
9	Boat
10	Large Ship
11	Stationary
12	Stunt Plane
13	Race Car
14	Train

Table 117: Vehicle Types

13.5.6 GPIO Configuration Packet

GPIO Configuration Packet				
Packet ID				188
Length				13
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u8	1	GPIO1 function. See **13.5.6.1 GPIO Functions**
3	2	u8	1	GPIO2 function. See **13.5.6.1 GPIO Functions**
4	3	u8	1	Auxiliary RS232 transmit function. See **13.5.6.2 Auxiliary Functions**
5	4	u8	1	Auxiliary RS232 receive function. See **13.5.6.2 Auxiliary Functions**
6	5		1	Reserved (set to zero)

Table 118: GPIO Configuration Packet

13.5.6.1 GPIO Functions

Value	Description	GPIO Function Type
0	Inactive	GPIO1, GPIO2
1	1PPS Output	
2	GNSS Fix Output	
3	Odometer Input	
4	Zero Velocity Input	
5	Pitot Tube Input	
6	NMEA Input	GPIO2
7	NMEA Output	GPIO1
8	Novatel GNSS Input	GPIO2
9	Topcon GNSS Input	
11	ANPP Input	
12	ANPP Output	GPIO1
13	Disable Magnetometers	GPIO1, GPIO2
15	Disable Pressure	GPIO1, GPIO2
16	Set Zero Orientation Alignment	
17	System State Packet Trigger	
18	Raw Sensors Packet Trigger	
19	RTCM Differential GNSS Corrections Input	GPIO2
20	Trimble GNSS Input	
21	u-blox GNSS Input	
22	Hemisphere GNSS Input	
23	Teledyne DVL Input	
24	Tritech USBL Input	
25	Linkquest DVL Input	

Value	Description	GPIO Function Type
26	Pressure Depth Transducer	GPIO1, GPIO2
27	Left Wheel Speed Sensor	
28	Right Wheel Speed Sensor	
29	1PPS Input	
30	Wheel Speed Sensor	
31	Wheel Encoder Phase A	
32	Wheel Encoder Phase B	
33	Event 1 Input	
34	Event 2 Input	
35	Linkquest USBL Input	GPIO2
38	GNSS Receiver Passthrough	GPIO1, GPIO2
39	TSS1 Output	GPIO1
40	Simrad 1000 Output	
41	Simrad 3000 Output	
42	Serial Port Passthrough	GPIO1, GPIO2
43	Gimbal Encoder Phase A	
44	Gimbal Encoder Phase B	
45	Odometer Direction, Forward Low	
46	Odometer Direction, Forward High	
51	Nortek DVL Input	GPIO2
53	Reverse Alignment, Forward Low	GPIO1, GPIO2
54	Reverse Alignment, Forward High	
55	Zero Angular Velocity Input	
59	Water Linked DVL Input	GPIO2
63	Nortek Nucleus DVL Input	
64	Nortek Nucleus DVL Output	GPIO1
65	Valeport SVS Input	GPIO2
66	Septentrio GNSS Input	GPIO2

Table 119: GPIO functions

13.5.6.2 Auxiliary Functions

Value	Description	Auxiliary Function Type
0	Inactive	Transmit, Receive
1	1PPS Output	Transmit
2	GNSS Fix Output	

Value	Description	Auxiliary Function Type
3	Odometer Input	Receive
4	Zero Velocity Input	
5	Pitot Tube Input	
6	NMEA Input	
7	NMEA Output	Transmit
8	Novatel GNSS Input	Receive
9	Topcon GNSS Input	
11	ANPP Input	
12	ANPP Output	Transmit
13	Disable magnetometers	Receive
14	Disable GNSS	Receive
15	Disable Pressure	
16	Set Zero Orientation Alignment	
17	System State Packet Trigger	
18	Raw Sensors Packet Trigger	
19	RTCM Differential GNSS Corrections Input	
20	Trimble GNSS Input	
21	u-blox GNSS Input	
22	Hemisphere GNSS Input	
23	Teledyne DVL Input	
24	Tritech USBL Input	
25	Linkquest DVL Input	
26	Pressure Depth Transducer	
27	Left Wheel Speed Sensor	
28	Right Wheel Speed Sensor	
29	1PPS Input	
30	Wheel Speed Sensor	
33	Event 1 Input	
34	Event 2 Input	
35	Linkquest USBL Input	
38	GNSS Receiver Passthrough	Transmit, Receive
39	TSS1 Output	Transmit
40	Simrad 1000 Output	
41	Simrad 3000 Output	

Value	Description	Auxiliary Function Type
42	Serial Port Passthrough	Transmit, Receive
45	Odometer Direction, Forward Low	Receive
46	Odometer Direction, Forward High	
51	Nortek DVL Input	
53	Reverse Alignment, Forward Low	
54	Reverse Alignment, Forward High	
55	Zero Angular Velocity Input	
59	Water Linked DVL Input	
63	Nortek Nucleus DVL Input	
64	Nortek Nucleus DVL Output	Transmit
65	Valeport SVS Input	Receive
66	Septentrio GNSS Input	Receive

Table 120: Auxiliary Functions

13.5.7 Magnetic Calibration Values Packet

This packet can be used to inspect the magnetic calibration values or write custom magnetic calibration values to the system. Under normal operation with automatic magnetic calibration the user should not need to use this packet.

Magnetic Calibration Values Packet				
Packet ID				189
Length				49
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	fp32	4	Hard iron bias X
3	5	fp32	4	Hard iron bias Y
4	9	fp32	4	Hard iron bias Z
5	13	fp32	4	Soft iron transformation XX
6	17	fp32	4	Soft iron transformation XY
7	21	fp32	4	Soft iron transformation XZ
8	25	fp32	4	Soft iron transformation YX
9	29	fp32	4	Soft iron transformation YY
10	33	fp32	4	Soft iron transformation YZ
11	37	fp32	4	Soft iron transformation ZX
12	41	fp32	4	Soft iron transformation ZY

Magnetic Calibration Values Packet				
13	45	fp32	4	Soft iron transformation ZZ

Table 121: Magnetic Calibration Values Packet

13.5.8 Magnetic Calibration Configuration Packet

Magnetic Calibration Configuration Packet				
Packet ID				190
Length				1
Read / Write				Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	13.5.8.1 Magnetic Calibration Actions

Table 122: Magnetic Calibration Configuration Packet

13.5.8.1 Magnetic Calibration Actions

Value	Description
0	Cancel magnetic calibration
2	Start 2D magnetic calibration
3	Start 3D magnetic calibration
4	Reset calibration to defaults

Table 123: Magnetic Calibration Actions

13.5.9 Magnetic Calibration Status Packet

Magnetic Calibration Status Packet				
Packet ID				191
Length				3
Read / Write				Read
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	13.5.9.1 Magnetic Calibration Status
2	1	u8	1	Magnetic calibration progress (%)
3	2	u8	1	Local magnetic error (%)

Table 124: Magnetic Calibration Status Packet

13.5.9.1 Magnetic Calibration Status

Value	Description
0	Magnetic calibration not completed
1	2D magnetic calibration completed
2	3D magnetic calibration completed

Value	Description
3	Custom values magnetic calibration completed
5	2D calibration in progress
6	3D calibration in progress
7	2D calibration error: excessive roll
8	2D calibration error: excessive pitch
9	Calibration error: sensor over range event
10	Calibration error: time-out
11	Calibration error: system error
12	Calibration error: interference error

Table 125: Magnetic Calibration Status

13.5.10 Odometer Configuration Packet

Odometer Configuration Packet				
Packet ID				192
Length				8
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u8	1	Automatic pulse measurement active
3	2		2	Reserved (set to zero)
4	4	fp32	4	Pulse length (m)

Table 126: Odometer Configuration Packet

13.5.11 Set Zero Orientation Alignment Packet

Set Zero Orientation Alignment Packet				
Packet ID				193
Length				5
Read / Write				Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u32	4	Verification sequence (set to 0x9A4E8055)

Table 127: Set Zero Orientation Alignment Packet

13.5.12 Reference Point Offsets Packet

The reference point offsets packet can be used to adjust the measurement point that all data is referenced to. By default all the values of this packet are zero and the measurement point that all data is referenced to is the centre of the Motus unit. The primary reference point offset can be used to adjust the measurement point to a different location on the vehicle, in the body co-ordinate frame. The primary reference point offset applies to all data output including NMEA etc as well as heave point 1.

The other heave point offsets do not apply to heave point 1 in the **13.4.37 Heave Packet**.

Reference Point Offsets Packet				
Packet ID				194
Length				49
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	fp32	4	Heave point 1 offset X (m) / Primary reference point offset X (m)
3	5	fp32	4	Heave point 1 offset Y (m) / Primary reference point offset Y (m)
4	9	fp32	4	Heave point 1 offset Z (m) / Primary reference point offset Z (m)
5	13	fp32	4	Heave point 2 offset X (m)
6	17	fp32	4	Heave point 2 offset Y (m)
7	21	fp32	4	Heave point 2 offset Z (m)
8	25	fp32	4	Heave point 3 offset X (m)
9	29	fp32	4	Heave point 3 offset Y (m)
10	33	fp32	4	Heave point 3 offset Z (m)
11	37	fp32	4	Heave point 4 offset X (m)
12	41	fp32	4	Heave point 4 offset Y (m)
13	45	fp32	4	Heave point 4 offset Z (m)

Table 128: Reference Point Offsets Packet

13.5.13 GPIO Output Configuration Packet

GPIO Output Configuration Packet				
Packet ID				195
Length				33
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u8	1	NMEA fix behaviour, see **13.5.13.1 NMEA Fix Behaviour**
3	2	u16	2	GPZDA Rates, see **13.5.13.2 GPIO Output Rates**
4	4	u16	2	GPGGA Rates, see **13.5.13.2 GPIO Output Rates**
5	6	u16	2	GPVTG Rates, see **13.5.13.2 GPIO Output Rates**
6	8	u16	2	GPRMC Rates, see **13.5.13.2 GPIO Output Rates**
7	10	u16	2	GPHDT Rates, see **13.5.13.2 GPIO Output Rates**
8	12	u16	2	GPGLL Rates, see **13.5.13.2 GPIO Output Rates**
9	14	u16	2	PASHR Rates, see **13.5.13.2 GPIO Output Rates**
10	16	u16	2	TSS1 Rates, see **13.5.13.2 GPIO Output Rates**
11	18	u16	2	Simrad Rates, see **13.5.13.2 GPIO Output Rates**
12	20	u16	2	GPROT Rates, see **13.5.13.2 GPIO Output Rates**
13	22	u16	2	GPHEV Rates, see **13.5.13.2 GPIO Output Rates**
14	24	u16	2	GPGSV Rates, see **13.5.13.2 GPIO Output Rates**
15	26	u16	2	PFECAtt Rates, see **13.5.13.2 GPIO Output Rates**
16	28	u16	2	PFECHve Rates, see **13.5.13.2 GPIO Output Rates**
17	30		3	Reserved (set to zero)

Table 129: GPIO Output Configuration Packet

13.5.13.1 NMEA Fix Behaviour

Value	Description
0	Normal
1	Always indicate 3D fix when the navigation filter is initialised

Table 130: NMEA Fix Behaviour

13.5.13.2 GPIO Output Rates

Value	Description
0-3	GPIO 1 rate, see **13.5.13.3 Port Output Rates**

Value	Description
4-7	Auxiliary RS232 Tx rate, see 13.5.13.3 Port Output Rates
8-15	Reserved (set to zero)

Table 131: GPIO output rates

13.5.13.3 Port Output Rates

Value	Bit 3	Bit 2	Bit 1	Bit 0	Description
0	0	0	0	0	Disabled
1	0	0	0	1	0.1 Hz
2	0	0	1	0	0.2 Hz
3	0	0	1	1	0.5 Hz
4	0	1	0	0	1 Hz
5	0	1	0	1	2 Hz
6	0	1	1	0	5 Hz
7	0	1	1	1	10 Hz
8	1	0	0	0	25 Hz
9	1	0	0	1	50 Hz
10	1	0	1	0	8 Hz

Table 132: Port Output Rates

13.5.14 User Data Packet

This packet is for storage of users data. The data itself is not used by Motus.

User Data Packet				
Packet ID				198
Length				64
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0		64	User Data

Table 133: User Data Packet

13.5.15 GPIO Input Configuration Packet

GPIO Input Configuration Packet	
Packet ID	199
Length	65
Read / Write	Read / Write

GPIO Input Configuration Packet				
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	fp32	4	Gimbal radians per encoder tick
3	5		60	Reserved (set to zero)

Table 134: GPIO Input Configuration Packet

13.5.16 Aiding Source Configuration Packet 1

Aiding Source Configuration Packet 1				
Packet ID				207
Length				243
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u16	2	**13.5.16.1 Aiding Source Bitmask 1**
3	3	f32[3]	12	Internal GNSS PVT - Position Offset
4	15	f32[3]	12	Internal GNSS Orientation - Orientation Offset
5	27	f32[3]	12	External GNSS PVT - Position Offset
6	39	f32[3]	12	External GNSS Orientation - Orientation Offset
7	51	f32[3]	12	External Position - Position Offset
8	63	f32[3]	12	External Odometer - Position Offset
9	75	f32[3]	12	External Heading - Orientation Offset
10	87	f32[3]	12	External Pressure - Position Offset
11	99	f32[3]	12	External Velocity - Position Offset
12	111	f32[3]	12	External Position Velocity - Position Offset
13	123	f32[3]	12	External Body Velocity - Position Offset
14	135	f32[3]	12	External Body Velocity - Orientation Offset
15	147	f32[3]	12	External Air Data - Position Offset
16	159	f32[3]	12	External Air Data - Orientation Offset
17	171	f32[3]	12	External Magnetometers - Orientation Offset
18	183	f32[3]	12	External LVS - Position Offset
19	195	f32[3]	12	External LVS - Orientation Offset
20-22	207		36	Reserved (set to zero)

13.5.16.1 Aiding Source Bitmask 1

Bit	Aiding Source
0	Internal GNSS PVT
1	Internal GNSS Orientation
2	Internal Magnetometers
3	Internal Pressure
4	External GNSS PVT
5	External GNSS Orientation
6	External Position
7	External Odometer
8	External Heading
9	External Pressure
10	External Velocity
11	External Position Velocity
12	External Body Velocity
13	External Air Data
14	External Magnetometers
15	External LVS

13.5.17 Aiding Source Configuration Packet 2

Aiding Source Configuration Packet 2				
Packet ID				208
Length				243
Read / Write				Read / Write
Field #	**Bytes Offset**	**Data Type**	**Size**	**Description**
1	0	u8	1	Permanent
2	1	u16	2	**13.5.17.1 Aiding Source Bitmask 2**
3	3	f32[3]	12	Internal Depth Sensor - Position Offset
4	15	f32[3]	12	External Subsonus - Position Offset
5	27	f32[3]	12	External Subsonus - Orientation Offset
6	39	f32[3]	12	External DVL Data - Position Offset
7	51	f32[3]	12	External DVL Data - Orientation Offset
8	63	f32[3]	12	External Depth -Position Offset
9	75	f32[3]	12	External USBL -Position Offset

Aiding Source Configuration Packet 2				
10	87		156	Reserved (set to zero)

13.5.17.1 Aiding Source Bitmask 2

Bit	Aiding Source
0	Internal Depth Sensor
1	External Subsonus
2	External DVL Data
3	External Depth
4	External USBL
5-15	Reserved (set to zero)